

**LOCAL GOVERNMENT UNIT 1: WEB-BASED GIS SYSTEM
FOR AUTOMATED ZONING COMPLIANCE
DEVELOPMENT PERMIT PROCESSING
IN LOCAL DEVELOPMENT**

A Capstone

Presented to the Faculty of
The College of Computer Studies
Bestlink College of the Philippines

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

**ALTHANIEL O. ABORDO
MA.CRISTINA D. BUENAFLO
JOSHUA CANCERAN
YUMIE MARGARETH N. DALAGAN
JUSTIN MICHAEL ESPINO
ROSARY MAE C. VELICARIA**

August 2026

APPROVAL SHEET

This capstone entitled **LOCAL GOVERNMENT UNIT 1: WEB-BASED GIS SYSTEM FOR AUTOMATED ZONING COMPLIANCE DEVELOPMENT PERMIT PROCESSING IN LOCAL DEVELOPMENT**, prepared and submitted by **Abordo, Althaniel O., Buenaflor, Ma. Cristina D., Canceran, Joshua, Dalagan, Yumie Margareth N., Espino, Justin Michael, Velicaria, Rosary Mae C.**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval for Oral Defense.

Ronald Roldan Jr.
Adviser

CAPSTONE REVIEW PANEL

Approved by the Committee on Oral Examination with a grade of ____

ENGR. JEAN LESTER M. CACHOLA	LASER RYAN V. LLENO, MSIT
Member	Member

DR. JUNNEL E. AVESTRO, DIT
Chairperson

Accepted and approved in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology.

ROMMEL J. CONSTANTINO, DIT
Dean, College of Computing Science

Date of Final Defense: August 26, 2026

ACKNOWLEDGEMENT

The researchers would like to express their heartfelt thanks and gratitude to the following persons who, in one way or another, has contributed much, and extended willingness and support needed to make this research possible:

Dr. Maria M. Vicente, President/CEO, Bestlink College of the Philippines, for her generosity and kind heart in establishing this institution and giving opportunities to those less fortunate students to continue their studies and pursue their dreams;

Dr. Glenn S. Cabacang, Executive Vice President, for providing the needed information to complete this research;

Dr. Ryan Ignacio, Vice President for Academic Affairs, for his support and encouragement to make this thesis writing possible;

Ms. Marian M. Vicente, Vice President for Administration and Finance, for his words of encouragement and motivation;

Dr. Joy Evelyn A. Ignacio, College Associate Research Head, for her good heart to extend her help needed by the researchers.

Dr. Rommel J. Constantino, Dean, College of Computing Studies of Bestlink College of the Philippines, for providing a guideline documentation in capstone project.

Dr. Rommel J. Constantino, Program Head, Bachelor of Science in Information Technology, for the constant supervision as well as providing

necessary information regarding the project and also for his support in completing this project

Ms. Kris Villaflores, Research Coordinator, for helping us in improving our research and guiding us in completing this project.

Mr. Ronald Roldan Jr., Capstone Adviser, for giving us suggestions and ideas to improve our research and guiding us in completing this project.

Panelists, **Engr. Jean Lester M. Cachola**, **Mr. Laser Ryan V. Llano**, and **Dr. Junnel E. Avestro**, who extended their effort and time to be able to constructively criticize this thesis and share their knowledge with them to deepen and widen their needed information.

Families and Friends, for all the financial and moral support that have enabled the researchers to triumph all the challenges, especially during the lowest time that served as their inspiration to complete this study; and

Above all, to the **Almighty God**, for the strength and knowledge that were used for the accomplishment of this research journey.

THE RESEARCHERS

DEDICATION

This research study is wholeheartedly dedicated first and foremost to the researchers, for their unwavering dedication, patience, and perseverance in bringing this study to completion despite the challenges encountered along the way.

To our beloved parents and families, whose love, sacrifices, and support have been our constant source of strength throughout this journey.

To our friends and classmates, for standing by us, listening, and encouraging us through every late night and difficult moment.

To Alyana Lagrante, for the quiet strength, unwavering support, and constant comfort and motivation that made the completion of this work possible.

To our research adviser and panel of professors, for their patience, invaluable guidance, constructive feedback, and generosity of time in shaping this study into what it has become.

Above all, to Almighty God, the source of our wisdom and strength, for the grace to complete this work. All that we have accomplished, we offer back to You.

THE RESEARCHERS

ABSTRACT

**Title: LOCAL GOVERNMENT UNIT 1: WEB-BASED GIS
SYSTEM FOR AUTOMATED ZONING COMPLIANCE
DEVELOPMENT PERMIT PROCESSING IN LOCAL
DEVELOPMENT**

**Authors: ALTHANIEL O. ABORDO
MA. CRISTINA D. BUENAFLORE
JOSHUA CANCERAN
YUMIE MARGARETH N. DALAGAN
JUSTIN MICHAEL ESPINO
ROSARY MAE C. VELICARIA**

Degree: Bachelor of Science of Information Technology

Major: Information Management, Information Security

Date of Completion:

The modernization of Local Government Units (LGUs) in the Philippines is hampered by traditional, manual, and paper-based workflows for zoning and permit processing, which are often prone to human error and significant administrative delays. This study addresses these inefficiencies through the development of a Web-Based GIS System for Automated Zoning and Permits. The system integrates Geographic

Information System (GIS) technology into the municipal approval framework to automate land-use compliance verification, replacing subjective manual checks with precise, data-driven assessments.

The project encompasses three functional pillars: an interactive GIS module for spatial visualization, an automated compliance engine for coordinate-based validation (including setbacks, height limits, and floor area ratios), and a post-approval monitoring dashboard to track construction adherence. Utilizing an Agile Scrum methodology, the system was developed in iterative sprints to ensure functional increments, such as automated zoning checks and secure database management, were rigorously tested and refined.

The findings demonstrate that the implementation of this digital infrastructure enhances operational efficiency, streamlines internal workflows for zoning officers, and fosters transparency for public applicants by providing real-time status tracking and high-precision spatial visualization. Ultimately, the system provides a reliable digital audit trail that minimizes administrative discretion, ensuring that municipal growth is sustainable and strictly compliant with the Comprehensive Land Use Plan (CLUP).

TABLE OF CONTENTS

	Page
TITLE PAGE	i
APPROVAL SHEET	ii
ACKNOWLEDGMENT	iii
DEDICATION	v
ABSTRACT	vi
TABLE OF CONTENTS	viii
LIST OF TABLES	xiii
LIST OF FIGURES	xv
LIST OF APPENDICES	xvii
CHAPTER	
1 Introduction	
1.1. Background of the Capstone Project	1
1.2. Context and Scope	3
1.3. Problem Statement	6
1.4. Objectives and Goals	8
1.5. Significance and Relevance	9
1.6. Structure of the Document	11

CHAPTER**2 Review of Related Literature**

2.1.	Agile Scrum Methodology Overview	13
2.2.	Enterprise Architecture Concepts	15
2.3.	Microservices Architecture	17
2.4	DevOps and CI/CD	19
2.5	Relevant Studies and Research	21
2.6	Integration of Information Systems in Enterprise Environment	28

CHAPTER**3 Methodology**

3.1.	Agile Scrum Methodology in the Project	31
3.2.	Roles (e.g., Scrum Master, Product Owner, Development Team)	33
3.3.	Sprint Cycles (e.g., Sprint Planning, Daily Standups, Sprint Review)	38
3.4.	Scrum Artifacts	41
3.5	Microservices Architecture	164
3.6	DevOps Implementation	168
3.7	Integration Approach for Information Systems	171
3.8	Introduction to TOGAF and the Four Architectural Domains	187

CHAPTER**4 Requirements Analysis**

4.1. Stakeholder Identification	191
4.2. Requirements Gathering Techniques	197
4.3. User Stories and Use Cases	200
4.4. Functional Requirements for Integration	204

5 Business Process Architecture

5.1. Identification of Business Processes	208
5.2. Business Process Diagrams	212
5.3. Alignment of Integrated System with Business Processes	216
5.4. Business Process Improvements	221

6 Application Architecture

6.1. Components of Application Architecture	227
6.2. Application Architecture Diagrams	231
6.3. Integration of Software Modules	238
6.4. Communication and Interaction Patterns	243

7 Data Architecture

7.1. Data Sources and Types	245
7.2. Data Flow Diagrams	250
7.3. Data Storage and Management	251
7.4. Data Synchronization Across Systems	260

8 Technology Architecture

8.1 Technology Stack and Infrastructure	263
8.2 Software Technologies	268
8.3. Scalability and Performance Considerations	273

9 Development Process

9.1. Agile Scrum Roles and Responsibilities	278
9.2. Sprint Planning and Backlog Management	280
9.3. Sprint Execution and Deliverables	282
9.4. Challenges Faced In the Development Process	284

10 Implementation

10.1. Technical Implementation Details	286
10.2. Tools and Technologies Used	289
10.3. Code Integration and Interoperability	296
10.4. Integration Testing and Debugging	298

11. Testing and Quality Assurance

11.1. Testing Strategies and Methodologies	301
11.2. Test Cases and Test Data	303
11.3. Test Results and Bug Reports	310
11.4. Quality Assurance Measures	317

12. Results and Evaluation

12.1. Project Outcomes and Deliverables	319
12.2. Alignment with Project Objectives	322

12.3 Stakeholder and User Feedback	324
12.4. Lessons Learned	325

CHAPTER

5 Conclusion

13.1. Key Takeaways and Summary	329
13.2. Project Achievements and Contributions	331
13.3. Future Work and Enhancements	335
13.4. Closing Remarks	338

REFERENCES

APPENDICES

CURRICULUM VITAE

LIST OF TABLES

Table no.

Page

1	Development Team	34
2	Roles and Responsibilities	38
3	Scrum Board	40
4	Product Backlog (User Stories)	104
5	Product Backlog for EIS Information Security	109
6	Product Backlog for EIS Standard	115
7	Product Backlog for EIS Integration	126
8	Product Backlog for Application System Analytics	128
9	Product Backlog for EIS Analytics	132
10	Sprint Backlog (Technical Execution)	157
11	Increment	164
12	Stakeholder Identification	196
13	User Stories	202
14	Use Cases	204
15	Roles and Responsibilities	280
16	Development Tools and Technologies	292
17	Deployment and Infrastructure Tools	293
18	Testing and Quality Assurance Tools	294

19	Collaboration and Project Management Tools	296
20	Sample Test Cases and Test Data	310
21	Test Results and Bug Reports	316

LIST OF FIGURES

Figure no.		Page
1	Agile Scrum Framework	31
2	Scrum Process	38
3	Sprint 1 Burndown Charts	157
4	Sprint 2 Burndown Charts	158
5	Microservices Diagram	165
6	Communication Pattern for Microservices	166
7	Data Flow Diagram for Microservices	167
8	CI/CD Pipeline	169
9	Infrastructure as a Code (IaC)	169
10	Monitoring and Alerting	170
11	Business Process Architecture 1	172
12	Business Process Architecture 2	174
13	API Gateway	176
14	Data Flow Diagram Level 0	177
15	Data Flow Diagram Level 1	178
16	Use Case	180
17	Applicant Sequence Diagram	181
18	Zoning Officer Sequence Diagram	182
19	Building Officer Sequence Diagram	183

20	Admin Sequence Diagram	183
21	Flowchart	185
22	Network Topology	186
23	Stakeholder Analysis	197
24	Admin/Zoning Officer BPA Diagram	212
25	Public Applicant BPA for Web-Based GIS System	213
26	Inspector BPA for Web-Based GIS System	215
27	Application Architecture	234
28	Architecture Diagram	234
29	Physical View of the Architecture of a Web Application	236
30	Understanding the Architecture 3-tier	237
31	Web application data flow	244
32	Rest API Architecture	244
33	DFD lvl 0	250
34	DFD lvl 1	251

LIST OF APPENDICES

Appendix A	Detailed Technical Documentation
Appendix B	Documents/URLs
Appendix C	Certificate of Originality
Appendix D	Certificate of Technical Adviser and Grammarian
Appendix E	Grammarian Curriculum Vitae
Appendix F	Technical Adviser Curriculum Vitae
Appendix G	Research Adviser Curriculum Vitae
Appendix H	Researcher's Curriculum Vitae

Chapter 1

INTRODUCTION

1.1 Background of the Capstone Project

The modernization of Local Government Units (LGUs) in the Philippines has become an urgent imperative as the intricacies of urban development and land management outpace traditional administrative capabilities. For decades, the enforcement of zoning ordinances has been anchored in manual, paper-based workflows, requiring applicants, zoning officers, and building officials alike to painstakingly cross-reference physical maps, Transfer Certificates of Title, and other permit documents by hand. This legacy approach is fundamentally inefficient, often characterized by protracted delays and a high susceptibility to human error. Without the benefit of modern spatial visualization, these manual methods frequently result in significant inconsistencies when implementing the Comprehensive Land Use Plan (CLUP), leading to disorganized urban expansion and regulatory bottlenecks.

Recognizing these systemic challenges, the conception of this project is rooted in the strategic integration of Geographic Information System (GIS) technology into the municipal approval framework. At its core, an automated rule engine performs point-in-polygon spatial validation, cross-referencing an applicant's plotted coordinates against digitized zoning

boundaries and CLUP-derived setback rules to instantly return a "Compliant" or "Non-Compliant" status, complete with a time-stamped digital audit trail and PDF summary receipt. Once an application clears zoning and building review, the platform guides the applicant through an integrated online payment gateway to settle the corresponding permit fee before final approval is released, while a dedicated AI Insights panel distills the system's generated reports into concise, natural-language summaries to support LGU decision-making.

By digitizing the zoning landscape in this way, the LGU can replace subjective manual checks with precise, data-driven assessments. This shift ensures that every permit application is evaluated against accurate, real-time spatial data, thereby eliminating the friction points inherent in paper-reliant systems.

Ultimately, the project titled "Web-Based GIS System for Automated Zoning and Permits" serves as a dedicated technological solution to the growing demands of local governance. Structured around role-based access control, the platform supports seven primary actors: the Applicant, who plots coordinates, settles the corresponding permit fee through an integrated online payment gateway, and tracks application status in real time; the Zoning Officer, who verifies whether a project falls within the correct zoning classification and approves or rejects applications strictly on zoning grounds; the Building Official, who reviews structural and

architectural plans, checks building code compliance, and issues the final permit once zoning clearance is granted; the Assessor, who accesses cleared applications to update property and parcel records for taxation purposes; the Inspector, who conducts on-site property inspections and submits inspection reports for applications assigned to them; the Admin, who oversees day-to-day application processing, generates reports, and manages user accounts; and the Super Admin, who retains full system-level control over platform configuration, permissions, and security.

By streamlining the intersection of urban planning and permit issuance across these roles, the system seeks to foster a more transparent and accountable regulatory environment. This modernization ensures that municipal growth is no longer a product of administrative guesswork but is instead a structured process aligned with established regulatory standards. Through this digital transformation, the LGU can guarantee that urban development is both sustainable and strictly compliant with the long-term vision of the community.

1.2 Context and Scope

The operational environment for this system is strategically situated within the administrative and regulatory framework of a municipal Zoning and Planning Office. These integrated features are engineered to streamline the internal workflows of Local Government Unit (LGU)

officers—reducing the time spent on manual map plotting—while simultaneously providing a transparent, user-friendly interface for public applicants to monitor their submissions.

To be focused and achievable in implementation, the project scope shall involve:

- **Interactive GIS Module for Spatial Visualization.**
 - An interactive mapping module that provides high-precision spatial visualization of land parcels and zoning boundaries, replacing static paper maps with a dynamic digital "ground truth" for both LGU officers and public applicants.
- **Automated Compliance Engine for Coordinate-Based Validation.**
 - An engine that automatically validates proposed land-use applications against existing zoning ordinances through coordinate-based checks—covering setbacks, height limits, and floor area ratios—reducing reliance on manual, subjective assessment.
- **Post-Approval Monitoring Dashboard.**
 - A dashboard that tracks construction adherence after permit approval, allowing LGU officers to monitor compliance with

approved plans and flag unauthorized deviations for further review.

- **Integrated Online Payment Gateway for Permit Fees.**

- A generic online payment gateway that allows applicants to settle the corresponding permit fee electronically once their application has cleared zoning and building review, with final approval released only after payment is confirmed.

- **AI-Assisted Reporting and Insights.**

- An AI Insights panel within the Reports module that generates a natural-language summary of key report metrics alongside the standard charts and tables, flagged as an AI-generated summary for LGU staff to verify against the underlying data.

Out of Scope:

- Replacement of on-site physical inspections; the system functions as an administrative and analytical tool and does not eliminate the need for LGU personnel to verify field conditions in person.
- Permitting phases and municipal functions outside the zoning and locational clearance process, keeping the automation focused on this specific regulatory intersection.

The project's primary objective remains strictly focused on the digital automation of the zoning and locational clearance phase of the permitting process. While the system provides the digital "ground truth" for land-use compliance, it acknowledges that human oversight and physical verification remain essential components of the broader municipal governance structure, paving the way for future scalability without overextending the initial technical requirements of the pilot area.

1.3 Problem Statement

The prevailing manual procedures for evaluating development permits within the Local Government Unit (LGU) are characterized by significant operational inefficiencies that stem from a reliance on physical, analog records. Zoning officers must painstakingly cross-reference paper-based Comprehensive Land Use Plan (CLUP) maps, Transfer Certificates of Title, and permit applications by hand, and the absence of a real-time spatial verification mechanism creates a systemic bottleneck that leads to prolonged turnaround times for permit approvals. These delays do more than just stall individual projects; they actively hinder local economic growth and slow down vital infrastructure development.

This procedural gap is not limited to the initial application phase but extends into the long-term oversight of urban spaces, significantly increasing the risk of unauthorized land use and illegal building expansions

that bypass regulatory standards. Specifically, this study seeks to address the following problems encountered by the Local Government Unit's Zoning and Planning Office:

1. Zoning officers manually cross-reference paper-based CLUP maps and permit applications, resulting in slow, error-prone, and inconsistent evaluation of land-use compliance.
2. The absence of a real-time spatial verification mechanism leads to prolonged turnaround times for permit approvals, delaying local economic growth and infrastructure development.
3. Fragmented, non-digital records create a transparency gap, making it difficult for both LGU officials and the general public to access accurate, up-to-date zoning and application-status information.
4. The lack of a centralized digital repository and automated compliance engine hampers the monitoring of post-approval construction, increasing the risk of unauthorized land use and illegal building expansions.
5. The absence of a unified digital audit trail makes it difficult to verify that permitting decisions and building activities strictly adhere to mandated urban planning ordinances, undermining accountability and long-term data integrity.

1.4 Objectives and Goals

The primary goal of this study is to fundamentally enhance the operational efficiency, data accuracy, and systemic transparency of the Local Government Unit's (LGU) urban planning processes through the strategic implementation of Geographic Information System (GIS) technology, replacing fragmented manual systems with a modernized governance framework guided by reliable data rather than administrative inertia. Specifically, this study aims to:

1. Develop a web-based module that automatically compares proposed land-use applications against existing zoning ordinances through algorithmic, coordinate-based validation.
2. Implement a secure, centralized database that manages and protects sensitive applicant records and spatial layers from unauthorized access or data corruption.
3. Create a high-precision, Leaflet.js-based interactive mapping interface that provides planners and the public with a visual "ground truth" for spatial analysis, replacing static paper maps.
4. Provide public applicants with a real-time status-tracking dashboard, allowing them to monitor their submissions from initial filing through final approval without physical follow-up visits.

5. Establish a reliable digital audit trail, complete with time-stamped logs and PDF summary receipts, that archives every step of the permitting process and minimizes human error and administrative discretion in zoning enforcement.

1.5 Significance and Relevance

The significance of this study extends across the entire urban development lifecycle, offering transformative benefits for a diverse range of stakeholders directly involved in the zoning and permitting process. By replacing fragmented, paper-based procedures with a unified, data-driven Geographic Information System (GIS), the research introduces a resilient and verifiable technological framework that strengthens the enforcement of local policies and streamlines the allocation of administrative resources.

Key beneficiaries include:

Local Government Unit (LGU): As an institution, the LGU gains a sophisticated digital infrastructure that safeguards its long-term vision for urban growth. Migrating from fragmented manual records to a unified GIS framework allows the municipal government to enforce zoning ordinances consistently and protect the integrity of the Comprehensive Land Use Plan (CLUP).

Zoning Officers: The automated compliance engine alleviates the burden of repetitive, manual cross-referencing of CLUP maps and

permit documents, reducing exhaustion-related errors. By automating the technical verification of setbacks, height limits, and land-use categories, officers are freed to focus on higher-level analytical duties and more nuanced urban planning assessments.

Building Officials: With access to consolidated, system-generated compliance reports and analytics, Building Officials can render final approval decisions more efficiently and with greater confidence, accelerating the issuance of building permits without compromising regulatory accuracy.

General Public and Permit Applicants: The system acts as a bridge toward greater transparency and civic accessibility. By providing a digitized platform for tracking application statuses and visualizing land-use restrictions in real time, it demystifies the regulatory process and reduces the time and financial costs typically associated with manual, in-person applications and follow-ups.

Future Researchers and IT Professionals: This study serves as a foundational framework for future researchers, IT practitioners, and urban planners interested in the intersection of spatial analytics and digital governance. It offers a scalable model for how modern software engineering principles can be applied to solve real-world challenges in sustainable city management.

1.6 Structure of the Document

This capstone research is systematically organized into five distinct chapters, each designed to provide a comprehensive and logical overview of the project's development lifecycle, placing this project in the larger context of research.

Chapter 1: Introduction - Presents the background of the capstone project, outlining the inefficiencies of manual, paper-based zoning enforcement within the Local Government Unit (LGU) and the rationale for integrating Geographic Information System (GIS) technology into the permitting process. This chapter also defines the study's context and scope, problem statement, objectives, and significance, establishing the foundation for the proposed Web-Based GIS System for Automated Zoning and Permits.

Chapter 2: Review of Related Literature - Examines the Agile Scrum methodology adopted for the project's development, alongside related studies and technologies relevant to GIS-based zoning and permitting systems. This chapter situates the research within existing literature, providing a scholarly basis for the system's design and technical approach.

Chapter 3: Methodology - Details the application of the Agile Scrum framework throughout the system's development, including the

Sprint cycles, defined team roles, and the structured ceremonies used to manage the project. This chapter explains how the development process was organized to incrementally build and refine the GIS mapping features, automated compliance checks, and security protocols.

Chapter 4: Requirements Analysis - Identifies the system's key stakeholders through Role-Based Access Control (RBAC), namely the Applicant, Zoning Officer, Building Official, Assessor, Inspector, Admin, and Super Admin, and describes the data-gathering techniques, such as document analysis, used to translate legal mandates and manual LGU workflows into functional system requirements.

Chapter 5: Conclusion and Recommendations - Provides a holistic summary of the study's findings, the resulting conclusions, and strategic recommendations for the system's continued development. This chapter also offers insights into the project's long-term impact on local governance and identifies potential directions for future enhancement, such as integration with other municipal permit types.

Chapter 2

REVIEW OF RELATED LITERATURE

2.1 Agile Scrum Methodology Overview

Agile Scrum is a leading iterative and incremental framework designed for managing complex software development projects, structured around short, time-boxed iterations called Sprints. Unlike traditional “Waterfall” models that follow a rigid and sequential flow, Scrum instead prioritizes adaptability, collaboration, and customer-driven development, allowing teams to respond quickly to shifting requirements. This methodology is particularly effective for this capstone project because local government zoning regulations, departmental routing, and spatial data integration present an evolving and complex environment, where the ability to refine requirements, adjust to new LGU inputs, and validate features quickly is crucial to the success of the Web-Based GIS System.

The three pillars of Scrum, according to Schwaber, K., & Sutherland, J. (2020), are transparency, inspection, and adaptation, which are upheld through structured events such as Sprint Planning, Daily Stand-ups, Sprint Reviews, and Sprint Retrospectives. Applied to this Web-Based GIS System, each sprint begins with the creation of a prioritized Product Backlog containing core features such as the automated GIS zoning

checks, department inspection requests, and the monitoring module, followed by Daily Stand-up meetings that maintain transparency and address technical bottlenecks as they arise. Upon completion, a Sprint Review evaluates functional deliverables—such as the map interface or permit workflow—while the subsequent Sprint Retrospective allows the team to reflect and continuously refine its development process. To execute this structure, the project also adapts the Scrum Master, Product Owner, and Development Team accountabilities identified by Wonohardjo et al. (2019), with the Project Manager facilitating the process, the Developer handling technical implementation, and Quality Assurance and Document Specialists ensuring the system's accuracy and documentation remain aligned with institutional standards.

Overall, Agile Scrum assisted us in ensuring that the Web-Based GIS System could evolve efficiently despite the academic time constraints and technical learning curve involved in integrating GIS technologies. Consistent with the literature identifying flexibility, adaptability, and risk reduction as core strengths of the framework (Sassa et al., 2023), Scrum gave the team the ability to adapt to changing permit requirements, reduce risks early, and maintain a strong focus on user-centered design for both applicants and LGU personnel. This dynamic and resilient structure

ultimately supported both the technical objectives and the practical, real-world applications of the automated zoning and permit system.

2.2 Enterprise Architecture Concepts

An organization's structure and operations are defined by its Enterprise Architecture (EA), a strategic framework that ensures alignment between its technology capabilities and business objectives. This project adopts The Open Group Architecture Framework (TOGAF) to guide its development, as it provides a structured foundation for aligning the Web-Based GIS System for Automated Zoning and Permits with the strategic goals of the Local Government Unit (LGU). By breaking the organization down into four architectural domains—Business, Application, Data, and Technology—TOGAF helps manage the inherent complexity of municipal governance and its supporting technology, ensuring that EA plays a crucial role in bridging the LGU's operational objectives, specifically efficient urban planning and accurate zoning verification, with the supporting technologies, processes, and systems.

The Business Architecture domain addresses the strategy, governance, and processes of the permitting workflow, outlining the step-by-step procedure from the initial submission of a permit application to the final issuance of a locational clearance. It defines the roles of the Applicant, Zoning Officer, Building Official, and System Administrator within this

workflow, ensuring that the automated system directly mirrors and supports the legal and procedural needs of the municipality rather than operating as a disconnected technical layer.

Complementing this is the Application Architecture, which defines the software systems and their interactions, specifically how the PHP-based controllers—such as the ApplicantController—and the interactive Leaflet.js mapping modules work together to handle applicant requests and automated zoning evaluations. The Data Architecture, in turn, focuses on the structure and management of information, using a centralized Database.php to organize cadastral parcels, machine-readable zoning ordinances, and applicant records into a cohesive spatial data model that protects sensitive land records against unauthorized modification.

Meanwhile, the Technology Architecture defines the underlying infrastructure and platforms, including the servers, network protocols, and web-based systems necessary to support real-time spatial analysis and secure access for both applicants and LGU planners. Taken together, these four domains ensure that the Web-Based GIS System for Automated Zoning and Permits is not merely a standalone application but a cohesive component of the LGU's broader enterprise architecture—one that is scalable, secure, and directly aligned with the strategic goals of the local government.

2.3 Microservices Architecture

A Microservices Architecture represents a sophisticated design approach where a complex application is structured as a collection of small, autonomous services that communicate through well-defined Application Programming Interfaces (APIs). In contrast to traditional monolithic systems, where the entire application is bundled into a single, inseparable unit, microservices allow each distinct component to be managed as an isolated entity. Within the context of this project, essential modules such as the User Authentication Service, the Spatial Data Service, and the Automated Rule Engine can be developed, deployed, and scaled independently without affecting the integrity of the broader system.

This modularity is particularly advantageous for a web-based GIS platform as it significantly enhances fault isolation. By decoupling the various system functions, the failure of a single service does not result in a total system outage, allowing the remaining administrative and mapping features to stay operational for the end-user. Furthermore, this architectural style grants developers the flexibility to utilize different technology stacks tailored to specific requirements. For instance, high-performance spatial processing can be handled by a specialized engine, while sensitive data management can be secured through a dedicated authentication service using the most robust encryption protocols available. To integrate these

decoupled services into a cohesive platform, the system relies on an API Gateway that serves as the centralized entry point for all backend communications, using REST and gRPC protocols to route requests to the appropriate service without compromising the isolation between them.

The implementation of microservices also facilitates a more efficient and agile development lifecycle, particularly when multiple developers are working on different system modules simultaneously. Because each service is independently deployable, updates and security patches can be pushed to the Spatial Data Service or the Automated Rule Engine without necessitating a full system reboot or a complete re-deployment of the application. This granular control reduces downtime during maintenance periods and ensures that the LGU's permit processing services remain available even as technical improvements are being integrated into the backend.

Furthermore, this architecture supports the long-term scalability and sustainability of the project as urban data requirements grow. As the LGU's spatial database expands or the volume of permit applicants increases, the system can scale specific high-traffic services—such as the Spatial Data Service—independently from lower-traffic administrative tools. This resource optimization ensures that the system remains responsive under heavy loads while maintaining a cost-effective infrastructure. By adopting

this decoupled approach, the project establishes a robust foundation that can easily adapt to future technological advancements or shifts in municipal zoning regulations.

2.4 DevOps and CI/CD

As modern software development has evolved, favored methodologies have increasingly prioritized adaptability, scalability, and swift delivery. Key practices such as DevOps and Continuous Integration/Continuous Deployment (CI/CD) have emerged as essential strategies for effectively managing complex systems like a Web-Based GIS. These concepts are deeply interconnected and are often implemented together to facilitate agile development, accelerate release cycles, and promote ongoing improvement throughout the project lifecycle.

The DevOps culture merges software development and operations into a cohesive practice, focusing on collaboration, automation, and shared accountability to achieve quicker and more dependable software delivery. This cultural and technical movement eliminates traditional barriers between development and deployment teams to foster institutional agility. Academic support for this movement highlights how DevOps practices enhance critical performance metrics, such as reducing lead time for new features, increasing deployment frequency, and improving the speed of recovery from system failures.

To implement these practices effectively, the project utilizes Continuous Integration and Continuous Deployment (CI/CD) pipelines. Continuous Integration involves regularly merging code changes into a shared repository, accompanied by automated builds and tests to identify errors at an early stage. Continuous Deployment extends this automation by ensuring that software updates are delivered swiftly and reliably to the production environment. This automation is vital for mitigating integration risks and ensuring that new spatial modules or security patches are functional before they reach the end-user.

For this capstone project, adherence to CI/CD principles is supported by the team's use of GitHub for source code management and collaborative version control throughout the Sprint cycles, which helps prevent the technical debt often referred to as "merge hell" by encouraging the frequent integration of code changes. Development and testing take place within a local XAMPP environment—bundling the Apache server, PHP, and a MariaDB/MySQL database—before validated changes are promoted to the scalable, cloud-hosted production environment. This systematic progression from local development to cloud deployment ensures that the GIS system remains in a deployable state throughout the entire development process, allowing the development team to respond

more swiftly to administrative requirements and maintain a high standard of software quality in a fast-paced municipal environment.

Furthermore, the synergy between DevOps and CI/CD allows for a sustained cycle of rapid innovation while providing the technical foundation for frequent and reliable releases. This integration has revolutionized how information technology projects are approached, allowing for a more responsive and transparent development process. By automating the build, test, and deployment phases, the project ensures that the automated zoning and permit system remains robust, secure, and ready to meet the dynamic needs of the Local Government Unit.

2.5 Relevant Studies and Research

The review of related literature provides an overview of existing studies, theories, and findings that are connected to the development of the Web-Based GIS System for Automated Zoning and Permits.

Local Studies

De Villa, Florencondia, and Aduna (2025), examined the digitalization of building permit systems in the Philippines, focusing on the effectiveness of ICT-driven permitting platforms such as the eBPLS and PBPLS in cities like Quezon City and Davao City. Their study found that digitalization improved processing time and transparency, but LGUs

continued to face challenges related to insufficient ICT infrastructure, uneven implementation readiness, and resistance to change among personnel. The researchers recommended the creation of a centralized national permit portal alongside structured training programs to support adoption. This aligns with the Permit Management Module of the proposed system, as consistent data handling and standardized digital workflows are critical to overcoming the implementation gaps identified in the study. The study matters because it demonstrates that digitalization alone is insufficient without institutional readiness and dedicated training support.

Rivas (2026), developed a Barangay Information Management System with automated mobile messaging in Zamboanga del Norte, Philippines, integrating resident profiling, records management, and real-time SMS notifications for announcements and emergency alerts. The study found that automated messaging significantly improved administrative efficiency and strengthened community engagement, with respondents rating real-time notification features and reduced manual workload especially favorably. This connects directly to the Notification Module of the proposed system, which similarly relies on automated alerts to keep applicants and inspectors informed of real-time status updates. The study matters because it demonstrates that automated communication

features meaningfully reduce follow-up burden and improve citizen-facing service delivery at the local government level.

Carrido et al. (2026), developed a geospatial framework that integrated land cover datasets with zoning maps to assess land-use and zoning compliance in a Philippine city. The study found that non-compliant land use areas increased significantly between 2017 and 2023, with "Not Allowed" areas growing by about 49.55% and "Conditionally Allowed" areas by about 80.51%, pointing to worsening encroachment into restricted zones despite existing monitoring frameworks. This aligns with the Automated Zoning Compliance Engine, as it highlights the need for continuous, automated spatial monitoring rather than periodic manual assessment to catch violations before they compound. The study is significant because it demonstrates that without systematic, technology-driven monitoring, land-use violations tend to worsen over time — reinforcing the case for GIS-based automation in Philippine LGUs.

Santillan and Heipke (2024), developed a web-based GIS application, built entirely on open-source geospatial software and earth observation datasets, to visualize the Philippines' progress on SDG 11.3.1 (sustainable urbanization). Their work demonstrated that free and open-source tools can be used to build and deploy a functioning, publicly accessible spatial monitoring platform at a national scale, without reliance

on costly proprietary GIS licenses. This aligns with the Technology Architecture of the project, where cost-efficiency is central to a sustainable government system. The study matters because it shows that open-source geospatial stacks are viable even for large-scale, government-facing applications — supporting the case for using similarly low-cost tools in an LGU-level system.

Lastly, **Felix et al. (2026)** conducted a nationwide assessment of institutional readiness among 666 Philippine LGUs to adopt geospatial land-use monitoring tools, evaluating indicators such as Comprehensive Land Use Plan status, GIS capacity, remote sensing capacity, and enforcement tools. The study found that while willingness to adopt such technologies was high among LGUs (84%), actual technical capacity remained low, with GIS and remote sensing capacities present in only 41.75% and 12.31% of surveyed LGUs, respectively. This connects directly to the Interactive Map and Monitoring Module, since it identifies both a documented capacity gap and a strong institutional appetite for exactly this kind of spatial monitoring tool. The study matters because it shows that the barrier to adopting web-based spatial monitoring in Philippine LGUs isn't a lack of willingness — it's a lack of accessible, ready-to-deploy systems, which is the gap this capstone project aims to address.

Foreign Studies

Abin, Shaju, Priya, Abhijith, and Jose (2021), examined the role of GIS combined with data analytics in transforming municipal administration in India, using drone-based surveys, geospatial digitization, and a centralized web portal to consolidate property, infrastructure, and socio-economic data. Their case studies showed that GIS-based digital repositories improved tax collection accuracy and administrative decision-making, with one municipality reporting a substantial revenue increase after previously unrecorded properties were identified through the system. This aligns with the GIS Map and Spatial Validation Module of the proposed system, as a centralized, web-based spatial repository similarly reduces reliance on fragmented manual records. The study matters because it demonstrates that GIS-based digitalization directly improves accuracy and administrative efficiency at the municipal level, not just data visibility.

Shahi, McCabe, and Shahi (2019), developed a framework for automated, model-based e-permitting systems for municipal jurisdictions, defining distinct levels of automation ranging from traditional paper-based practices (Level 0) to a fully integrated, model-based city planning tool (Level 3). Their framework, validated by permitting experts, found that specific elements of lower automation levels had already been implemented by municipalities worldwide with reported productivity

improvements, while also identifying key barriers to reaching higher levels of integration. This has direct relevance to the Automated Compliance Engine of the project, which represents a coordinate-based automation approach positioned toward the higher levels of this framework. The study is significant because it provides a structured way to understand where a system like this one sits on the broader spectrum of permitting automation maturity.

Khanna, Singh, and Angra (2026), proposed a blockchain-enabled e-governance framework for land record management, modeling distinct administrative roles with role-aware access control and combining attribute-based encryption for long-lived, role-driven document access with identity-based encryption for short, time-bound disclosures. Their benchmark testing demonstrated that this role-aware governance model could sustain high transaction throughput while maintaining strict access traceability across administrative roles. This strongly supports the System Security and RBAC Module of the proposed system, where restricting spatial and permit data access according to defined user roles is central to protecting the integrity of LGU zoning data. The study is significant because it demonstrates that structuring access control around real administrative role hierarchies, not just generic permission levels, is key to both institutional trust and data security.

Cina, Elbasi, Elmazi, and AlArnaout (2025), reviewed how AI-powered predictive modeling supports sustainable urban development, focusing on applications including infrastructure optimization, resource management, and environmental monitoring. Their review found that AI-driven predictive tools allow cities to analyze large-scale datasets, anticipate infrastructure and resource demands ahead of time, and shift from reactive to proactive planning. This aligns with the Analytics and Reporting Module of the project, which similarly emphasizes forward-looking, data-driven reporting to support efficient urban planning rather than reactive record-keeping. It matters to the module because it confirms that predictive, AI-assisted forecasting is an established international direction for urban planning systems — grounding the analytics module in current, recognized research rather than a purely local design choice.

Finally, **Osakwe et al. (2025)** conducted a systematic literature review of 93 studies (2018–2024) on the integration of data analytics and GIS in smart city development. Their review found that most existing research on this integration comes from developed countries, and identified a significant gap in studies from developing regions — despite the pressing need for such tools there — along with recurring challenges around data privacy, system interoperability, and stakeholder engagement. This connects to the Public Dashboard and Field Monitoring features of the

project, which represent exactly the kind of data-analytics-and-GIS integration the review found underexplored in developing-country contexts. The study matters because it positions a Philippine LGU system like this one as addressing a documented research and implementation gap, rather than replicating existing solutions.

2.6 Integration of Information Systems in Enterprise Environment

Enterprise Application Integration (EAI) refers to the strategic process of enabling previously isolated software systems, databases, and platforms to communicate, share data, and function together as a unified whole. In a typical enterprise or government setting, individual departments often rely on separate tools for records management, communications, and spatial analysis, which over time creates fragmented workflows and inconsistent data. Addressing this fragmentation is central to the Web-Based GIS System for Automated Zoning and Permits, as the platform must unify GIS mapping, permit processing, notifications, and administrative reporting into a single, cohesive enterprise solution rather than a set of disconnected tools.

Several methods exist for achieving this level of integration, ranging from simple point-to-point connections to more scalable middleware-based

approaches such as Enterprise Service Buses (ESBs) and Application Programming Interfaces (APIs). This project favors an API-driven strategy, consistent with the Microservices Architecture discussed earlier, wherein the centralized API Gateway routes requests between independent services such as the Spatial Data Service, the Automated Rule Engine, and the Notification Module. This approach allows the GIS mapping component to remain synchronized with the permit database in real time, ensuring that any change in zoning status or application progress is immediately reflected across every module that the Applicant, Zoning Officer, or System Administrator interacts with.

Despite its benefits, system integration within an enterprise environment presents notable challenges, particularly for local government units transitioning from legacy, paper-based, or siloed digital processes. Differences in data formats, inconsistent recordkeeping practices across departments, and the need to maintain security while allowing multiple services to exchange sensitive information all complicate the integration effort. For the LGU adopting this system, these challenges are addressed by standardizing data exchange through the centralized Database.php structure and enforcing role-based access control, which together ensure that spatial records, permit documents, and applicant information remain consistent and protected as they move between integrated modules.

Ultimately, successful integration transforms a collection of independent tools into a dependable enterprise system capable of supporting the LGU's day-to-day operations. By linking the GIS mapping, permit workflow, and notification systems through well-defined APIs and consistent data structures, the project ensures that zoning officers, applicants, and administrators experience the platform as a single, coherent service rather than a patchwork of unrelated applications. This level of integration is what ultimately allows the Web-Based GIS System for Automated Zoning and Permits to function reliably within the broader enterprise environment of the local government.

Chapter 3

METHODOLOGY

3.1. Agile Scrum Methodology in the Project

The development of the Web-Based GIS System for Automated Zoning and Permits was managed using the Agile Scrum framework. This methodology was selected to handle the complexity of integrating spatial data with administrative workflows through an iterative approach. By breaking the development process into manageable cycles, the project remained flexible, allowing for continuous refinement of the GIS mapping features, automated compliance checks, and security protocols based on ongoing testing and evaluation.

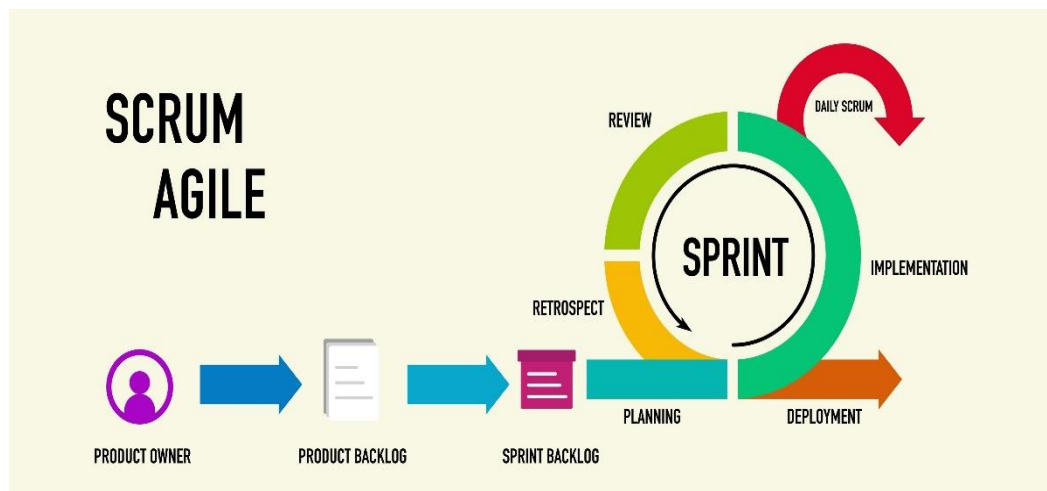


Figure no. 1 Agile Scrum Framework

To execute this effectively, the system's requirements were translated into a prioritized Product Backlog. Development was divided into

targeted, time-boxed Sprints, with each cycle focused on delivering a functional, demonstrable module derived from the overall system flowchart. Early sprints concentrated on establishing the foundational Application Submission portal and the parallel routing logic for Departmental Inspection Requests. Subsequent iterations tackled the system's core technical challenge: the Automated GIS Zoning Check. By dedicating specific sprints to the zoning consolidation algorithm and the post-approval Monitoring Module, the team ensured that the spatial compliance logic was built, tested, and refined incrementally before being integrated into the final permit decision workflow.

The structured ceremonies and defined roles within the Scrum framework were critical to maintaining project momentum and alignment. Sprint Planning and Daily Stand-ups, facilitated by the Project Manager, allowed the team to track progress and quickly remove technical bottlenecks, enabling the Developer to focus on coding the map integrations and backend logic. At the conclusion of each sprint, Sprint Reviews and Retrospectives provided a platform for the Quality Assurance Lead to present system testing results—such as validating the accuracy of the zoning output—while the Document Specialists systematically updated the architectural diagrams and technical manuscript. This collaborative, cyclical approach guaranteed that the final system met its technical

objectives while remaining adaptable to the practical demands of urban planning operations.

3.2 Roles and Responsibilities

Name	Role	Responsibilities
Mr. Ronald Roldan Jr.	Capstone Adviser	Provides guidance and mentorship to the team, reviews progress, ensures academic requirements are met, gives constructive feedback, and aligns project outcomes with learning objectives.
Ma.Cristina D. Buenaflor	Project Manager	Oversees the overall project execution and lifecycle. Manages Agile/Scrum sprints, coordinates team tasks, tracks progress against deadlines and leads team communication.
Ma.Cristina D. Buenaflor Rosary Mae C. Velicaria	Quality Assurance	Conducts rigorous system testing (unit, integration, and user acceptance) on the GIS mapping and permit routing flows. Identifies

		bugs and ensures the system meets the defined objectives.
Yumie Margareth N. Dalagan	Full-Stack Developer	Responsible for writing the code for both the frontend and backend of the system. Integrates GIS mapping features, builds the automated zoning logic, and manages the database architecture.
Althaniel O. Abordo Joshua Canceran Justin Michael Espino	Document Specialist	Assists in writing, editing, and formatting the capstone manuscript. Ensures all system documentation, diagrams, and literature reviews meet institutional standards.

Table no. 1 Development Team

Product Owner: Responsible for defining and prioritizing the product backlog and ensuring the development team understands the product vision, which includes features such as:

Name of Product/Module	Role/Responsibilities
Zoning Officer Dashboard	• Define the key performance indicators, metrics, and visualizations (e.g., pending permits, approved clearances, map alerts) to provide a clear summary of daily LGU operations. • Ensure the dashboard delivers actionable insights that help zoning administrators make informed decisions. • Prioritize which information is most critical to display, balancing clarity with comprehensive spatial data. • Continuously review and adjust dashboard features based on LGU staff feedback to maintain relevance and usability.
Public Applicant Portal	• Approve the fields and workflow for registering new permit applications to ensure that all essential

	applicant details and land documents are captured accurately. • Ensure the registration and submission process is user-friendly for the public and aligns with the LGU's legal requirements. • Validate that applicant data is structured securely, adhering to information security standards. • Review and authorize updates to the portal to accommodate changes in municipal regulations.
GIS Map Engine & Spatial Validation	• Define the requirements for interactive map layers (e.g., land-use zones, property boundaries, hazard maps). • Ensure the automated compliance logic accurately reflects the LGU's actual zoning ordinances and rules. • Work with the development team to ensure map visualizations are clear, responsive, and useful for spatial analysis. • Approve updates or enhancements to the mapping tools (like polygon drawing or coordinate tracking) to reflect stakeholder needs.

Permit Tracking & Workflow	• Prioritize the creation, assignment, and routing of permit applications through the different LGU approval stages. • Define what tracking data should be captured, such as submission dates, assigned evaluating officers, and compliance statuses. • Approve enhancements to task management features, such as sorting, filtering, or progress tracking, to meet the zoning staff's needs. • Work closely with LGU departments to ensure the system supports accountability and reduces bottlenecks in permit issuance.
System Administration & Security	• Set guidelines for role-based access control (RBAC), ensuring that only authorized personnel can alter sensitive zoning maps or approve permits. • Approve the reporting formats for system audits and user activity logs. • Monitor that all data management aligns with strict information security and compliance requirements.

	• Prioritize security features and review audit reports regularly to ensure the system's integrity remains uncompromised.
--	---

Table no. 2 Roles and Responsibilities

3.3 Sprint Cycles (Planning, Standups, Review)

The development of the web-based GIS system was executed through a series of time-boxed iterations known as Sprints, typically lasting two weeks each. Serving as the project's heartbeat, these fixed-length cycles established a regular rhythm for planning, execution, and continuous learning. By organizing the workload into these distinct intervals, the team could focus on delivering functional increments of the system—such as the automated coordinate validation or the user dashboard—without being overwhelmed by the platform's overall complexity.



Figure no. 2 Scrum Process

To ensure the team built the right features, adapted to technical challenges, and continuously improved, the sprint cycle was powered by four key Agile ceremonies. These structured events facilitated continuous collaboration among the Project Manager, Developer, Quality Assurance, and Document Specialists throughout the project's lifespan.

To Do	In Progress	Done
Integration of 3D Cadastral Mapping Extensions	Online Payment Gateway Integration for Permit Fees	Multi-Step Applicant Registration Wizard & ID Uploads
Mobile Push Notifications for Permit Approvals & Deficiencies	Auto-Drafting Watermarked PDF Zoning Certificates	Mandatory Email 2FA One-Time-Password Verification
Predictive AI Models for Future Urban Growth & Permit Demand	Mobile Viewport Optimization	Interactive Leaflet.js Mapping Canvas & Coordinate Picker
		Point-in-Polygon Automated Zoning Compliance Engine

		Protected Hazard Band & Easement Intersection Warnings
		Role-Based Access Control (RBAC) for 7 Primary User Roles 40 Centralized Inspection Calendar & Dynamic Scheduling
		Geo-fenced Mobile Inspector Field Report & Violation Intake
		Unalterable System Audit Logging & Severity Tagging
		CSRF Token Validation & SQL Transaction Rollback Defenses

Table no. 3 Scrum Board

3.4 Scrum Artifacts (Product Backlog, Sprint Backlog)

The development of the Web-Based GIS System for Automated Zoning and Permits was guided by three primary artifacts: the Product Backlog, the Sprint Backlog, and the increments of the system. These artifacts provided a transparent roadmap, ensuring that the development remained organized and focused on high-priority goals.

3.4.1 Product Backlog (User Stories)

The Product Backlog serves as a comprehensive, prioritized list of all required system functionalities. Each feature is defined as a User Story to ensure that development remains centered on the needs of LGU administrators and the public.

User Story No.	Features/ Tasks	User Stories	Priority	Status
Registration & Identity Verification				
F1	Multi-Step Registration Form	As an Applicant, I want a clean, 3-step registration wizard (Personal Info,	High	Done

		Account Security, Address & Verification) so that the sign-up process feels structured and not overwhelming.		
F2	Age Validation Control	As an Admin, I want the system to reject registration inputs if the applicant's age is under 18 years old to comply with legal capacity rules.	High	Done
F3	Password Complexity Enforcer	As a Security Admin, I want the system to require passwords to be at least 8 characters long and combine uppercase, lowercase, and	High	Done

		numbers to prevent weak accounts.		
F4	Registration Valid ID Uploads	As a Zoning Officer, I want applicants to select a Valid ID type (PhilSys, Driver's License, Passport) and upload separate clear images of the ID Front and ID Back for identity audit.	High	Done
F5	Real-time Password Strength Meter	As an Applicant, I want a visual strength bar with color-coded feedback below the password input so that I can gauge the security of my	Medium	Done

		chosen password in real time.		
F6	Text Input Capitalization Standard	As a Data Administrator, I want fields like First Name, Last Name, Street, Barangay, and City to be automatically properly formatted (Title Case) upon form submission to maintain database consistency.	Low	Done
F7	Automated Secure File Renaming and Storage	As a System Administrator, I want uploaded validation files to be programmatically renamed using the format	High	Done

		[username]_[SIDE]_[timestamp].[extension] and saved securely in a specific file directory to avoid server collisions.		
F8	Role Allocation Defaulting	As a System Developer, I want all newly verified self-registrants to be permanently assigned the database security role of applicant by default to enforce proper privilege scoping.	Critical	Done
F9	Prevent Duplicate	As a Database Clerk, I want the system to reject	High	Done

	Registratio ns	registration attempts if the requested username or email matches an existing record to preserve record uniqueness.		
F10	Dual State Viewport Rendering	As an Applicant, I want the user interface to dynamically toggle between the registration input wizard layout and the 6-digit verification code form based on my current verification state.	Medium	Done
Login & Two-Factor Authentication (2FA)				

F11	Dual Identifier Login	As a User, I want to log in using either my Username or Email along with my password so that I don't have to remember which specific identifier I registered with.	High	Done
F12	Auto-Redirect Unverified Registration	As an Unverified User, if I attempt to log in but my account hasn't completed email verification, the system must automatically route me back to the registration OTP verification view.	High	Done

F13	Secure Session Handling	As a System Developer, I want the session ID to be regenerated (session_regenerate_id(true)) and temporary credentials cleared upon a successful login to mitigate session fixation attacks.	Critical	Done
F14	Mandatory 2FA Login OTP	As a User, after entering valid credentials, I want a secure 6-digit One-Time Password (OTP) sent to my email that must be verified before I am	Critical	Done

		granted access to my dashboard.		
F15	Short-lived OTP Expiry Lock	As a Security Admin, I want all generated login and registration OTP codes to strictly expire within 1 minute to prevent interception and brute-force re-use.	Critical	Done
F16	Integrated Resend OTP Control with Cooldown	As a User, I want a "Resend OTP" option directly within the login/registration views that enforces a 60-second locking cooldown timer to prevent mail server abuse.	High	Done

F17	Password Input Visibility Toggle	As a User, I want an eye icon button inside the password field to toggle the visibility of my input characters so that I can verify my entry while maintaining privacy on mobile screens.	Medium	Done
F18	Post-Verification Database Cleansing	As a Security Developer, I want OTP parameters (otp_code, otp_expiry) inside the database row to be completely nullified immediately after execution to block replay attacks.	Critical	Done

F19	Authenticated State Interception	As an Authenticated User, if I attempt to manually browse to login or registration form routes, I want the application to automatically redirect me directly back to my dashboard workspace.	High	Done
F20	Dynamic Client-Side Countdown Sync	As a User, I want the visual 60-second resend cooldown timer to actively synchronize its initial anchor timestamp via client-side storage (sessionStorage) to	High	Done

		persist accurately during page reloads.		
F21	Stateful Redirect Flags	As an Applicant, I want the application to read query parameter tracking flags (e.g., ?otp=1, ?resend=success) to display contextual bootstrap confirmation windows directly on screen.	Medium	Done
System Interfaces & Alerts (UX/UI Standard Compliance)				
F22	Modern Custom Alert Overlay	As a User, I want error and success messages (such as password mismatches, invalid logins, or expired OTPs) displayed via	Medium	Done

		a modern, blurred modal overlay (#customAlertOverlay) instead of default browser blocks.		
F23	Dual-Language Content Placeholders	As an Applicant, I want system fields and headers to support translation attributes (data-en / data-tl) for a seamless localized English and Tagalog user interface.	Medium	Done
F24	Mobile Responsive Viewports (Authentication)	As a Mobile User, I want the login and registration cards to smoothly adapt layout scaling, element margins, and input spacing	High	Done

		exactly at 768px, 480px, and 320px responsive breakpoints.		
F25	Compact 6-Digit Entry Fields	As a Mobile User, I want the 6 distinct OTP input boxes to automatically scale down to a tight, swipeable format at 320px widths to ensure zero horizontal container spill.	High	Done
F26	Multi-Device Element Rescaling	As a Mobile User, I want multi-column fields (such as Street, Barangay, City) and structural headings to stack vertically at widths	High	Done

		under 480px to preserve comfortable input tap targets.		
F27	Dynamic Date Format Normalization	As a System Administrator, I want the application to convert user-defined calendar setting keys (e.g., M/D/YYYY, YYYY-MM-DD) into valid tokens to enforce consistent localized dates across the dashboard workspace.	Low	Done
Dashboard Metrics & Analytics				
F28	Real-Time Operational	As an LGU Administrator, I want to view real-time	High	Done

	Performance Indicators	data on Average Processing Time, Top Performing Barangay, and Upcoming Inspections so that I can monitor the efficiency of our zoning operations.		
F29	System-Wide Application Counter Cards	As an Admin/Zoning Officer, I want a grid of color-coded statistical cards showing the total, pending, approved, and rejected applications to see the exact daily workload at a glance.	High	Done

F30	Live Dynamic System Clock and Localizatio n	As a System User, I want the dashboard header to render a live ticking clock and localized calendar date dynamically matching the system's timezone and format settings.	Medium	Done
F31	Automatic Overdue SLA Alert System	As a Zoning Administrator, I want an active alert banner to appear automatically if any application remains unreviewed for more than 3 days, providing a direct link to prioritize it.	High	Done
F32	Inspector- Specific	As an Inspector, I want the dashboard	Critical	Done

	Scope Restriction	to strictly filter and show only the applications and inspection schedules explicitly assigned to my account to avoid information clutter.		
F33	Inspector Quick- Action Reporting	As an Inspector, I want a list of my recent inspection schedules with clear dates and a "Go to Report" action button to smoothly submit my field evaluation reports.	High	Done
F34	Multi- Parameter Application	As a Staff Member, I want to search recent applications by application	High	Done

	Search and Filter	number, project name, or applicant name, and filter them by status without reloading the page layout.		
F35	Interactive Application Status Distribution Chart	As an LGU Executive, I want to view a doughnut chart dividing applications into Approved, Pending, Under Review, Revision, and Rejected segments to track case management balances.	Medium	Done
F36	Project Category	As an Urban Planner, I want an interactive pie chart	Medium	Done

	Breakdown Analytics	rendering total application metrics grouped by explicit project classifications to identify local zoning demand shifts.		
F37	Top Barangay Volumetric Bar Chart	As an Admin, I want an interactive bar graph displaying total volumes across the top 10 most active barangays with structured border radiuses for immediate visual categorization.	Medium	Done
F38	Annual Application Growth Trend Line	As a Management Officer, I want a smooth, curved line graph (tension: 0.4)	Medium	Done

		with a tinted area fill showing the running volume of monthly application submissions across the current calendar year.		
F39	Role-Based Dashboard Redirection	As a Security Officer, I want the application to dynamically toggle dashboard views so that inspectors only render specific inspection timelines while admin/staff accounts receive general processing lists.	High	Done
F40	Intermittent Table	As a Mobile User, I want the recent	High	Done

	Column Hiding	applications table to selectively hide secondary information columns (like Applicant and Date) at 768px, 480px, and 320px breakpoints to avoid layout text clipping.		
F41	Full-Width Mobile Action Adapters	As a Mobile User, I want priority action controls (like the Overdue SLA Review button) to stretch across the entire horizontal space below 480px for optimal screen utility and ease of input tapping.	Medium	Done
User Workspace & Calendar Management				

F42	Secure Session Profile Synchroni zation	As an Applicant, I want the system to execute a fresh data check upon dashboard loading to ensure my cached session information matches my latest account record.	High	Done
F43	Dynamic User Preference s Loader	As a User, I want the workspace to automatically load and parse my customized date formats and time configurations (12h or 24h) from the database to map system outputs to my reading style.	Medium	Done

F44	Native Multi- Language Dictionary Mapping	As a User, I want a robust localization layer (dt()) supporting clean translation switches between English (en_PH) and Filipino (fil) for all buttons, labels, and text fields.	Medium	Done
F45	Automated Office Service Status Monitor	As an Applicant, I want to see a live visual indicator detailing whether the office is currently open or closed based on operating hours (8:00 AM - 5:00 PM), weekends, and	High	Done

		standard Philippine holidays.		
F46	Interactive Inspection Calendar View	As an Applicant, I want a monthly calendar grid that explicitly highlights dates with active inspection appointments (. has-appt) and frames the current day (. today).	High	Done
F47	Contextual Inspection Quick- Action Panel	As an Applicant, I want to click any highlighted date on the calendar to instantly reveal specific project inspection data and action triggers in an	High	Done

		on-screen side details panel.		
F48	Reschedule Appointment Workflow Modal	As an Applicant, I want an on-demand modal pop-up containing validation blocks that restrict selection to future dates and mandate an explicit textual explanation before filing a reschedule request.	High	Done
F49	Preference - Synchronized Client Ticking Clock	As a User, I want a live ticking clock inside the dashboard header that dynamically formats itself to my selected 12-hour or 24-hour visual	Medium	Done

		preferences at 1-second intervals.		
Application Submission & Progress Tracking				
F50	Applicant- Role Security Guard	As an Applicant, I want the system to validate my specific role permissions before exposing historical pipelines or submission forms to block unauthorized access endpoints.	High	Done
F51	Localized Historical Application Registry	As an Applicant, I want a comprehensive table displaying all my submitted tracking numbers, project names, document counts,	High	Done

		and submission dates translated seamlessly into my localized language preference.		
F52	Adaptive Viewport Layout Switcher	As a Mobile User, I want the system to dynamically collapse the complex application table into a scannable, stacked-card layout on screens $\leq 480\text{px}$ to facilitate easy thumb-scrolling and asset tracking.	Medium	Done
F53	Comprehensive Permit Application intake	As an Applicant, I want to fill out a structured, multiparametric intake form	High	Done

		capturing detailed Project Profiles (name, type, description) alongside precise Land Parcel Identities (PIN, lot, block, street, barangay).		
F54	Leaflet GIS Spatial Coordinate Picker	As an Applicant, I want an integrated Leaflet map canvas that enables me to interactively click or position a draggable pin to automatically populate high-precision latitude and longitude coordinates into my form.	High	Done

F55	Dynamic Multi- Document Asset Uploader	As an Applicant, I want to dynamically append or remove file upload slots on demand, designating structural file types (Site Plan, Lot Plan, Proof of Ownership, Building Plan) directly into my payload.	High	Done
F56	Real-Time Application Tracking Pipeline	As an Applicant, I want to view a linear, color-coded progress timeline (Pending, Under Review, Site Inspection, Approved/Rejected) so that I know	High	Done

		exactly which office is currently holding my file.		
F57	Contextual Revision Defect Flagging	As an Applicant, I want to see explicit, typed-out remarks next to specific rejected documents if my file status drops to "Needs Revision" so that I can correct and re-upload the exact missing assets.	High	Done
Application Processing & Validation				
F58	Centralize d Master Application s Console	As a Staff Member, I want a unified queue listing all incoming municipal applications with quick-glance status	High	Done

		indicators so that I can manage daily files systematically.		
F59	Single-View Evaluation Workspace	As an Admin Reviewer, I want a single workspace layout that presents applicant details, land parcel identities, uploaded documents, and action forms on one screen to optimize review speeds.	High	Done
F60	Multi-Role Dynamic Officer Assignment	As an Administrative Supervisor, I want to assign specific files to available field inspectors or secondary zoning officers directly via	High	Done

		an embedded dropdown menu.		
F61	Dropdown Caseload Load Balancing Indicator	As an Administrative Supervisor, I want the active application count displayed next to each officer's name in the routing dropdown so that I can distribute tasks evenly.	Medium	Done
F62	Interactive Document Audit Framework	As a Zoning Officer, I want to review, download, and check off individual uploaded legal files (e.g., Land Title, Lot Plan) inside an evaluation tray to track file validity.	High	Done

F63	Walk-In Queue Generatio n Engine	As an Office Encoding Clerk, I want to manually register walk-in applications into a custom administrative form that automatically builds a standard queue index for off- line tracking.	High	Done
Spatial Verification & Automated GIS				
F64	Admin GIS Layer Control Toggling	As an Admin Reviewer, I want to toggle individual map overlays (Zoning Boundaries, Buffer Zones, Protected Areas) on the Leaflet map	High	Done

		layout to confirm land use matches.		
F65	Pin Adjuster Modal Wrapper (#btn- select- map)	As an Admin / Zoning Officer, I want to adjust the geographic pin location inside a modal viewport if a walk-in client's coordinate parameters require manual calibration.	High	Done
F66	Responsiv e Map Area Recalculati on (invalidate Size)	As a Mobile Admin User, I want the Leaflet map container to force a bounding box recalculation upon modal open to ensure the interactive map	High	Done

		scales across smaller screens without clipping.		
F67	Automated Buffer Intersection Warning	As a Zoning Officer, I want the system to automatically flag any property coordinate pin that falls into protected hazard bands or restricted easements before I sign off.	Critical	Done
F68	Spatial Verification Snapshot Attachment	As an Admin Reviewer, I want to generate a static snapshot of the verified property pin layout overlay to attach directly to the digital record ledger.	Medium	Done

F69	Quezon City Geofence Viewport Enforcer	As an Application Evaluator, I want the map interface to strictly restrict panning and zooming actions within the official geographical boundaries of Quezon City (maxBounds) so that I do not accidentally lose sight of local zoning assets.	High	Done
F70	Custom FAB Map Layer Switcher	As a System User, I want an accessible Floating Action Button (FAB) layer control menu styled in a clean Google	Medium	Done

		Maps aesthetic to rapidly toggle between Road, Satellite, Hybrid, and Terrain tile layers.		
F71	Dynamic GIS Dual-Language Localization Engine	As an internal staff member, I want all user interface elements, tooltips, and control labels inside the GIS viewer to switch dynamically between English and Filipino using a frontend lookup dictionary matrix (t_map()).	Medium	Done
Zoning Compliance Guardrails & Core Security				

F72	Automated Evaluation Form Locking	As a Head Zoning Administrator, I want the final "Approved" and "Rejected" status triggers to stay completely locked out until the compliance and impact assessments are marked as complete.	High	Done
F73	Stateful Processin g CSRF Shielding	As a Security Admin, I want all status modifications, assignments, and database writes to run cryptographic token validations to shield administrative modules from external forgery.	Critical	Done

F74	Multi- Table Transaction Failure Rollback	As a Database Developer, I want any file update failures in application_status_history to instantly trigger a SQL rollback across parent compliance records to preserve system truth.	Critical	Done
F75	System-Wide Administrative Audit Logs	As an Auditor, I want an unalterable log table tracking every instance of administrative file re-routing, parameter adjustments, and final evaluation decisions.	Critical	Done

Field Logistics & Site Inspection Management				
F76	Dynamic Inspection Scheduling Engine	As a Zoning Officer, I want to open an on-screen scheduling tool to book exact dates and field time slots for properties moving past initial document screens.	High	Done
F77	Automated Calendar Conflict Guard	As an Administrative Clerk, I want the booking system to reject scheduling entries if an inspector's allocation limits for that specific date calendar are maxed out.	High	Done

F78	Mobile Field Evaluation Report Intake	As an Inspector working on-site, I want an optimized mobile submission view to log real-time structural assessments, attach verification notes, and sign off field tasks.	High	Done
F79	Geo- Fenced Inspection Location Validation	As a Zoning Administrator, I want the mobile inspection module to verify if the inspector's device coordinates match the property entry within a reasonable buffer.	Medium	Done

F80	Centralize d FullCalendar ar Monitoring Matrix	As an Inspector / Administrator, I want a dual-pane calendar grid dashboard powered by FullCalendar showing dynamically color-coded inspection event banners so that I can see scheduled and passed cases visually at a glance.	High	Done
F81	Searchabl e Input Dropdown Selection Engine	As an Administrative Clerk, I want a custom-engineered filterable dropdown menu with built-in instant search (initSearchableDrop down) to smoothly	Medium	Done

		select approved applications and inspectors during scheduling.		
F82	Multi-Tier Integrated Zoning Compliance Checklist	As an Inspector, I want a mandatory digital evaluation form covering Land Use Classification, Plan & Setback Consistency, and Expansion Controls to enforce rigorous field validation standards.	Critical	Done
F83	Real-time Category Status Filter Tabs	As an Administrator, I want interactive navigational tab pills (All, For Inspection, Scheduled, Completed) to	Medium	Done

		seamlessly slice the data table views on-the-fly without refreshing the page workspace.		
F84	Mobile Layout Viewport Breakpoint Adjusters	As a Mobile User, I want the system-wide application cards, inspection schedules, checkboxes, and textareas to dynamically scale down, stack vertically, or intelligently hide non-critical metadata columns at 768px, 480px, and 320px screens.	High	Done

F85	Secure Role- Based Inspector Router	As an Inspector, I want the system to enforce strict session authorization safeguards (`requireRole('inspe ctor')`) so that unauthorized users cannot view my field verification queues.	Critical	Done
F86	Inspector Chronologi cal Task Matrix	As an Inspector, I want a centralized dashboard view (`My Inspection Tasks`) that aggregates my active assigned properties by their scheduled timestamps so that I	High	Done

		can efficiently plan my site visits.		
F87	Contextual Field Verification Quick Actions	As an Inspector, I want each project assignment card to feature direct action links to view the application data profile or jump straight to the field evaluation intake form.	High	Done
Administrative Reporting & Output Systems				
F88	Automated Zoning Certificate Drafting	As a Head Zoning Administrator, I want the application to automatically populate municipal boilerplate templates with applicant details	High	Done

		once a file hits "Approved" status.		
F89	Secure System Digital Seal Generatio n	As a System Administrator, I want generated digital zoning documents to embed an official tracking code and system watermark stamp to counter fraudulent paper duplicates.	High	Done
F90	Comprehe nsive Multi- Parameter Batch Export	As an Urban Planner, I want to extract custom data slices (grouped by Barangay, Project Type, or Application Status) into spreadsheet formats	High	Done

		for municipal audit reviews.		
F91	Automated System Database Backup Trigger	As a System Administrator, I want to initiate complete system data dumps and structural record archival transfers directly from the backend settings view.	Critical	Done
F92	Immediate Digital Enforcement Notice of Violation	As an Inspector, I want a secure legal submission panel with a dropdown containing critical infractions (e.g., Deviation from Plan, Encroachment) to rapidly generate a Notice of Violation.	High	Done

F93	Multi-Part Multi- Layer File Evidence Pipeline	As an Inspector, I want the violation report module to handle complex multipart/form-data multi-format file streams to securely attach digital photo proof directly into the compliance record storage.	High	Done
User Profile				
F94	Edit Personal Profile	As a User, I want to view and edit my personal information (name, email, phone, address) so that my account details stay accurate.	High	Done

F95	Change Own Password	As a User, I want to change my account password with a strength indicator so that I can keep my account secure.	High	Done
F96	Upload Profile Avatar	As a User, I want to upload a profile photo so that my account is personalized and easily recognizable to others.	Low	Done
System Settings & Administration				
F97	Site-wide Announce ment Banner	As an Admin, I want to configure a site- wide announcement banner (message, type, enabled state) with a live preview so that I can	Medium	Done

		communicate important notices to all users.		
F98	Role Permissions Matrix	As a Super Admin, I want to manage a permissions matrix per role so that I can control exactly which features each role can access.	High	Done
F99	Personal Login Activity Log	As a User, I want to view my recent login activity (status, browser/device, IP, date/time) so that I can verify no unauthorized access has occurred on my account.	Medium	Done
F100	Secure Database	As a Super Admin, I want to download a	High	Done

	Backup Download	full SQL backup of the database after verifying my password and stating a reason so that I have a secure, audited copy for disaster recovery.		
F101	Restricted Settings Access for Staff	As a Staff member, I want to access only the Language & Locale tab in Settings while other tabs remain locked so that system-wide configuration stays limited to Admins.	Medium	Done
User Management Module				
F102	Restricted User	As a Super Admin, I want to restrict the User Management	High	Done

	Managem ent Access	page to Admin and Super Admin roles so that only authorized staff can manage user accounts.		
F103	Create New User Account	As an Admin, I want to create new user accounts with role assignment so that I can onboard staff and applicants into the system.	High	Done
F104	Edit User Account Details	As an Admin, I want to edit an existing user's name, username, email, phone, and role so that account records stay accurate.	High	Done

F105	Optional Password Reset on Edit	As an Admin, I want to optionally set a new password when editing a user so that I can reset credentials without recreating the account.	Medium	Done
F106	Password Strength Enforceme nt	As a System, I want to enforce password complexity rules (8+ characters, uppercase, and a number) so that user accounts remain secure.	High	Done
F107	Live Password Strength Indicator	As an Admin, I want to see a live strength meter while typing a password so that I can create	Low	Done

		or reset a compliant password.		
F108	Toggle Password Visibility	As an Admin, I want to toggle password field visibility so that I can verify what I typed before submitting.	Low	Done
F109	Quick Activate/Deactivate Action	As an Admin, I want to activate or deactivate a user account with a confirmation prompt so that I can control account access without opening the full edit form.	High	Done
F110	Search and Filter User List	As an Admin, I want to search users by name or email and filter by role so that I	Medium	Done

		can quickly locate specific accounts.		
F111	Paginated User Table	As an Admin, I want the user list paginated so that I can browse large numbers of accounts without performance issues.	Medium	Done
F112	Online/Offline Status Indicator	As an Admin, I want to see an online or offline indicator per user so that I know who is currently active in the system.	Low	Done
F113	View Identity Verification Documents	As an Admin, I want to view an applicant's uploaded ID front and back images so that I can verify their identity.	High	Done

F114	Zoom ID Image Preview	As an Admin, I want to click an ID image to zoom in so that I can inspect the details clearly before making a decision.	Low	Done
F115	Approve or Reject Identity Verification	As an Admin, I want to approve or reject an applicant's identity verification with a selectable reason so that I can control who is verified in the system.	High	Done
F116	Custom Rejection Reason	As an Admin, I want to type a custom rejection reason when the standard list doesn't apply so that applicants	Medium	Done

		receive an accurate explanation.		
F117	Automatic Verification Notification	As an Applicant, I want to receive a system message when my identity is approved or rejected so that I know the outcome without checking manually.	High	Done
F118	View User Activity Logs	As an Admin, I want to view a user's last login and submitted applications so that I can review their engagement history.	Medium	Done
F119	Secure CSV Export Verification	As an Admin, I want to re-enter my password and state a purpose before exporting the user	High	Done

		list to CSV so that sensitive data exports are properly authorized.		
F120	One-Time Export Token	As a System, I want CSV exports of the user table to require a one-time verification token so that exported downloads cannot be replayed or accessed without going through verification.	Medium	Done
F121	Multilingual User Management Interface	As a User, I want the User Management page labels and messages available in English or Filipino	Medium	Done

		so that I can use the interface in my preferred language.		
Audit & Compliance Logging				
F122	Restricted Audit Log Access	As a Super Admin, I want only Admin and Super Admin roles to access the Audit Logs page so that sensitive activity records stay protected from unauthorized staff.	High	Done
F123	Automatic Severity Classification	As an Admin, I want the system to automatically tag each log entry as Critical, Warning, or Info based on the action performed so that I can quickly	High	Done

		spot high-risk activities.		
F124	Filter Logs by Action and Date	As an Admin, I want to filter audit logs by action type and date range so that I can narrow down records for a specific investigation.	High	Done
F125	Paginated Audit Log Table	As an Admin, I want the audit logs to be paginated so that I can browse large volumes of records without performance issues.	Medium	Done
F126	Log Detail Modal	As an Admin, I want to click a log entry to view full details including performer,	High	Done

		IP address, timestamp, and device or browser info so that I can investigate an action thoroughly.		
F127	Device and Browser Detection	As an Admin, I want the system to parse the user agent string into a readable device and browser name so that I can identify where an action originated from.	Low	Done
F128	Configurable Log Retention Purge	As a Super Admin, I want to purge audit logs older than a selected retention period (1, 2, 3, or 5 years) so that the	Medium	Done

		database stays within storage and compliance requirements.		
--	--	--	--	--

Table no. 4 Product Backlog (User Stories)

3.4.2 Product Backlog for EIS Information Security

EIS No.	EIS User Stories	EIS IS Priority	Revision Priority	Status
IS-1	As a Security Admin, I want the system to require passwords to be at least 8 characters long and combine uppercase, lowercase, and numbers to prevent weak accounts.	2	2	Done
IS-2	As a System Developer, I want all newly verified self-registrants to be permanently assigned the database security role of applicant by	1	2	Done

	default to enforce proper privilege scoping.			
IS-3	As a System Developer, I want the session ID to be regenerated (session_regenerate_id(true)) and temporary credentials cleared upon a successful login to mitigate session fixation attacks.	1	2	Done
IS-4	As a Security Admin, I want all generated login and registration OTP codes to strictly expire within 1 minute to prevent interception and brute-force re-use.	1	1	Done
IS-5	As a Security Developer, I want OTP parameters (otp_code, otp_expiry) inside the database row to be completely nullified	1	2	Done

	immediately after execution to block replay attacks.			
IS-6	As an Inspector, I want the dashboard to strictly filter and show only the applications and inspection schedules explicitly assigned to my account to avoid unauthorized data exposure.	1	2	Done
IS-7	As an Applicant, I want the system to validate my specific role permissions before exposing historical pipelines or submission forms to block unauthorized access endpoints.	2	3	Done
IS-8	As a Security Admin, I want all status modifications, assignments, and database writes to run cryptographic token validations to shield	1	1	Done

	administrative modules from Cross-Site Request Forgery.			
IS-9	As an Auditor, I want an unalterable log table tracking every instance of administrative file re-routing, parameter adjustments, and final evaluation decisions.	1	2	Done
IS-10	As an Inspector, I want the system to enforce strict session authorization safeguards (requireRole('inspector')) so that unauthorized users cannot view my field verification queues.	1	2	Done
IS-11	As a Super Admin, I want to manage a permissions matrix per role so that I can control exactly which features each role can access.	2	2	Done

IS-12	As a Super Admin, I want to restrict the User Management and Audit Log pages strictly to Admin and Super Admin roles to protect sensitive records.	2	2	Done
IS-13	As a System, I want CSV exports of the user table to require a password challenge and a one-time verification token so that sensitive data downloads cannot be replayed or leaked.	3	2	Done
IS-14	As a Security Admin, I want the system to automatically terminate a user's session after 2 minutes of inactivity so that unattended devices do not expose sensitive LGU or applicant data.	2	2	Done

IS-15	As a System, I want exported PDF reports to be password-protected and marked confidential so that sensitive LGU analytics cannot be viewed if the file is intercepted or shared without authorization.	3	2	Done
-------	--	---	---	------

Table no. 5 Product Backlog for EIS Information Security

3.4.3 Product Backlog for EIS Standards (UI/UX)

The expanded backlog below details all the user interface and user experience standards required across the system. It maps the basic layout rules directly to your core front-end architectural requirements, ensuring mobile responsiveness, custom-engineered UI overlays, real-time feedback loops, and multi-language support.

EIS Standard	EIS Standard User Stories	EIS Standard Priority	Revision Priority	Status
UI-1	As a Product Owner, I want a clean, minimalist	1	1	Done

	dashboard so that users can navigate features without confusion.			
UI-2	As a Product Owner, I want the GIS map to be fully responsive so that it can be viewed on tablets and mobile devices during field inspections.	1	1	Done
UI-3	As a Product Owner, I want color-coded status badges (e.g., Green for Compliant, Red for Non-Compliant) for instant data assessment.	1	2	Done
UI-4	As an Applicant, I want a clean, 3-step registration wizard interface so that the user sign-up workflow feels structured and visually bounded.	2	2	Done

UI-5	As an Applicant, I want a visual password strength bar with color-coded feedback below inputs to gauge account security in real time.	3	3	Done
UI-6	As a User, I want an eye icon button inside password forms to easily toggle the visibility of hidden characters on constrained screens.	3	3	Done
UI-7	As a User, I want validation error and success banners displayed via a custom, blurred modal overlay (#customAlertOverlay) instead of standard default browser alert blocks.	2	2	Done

UI-8	As an Applicant, I want layout headers, buttons, and system metrics fields to support dictionary-mapped translation data attributes (data-en / data-tl) for dynamic localization.	2	2	Done
UI-9	As a Mobile User, I want the authentication cards to smoothly adapt spacing and element scaling exactly at 768px, 480px, and 320px responsive breakpoints.	1	1	Done
UI-10	As a Mobile User, I want the 6 distinct 2FA verification boxes to scale down to a tight format at 320px screen widths to prevent any container spill.	1	1	Done

UI-11	As a Mobile User, I want complex, multi-column input sets to stack vertically under 480px to preserve comfortable touch and tap targets.	1	1	Done
UI-12	As a Mobile User, I want the recent applications table data to selectively drop and hide secondary metadata columns at lower viewports to block text clipping.	2	2	Done
UI-13	As a Mobile User, I want priority call-to-actions and buttons to span full horizontal widths below 480px for optimal screen utility.	2	2	Done
UI-14	As an Applicant, I want an interactive monthly	2	2	Done

	calendar view powered by FullCalendar that visually frames the current day (.today) and highlights inspection dates (.has-appt).			
UI-15	As a Mobile User, I want the complex application grid records to dynamically collapse into a scannable, stacked-card layout on viewports $\leq 480\text{px}$ to maximize thumb-scrolling.	2	2	Done
UI-16	As a Mobile Admin User, I want the Leaflet map container to execute an explicit size recalculation (invalidateSize) upon modal trigger to ensure it scales without clipping.	1	1	Done

UI-17	As a System User, I want a Floating Action Button (FAB) layer selector menu styled in a clean modern aesthetic to quickly toggle between structural tile layers.	3	3	Done
UI-18	As an Administrative Clerk, I want selection elements equipped with a custom-engineered filterable dropdown menu and instant search (initSearchableDropdown) to ease operational inputs.	2	2	Done

Table no. 6 Product Backlog for EIS Standard

3.4.4 Product Backlog for EIS Integration

EIS Integration No.	EIS Integration User Stories	EIS Integration Priority	Revision Priority	Status
INT-1	As an Admin, I want the Map Module to pull real-time data from the Permit Database so that approved structures appear on the public map.	1	2	Done
INT-2	As a Zoning Officer, I want the Compliance Engine to automatically pull the "Zoning Classification" from the GIS layer to validate applications.	1	1	Done

INT-3	As a User, I want the Notification Center to trigger alerts based on status updates made in the Permit Management Module.	1	2	Done
INT-4	As an Applicant, I want my interactive Leaflet Map pin placement coordinates (<i>Latitude, Longitude</i>) to instantly synchronize and populate the text input fields of the intake form in real-time.	1	2	Done
INT-5	As an Administrative Supervisor, I want	2	3	High

	the Officer Routing dropdown menu to run an active database count query against the application table to display live caseload numbers next to each staff name.			
INT-6	As a Zoning Officer, I want the property coordinate pin to run an automated spatial intersection check against protected hazard bands and easement layers, throwing a critical system-wide flag upon violation.	1	1	Done

INT-7	As an Admin Reviewer, I want the Map interface to automatically bind a static snapshot of the verified property pin layout directly to the digital record ledger on transaction save.	2	3	Done
INT-8	As a Security Developer, I want all status modifications and routing assignments to explicitly validate a cryptographic CSRF token before executing updates on the administrative database tables.	1	1	Done

INT-9	As a Database Developer, I want a failure in the status history recording step (application_status_history) to trigger an immediate programmatic SQL rollback across parent compliance records to prevent data desynchronization.	1	1	Done
INT-10	As an Auditor, I want an unalterable system log engine that automatically intercepts every instance of administrative routing, coordinate	1	1	Done

	adjustment, and final evaluation to pipe them into the master audit trail.			
INT-11	As an Admin/Zoning Officer, I want the central dashboard metrics to pull direct aggregate statistics from the applications table to dynamically update the color-coded counter grid and SLA overdue alerts.	2	2	Done
INT-12	As an LGU Executive, I want the Chart.js visual graphs (doughnut, pie, bar, and trend line charts) to query real-time	2	3	Done

	transactional status and barangay volume counts for up-to-the-minute analytical rendering.			
INT-13	As an Inspector working on-site, I want the Mobile Field Inspection Module to run a real-time comparison between my device's GPS coordinates and the property's recorded GIS coordinates to validate our geo-fenced proximity buffer before letting me submit data	2	3	Done
INT-14	As an Inspector, I want the	1	2	Done

	FullCalendar monitoring interface to synchronize directly with active database inspection schedules to dynamically render color-coded event banners (.has-appt, .today).			
INT-15	As a Head Zoning Administrator, I want the system to successfully check off all items on the multi-tier digital evaluation compliance checklist before unlocking the final "Approved" or	1	1	Done

	"Rejected" database transaction triggers.			
INT-16	As a Head Zoning Administrator, I want an "Approved" database status change to instantly trigger a municipal boilerplate template compile engine to auto-draft a secure, watermarked digital Zoning Certificate	1	2	Done
INT-17	As an Urban Planner, I want the comprehensive batch export engine to map and filter database structures by Barangay, Project Type, and Status into	2	2	Done

	cleanly formatted spreadsheet schemas for immediate local auditing.			
INT-18	As an Inspector, I want the digital Notice of Violation panel to accept multi- part file streams and attach multi-format image proofs directly onto the specific property record during field audits.	1	2	Done
INT-19	As an Applicant, I want the system to integrate with an online payment gateway so that I can settle my permit fee	1	2	Done

	electronically once my application is cleared, before final approval is granted.			
--	--	--	--	--

Table no. 7 Product Backlog for EIS Integration

3.4.5 Product Backlog for Analytics

3.4.5.1 Application System Analytics

EIS Integration No.	EIS Integration User Stories	EIS Integration Priority	Revision Priority	Status
ASA-1	As an Admin, I want a unified dashboard with KPIs from the GIS and Permit modules to monitor system load.	1	2	Done
ASA -2	As a Zoning Officer, I want to	1	3	Done

	view a report on permit approval rates to identify bottlenecks in the workflow.			
ASA -3	As an Admin, I want to analyze the frequency of zoning violations detected by the automated compliance engine.	1	2	Done
ASA-4	As a Planning Manager, I want to view the average time taken for an application to move from	1	2	Done

	"Submitted" to "Issued."			
ASA-5	As an Admin, I want an AI Insights panel that generates a natural-language summary of key report metrics alongside the standard charts, to quickly interpret report trends.	1	3	Done

Table no. 8 Product Backlog for Application System Analytics

3.4.5.2 EIS Analytics

This module provides high-level visualizations for LGU executives to aid in long-term urban planning.

EA No.	EIS Analytics Stories	EIS Analytics Priority	Revision Priority	Status
EA-1	As an LGU Executive, I want an interactive, structured bar graph displaying the volumetric distribution of total permits issued per barangay or district to easily track regional growth shifts.	1	1	Done
EA-2	As a Planning Director, I want a real-time spatial density heatmap overlay rendered on the Leaflet map layout to dynamically identify areas experiencing	1	1	Done

	the highest local construction activity.			
EA-3	As an Admin, I want an interactive pie and doughnut chart summary of land-use classification distribution (e.g., Residential, Commercial, Industrial) currently recorded in the database to cross-verify master plan compliance.	1	1	Done
EA-4	As an LGU Executive, I want to view a smooth, curved trend line graph detailing the running totals of municipal revenue generated from locational clearance fees over a selected calendar period.	1	1	Done

EA-5	As an Urban Planner, I want a comparative analytical chart showing the ratio of approved permits versus flagged zoning violations per district to assess local zoning law enforcement trends.	2	2	Done
EA-6	As a Planning Director, I want a filtered seasonal trend report that projects future application surges based on historical annual submission peaks to optimize office staff allocation.	2	3	Done
EA-7	As a Super Admin, I want a clean dashboard matrix displaying system-wide data integrity metrics,	2	2	Done

	including total unalterable audit trails generated and database backup frequencies, to present at executive compliance reviews.			
--	---	--	--	--

Table no. 9 Product Backlog for EIS Analytics

3.4.6 Sprint Backlog (User Stories)

The project followed an 11-sprint cycle. Each sprint lasted a specific number of hours and involved the collaborative efforts of the development and documentation team.

Task No.	User Story No.	User Stories	Tasks	Timelin e	Responsi ble Team Members
SPRINT 1 System Identity, Core Security & Registration Engine					
S1_1	F1	As an Applicant, I want a clean, 3-step registration wizard (Personal Info, Account	PLANNING DESIGN	10 Hours	Developer , UI/UX Designer, Document Specialist

		Security, Address & Verification) so that the sign-up process feels structured and not overwhelming.	CODE DOCUME NTATION		
S1_2	F2	As an Admin, I want the system to reject registration inputs if the applicant's age is under 18 years old to comply with legal capacity rules.	PLANNING CODE DOCUME NTATION	8 Hours	Developer , QA Tester
S1_3	F3, IS-1	As a Security Admin, I want the system to require passwords to be at least 8	PLANNING CODE	8 Hours	Developer , QA Tester

		characters long and combine uppercase, lowercase, and numbers to prevent weak accounts.	DOCUMENTATION		
S1_4	F4	As a Zoning Officer, I want applicants to select a Valid ID type (PhilSys, Driver's License, Passport) and upload separate clear images of the ID Front and ID Back for identity audit.	PLANNING DESIGN CODE DOCUMENTATION	8 Hours	Developer , Document Specialist
S1_5	F8, IS-2	As a System Developer, I want	PLANNING	8 Hours	Developer

		all newly verified self-registrants to be permanently assigned the database security role of applicant by default to enforce proper privilege scoping.	CODE DOCUME NTATION		
SPRINT 2 GIS Map Foundations & Interactive Layering					
S2_1	F54, UI-2	As an Applicant, I want an integrated Leaflet map canvas that enables me to interactively click or position a draggable pin to automatically populate high-precision latitude	PLANNING DESIGN CODE DOCUME NTATION	8 Hours	Developer

		and longitude coordinates into my form.			
S2_2	F69	As an Application Evaluator, I want the map interface to strictly restrict panning and zooming actions within the official geographical boundaries of Quezon City (maxBounds) so that I do not accidentally lose sight of local zoning assets.	PLANNING CODE DOCUMENTATION	5 Hours	Developer
S2_3	F70, UI-17	As a System User, I want an accessible	PLANNING	5 Hours	UI/UX Designer, Developer

		Floating Action Button (FAB) layer control menu styled in a clean Google Maps aesthetic to rapidly toggle between Road, Satellite, Hybrid, and Terrain tile layers.	DESIGN CODE DOCUME NTATION		
S2_4	INT-1	As an Admin, I want the Map Module to pull real-time data from the Permit Database so that approved structures appear on the public map.	PLANNING CODE DOCUME NTATION	5 Hours	Developer

S2_5	INT-2	As a Zoning Officer, I want the Compliance Engine to automatically pull the "Zoning Classification" from the GIS layer to validate applications.	PLANNING CODE DOCUMENTATION	5 Hours	Developer
SPRINT 3 Zoning Rules Engine & Secure Intake Form Controls					
S3_1	F53	As an Applicant, I want to fill out a structured, multiparametric intake form capturing detailed Project Profiles (name, type, description) alongside precise	PLANNING DESIGN CODE DOCUMENTATION	7 Hours	Developer

		Land Parcel Identities (PIN, lot, block, street, barangay).			
S3_2	F55	As an Applicant, I want to dynamically append or remove file upload slots on demand, designating structural file types (Site Plan, Lot Plan, Proof of Ownership, Building Plan) directly into my payload.	PLANNING DESIGN CODE DOCUMENTATION	7 Hours	Developer, QA Tester
S3_4	F67, INT-6	As a Zoning Officer, I want the	PLANNING	6 Hours	Developer

		system to automatically flag any property coordinate pin that falls into protected hazard bands or restricted easements before I sign off.	CODE DOCUME NTATION		
S3_5	F72	As a Head Zoning Administrator, I want the final "Approved" and "Rejected" status triggers to stay completely locked out until the compliance and impact	PLANNIN G CODE DOCUME NTATION	6 Hours	Developer , QA Tester

		assessments are marked as complete.			
SPRINT 4 Admin Dashboard Controls & Operational Counters					
S4_1	F28, F29	As an LGU Administrator, I want to view real-time data on Average Processing Time, Top Performing Barangay, and color-coded counter grid cards showing total, pending, approved, and rejected applications to monitor operations.	PLANNING DESIGN CODE DOCUMENTATION	7 Hours	UI/UX Designer, Developer

S4_2	F31	As a Zoning Administrator, I want an active alert banner to appear automatically if any application remains unreviewed for more than 3 days, providing a direct link to prioritize it.	PLANNING DESIGN CODE DOCUMENTATION	7 Hours	Developer , QA Tester
S4_3	F32, IS-6	As an Inspector, I want the dashboard to strictly filter and show only the applications and inspection schedules	PLANNING CODE DOCUMENTATION	8 Hours	Developer

		explicitly assigned to my account to avoid information clutter.			
S4_4	F33, F86	As an Inspector, I want a centralized workspace ("My Inspection Tasks") aggregating active assigned properties by their scheduled timestamps with direct quick- action links to submit field reports.	PLANNING DESIGN CODE DOCUMENTATION	8 Hours	Developer
SPRINT 5 GIS Data Sync & Responsive Viewport Engineering					

S5_1	INT-4	As an Applicant, I want my interactive Leaflet Map pin placement coordinates (Latitude, Longitude) to instantly synchronize and populate the text input fields of the intake form in real-time.	PLANNING CODE DOCUMENTATION	9 Hours	Developer
S5_2	F68, INT-7	As an Admin Reviewer, I want the Map interface to automatically generate and bind a static snapshot of the	PLANNING CODE DOCUMENTATION	9 Hours	Developer

		verified property pin layout directly to the digital record ledger on transaction save.			
S5_3	F66, UI-16	As a Mobile Admin User, I want the Leaflet map container to force a bounding box recalculation (invalidateSize) upon modal open to ensure the interactive map scales across smaller screens without clipping.	PLANNIN G CODE DOCUME NTATION	9 Hours	Developer , QA Tester
S5_4	F52, UI-15	As a Mobile User, I want the complex	PLANNIN G	9 Hours	UI/UX Designer, Developer

		application grid records to dynamically collapse into a scannable, stacked-card layout on viewports $\leq 480\text{px}$ to facilitate easy thumb-scrolling and asset tracking.	DESIGN CODE DOCUME NTATION		
SPRINT 6 Chart.js Analytics & Document Output Systems					
S6_1	F35, F38, EA-4	As an LGU Executive, I want interactive Chart.js graphs (doughnut, pie, bar, and curved trend lines) rendering running	PLANNING DESIGN CODE	7 Hours	Developer , UI/UX Designer

		totals of application volumes and locational clearance revenues.	DOCUMENTATION		
S6_2	F88, F89	As a Head Zoning Administrator, I want the system to auto-draft digital Zoning Certificates embedding an official tracking code and digital seal watermark once an application reaches	PLANNING DESIGN CODE DOCUMENTATION	6 Hours	Developer

		"Approved" status.			
S6_3	ASA-5	As an Admin, I want an AI Insights panel that generates a natural-language summary of key report metrics alongside standard charts so that I can quickly interpret trends without manually reading every table.	PLANNING DESIGN CODE DOCUMENTATION	6 Hours	Developer
SPRINT 7 Two-Factor Authentication (2FA) & Workspace Logic					
S7_1	F14, F15, IS-4	As a User, after entering valid credentials, I want a mandatory	PLANNING	7 Hours	Developer

		6-digit OTP sent to my email with a strict 1-minute expiration limit to verify access to my dashboard securely.	DESIGN CODE DOCUME NTATION		
S7_2	F18, IS-5	As a Security Developer, I want OTP parameters (otp_code, otp_expiry) inside the database row to be completely nullified immediately after execution to block replay attacks.	PLANNIN G CODE DOCUME NTATION	7 Hours	Developer

S7_3	F16, F20	As a User, I want a "Resend OTP" button enforcing a 60-second locking cooldown timer synced via client-side sessionStorage to persist accurately during page reloads.	PLANNING DESIGN CODE DOCUMENTATION	5 Hours	Developer , QA Tester
SPRINT 8 Advanced User Management & ID Verification					
S8_1	F102, IS-12	As a Super Admin, I want to restrict the User Management page strictly to Admin and Super Admin roles so that only authorized staff	PLANNING CODE DOCUMENTATION	7 Hours	Developer

		can manage user accounts.			
S8_2	F113, F114	As an Admin, I want to view an applicant's uploaded ID front and back images inside a dedicated verification tray with interactive modal zoom features before approving account credentials.	PLANNING DESIGN CODE DOCUMENTATION	6 Hours	Developer , UI/UX Designer
S8_3	F115, F116	As an Admin, I want to approve or reject an applicant's identity	PLANNING DESIGN	6 Hours	Developer , QA Tester

		verification with pre-selected dropdown reasons or custom textual notes to communicate clear verification outcomes.	CODE DOCUME NTATION		
SPRINT 9 Cryptographic Defenses & Compliance Audit Logging					
S9_1	F75, IS-9	As an Auditor, I want an unalterable log table tracking every instance of administrative file re-routing, parameter adjustments, and final evaluation decisions	PLANNING CODE DOCUME NTATION	8 Hours	Developer

		automatically classified by severity.			
S9_2	F73, IS-8	As a Security Admin, I want all status modifications, assignments, and database writes to run cryptographic token validations (CSRF protection) to shield administrative modules from forgery.	PLANNIN G CODE DOCUME NTATION	7 Hours	Developer
S9_3	F74, INT-9	As a Database Developer, I want any file update	PLANNIN G	7 Hours	Developer

		failures in application_status_history to instantly trigger a SQL rollback across parent compliance records to preserve system truth.	CODE DOCUMENTATION		
SPRINT 10 Field Logistics & Enforcement Notice Channels					
S10_1	F79, INT- 13	As a Zoning Administrator, I want the mobile inspection module to verify if an inspector's device GPS coordinates match recorded property anchors	PLANNING CODE DOCUMENTATION	9 Hours	Developer

		within a strict geofenced buffer before permitting data uploads.			
S10_2	F92, F93	As an Inspector, I want a secure legal submission panel to log standardized municipal infractions (Notice of Violation) supported by a multi-part file uploader to attach digital photo proof directly into field evidence records.	PLANNING DESIGN CODE DOCUMENTATION	10 Hours	Developer

SPRINT 11 Secure Export Systems, Database Archiving & Deployment					
S11_1	F119, F120	As an Admin, I want CSV/Excel data exports to require a password re-challenge, an explicit statement of purpose, and a single-use token to prevent unauthorized data extractions and replay attacks.	PLANNING DESIGN CODE DOCUMENTATION	8 Hours	Developer
S11_2	F91, F100	As a Super Admin, I want to execute automated system database	PLANNING CODE	12 Hours	Developer

		backup dumps and configure log retention purge limits directly from backend settings for disaster recovery readiness.	DOCUME NTATION		
--	--	--	-------------------	--	--

Table no. 10 Sprint Backlog (Technical Execution)

3.4.6.1 Sprint Burndown Charts

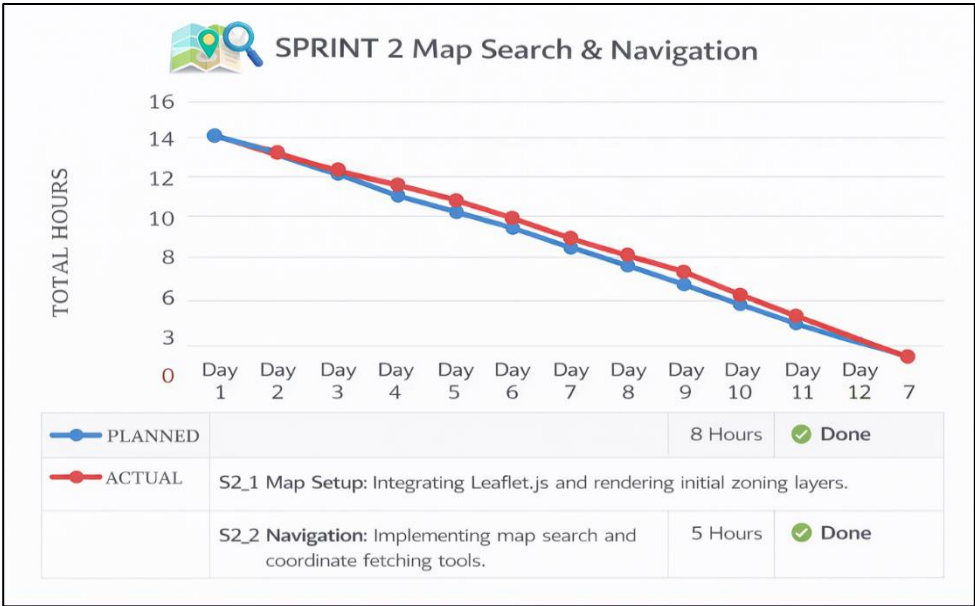


Figure no. 3 Sprint 1 Burndown Charts

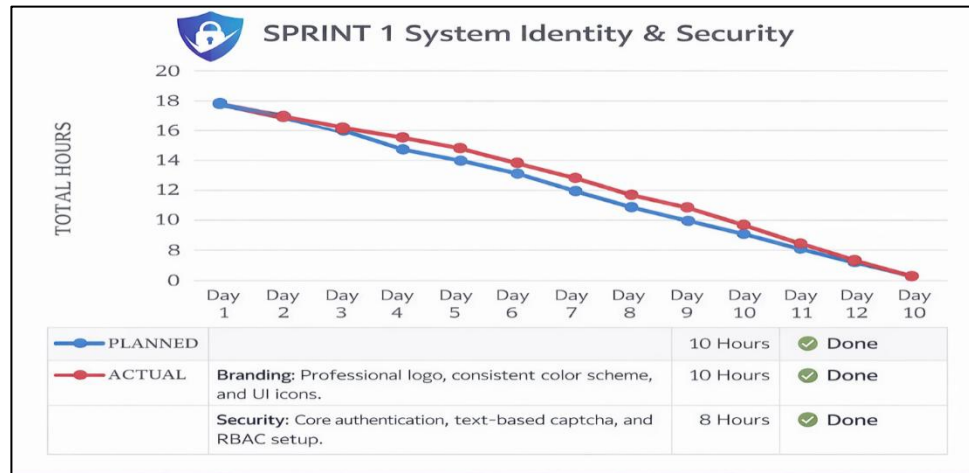


Figure no. 4 Sprint 2 Burndown Charts

3.4.7 Increment

Sprint No.	Increment / Feature Delivered	User Story / Backlog Reference	Definition of Done (DoD) Criteria	Status	Remarks
Sprint 1	System Identity & Core Registration Engine	F1, F2, F4, F8, IS-1, IS-2	- Code completed - Unit tested - Integrated with DB - Documentation updated	Done	Core authentication and user registration wizard operational.
Sprint 1	Password Security & Identity Verification Setup	F3, F7, F9	- Security rules configured - DB constraints validated - Tested with sample inputs - Documentation updated	Done	Password complexity rules and secure upload handling active.

Sprint 2	Leaflet GIS Mapping Engine & QC Geofence	F54, F69, UI-2	- Leaflet canvas integrated - Coordinate picker functional - Map boundaries locked to QC - Documentation updated	Done	Spatial map canvas with draggable coordinate pin ready.
Sprint 2	GIS Layer Control & Permit DB Integration	F70, INT-1, INT-2, UI-17	- Tile layers toggleable - DB queries integrated - Tested on staging - Documentation updated	Done	Layer FAB switcher and compliance layer sync working.
Sprint 3	Multiparametric Intake Form & Asset Uploader	F53, F55	- Form controls built - Dynamic upload slots tested - Input validation passed - Documentation updated	Done	Structured permit intake payload submission ready.
Sprint 3	Automated Hazard Buffer Flagging & Guardrails	F67, F72, INT-6	- Spatial intersection logic coded - Status triggers locked - Unit tested - Documentation updated	Done	Automated zoning violation flags and evaluation lock active.

Sprint 4	Real-Time Admin Operational Dashboard	F28, F29, F31	- KPI cards rendered - Overdue SLA banner functional - Tested with live DB records - Documentation updated	Done	Workload counters and dynamic SLA alerts active.
Sprint 4	Inspector Task Workspace & Scope Restriction	F32, F33, F86, IS-6	- Role-based scope enforced - Inspection workspace built - Tested with sample inspector accounts - Documentation updated	Done	Field tasks correctly isolated per inspector account.
Sprint 5	GIS Form Auto-Sync & Viewport Engineering	F52, F66, INT-4, UI-15, UI-16	- Coordinate auto-fill active - Map resize trigger fixed - Responsive grid cards tested on mobile - Documentation updated	Done	Mobile stacked-card view and instant coordinate sync fully functional.
Sprint 5	Spatial Verification Snapshot Attachment	F68, INT-7	- Snapshot generation coded - Ledger binding verified - Unit tested - Documentation updated	Done	Static map layout snapshots bound to record ledger.

Sprint 6	Chart.js Executive Analytics Engine	F35, F38, EA-4, INT-12	- Chart views rendered - Revenue trend lines mapped - Performance tested - Documentation updated	Done	Interactive graphs and revenue analytics rendering.
Sprint 6	Digital Zoning Certificate Generator & AI Insights	F88, F89, ASA-5, INT-16	- Certificate template compiled - Digital seal watermarked - AI summary panel functional - Documentation updated	Done	Auto-drafting digital clearances and AI summary panel working.
Sprint 7	Two-Factor OTP Authentication Gateway	F14, F15, F16, F20, IS-4	- Email OTP integrated - 1-minute expiration locked - Resend cooldown timer synced - Documentation updated	Done	2FA verification gate active across user logins.
Sprint 7	Post-Verification Security Cleansing	F18, IS-5	- Nullification logic implemented - Replay attack tests passed - Code reviewed - Documentation updated	Done	Database row OTP parameters cleared instantly after use.

Sprint 8	Restricted User Management & ID Verification Tray	F102, F113, F114, IS-12	- RBAC routes secured - Image zoom modal built - UAT passed - Documentation updated	Done	Administrative tray for ID verification operational.
Sprint 8	Identity Verification Workflow & Custom Notes	F115, F116, F117	- Approval/rejection triggers active - System alerts mapped - Tested with sample accounts - Documentation updated	Done	Automated feedback notifications sent upon identity review.
Sprint 9	System-Wide Compliance Audit Logging	F75, IS-9, INT-10, F123	- Audit table created - Event interception active - Severity tagging verified - Documentation updated	Done	Unalterable log ledger tracking system modifications.
Sprint 9	CSRF Token Validation & SQL Transaction Rollback	F73, F74, IS-8, INT-8, INT-9	- Cryptographic tokens validated - Rollback logic tested - Security audit passed - Documentation updated	Done	Transactional safety and token defenses fully active.
Sprint 10	Geo-Fenced Mobile	F79, INT-13	- Mobile GPS distance	Done	Inspector location

	Inspection Module		validation coded - Geofence buffer tested - Field tested - Documentation updated		validated before field report upload.
Sprint 10	Field Notice of Violation & Multi-Part File Stream	F92, F93, INT-18	- Infraction panel built - Multipart file stream functional - End-to-end field test passed - Documentation updated	Done	Legal violation notices logging and evidence upload operational.
Sprint 11	Password-Protected CSV/Excel Export Verification	F119, F120, IS-13	- Re-authentication form built - One-time token generation active - Download security tested - Documentation updated	Done	Secure export verification gate preventing unauthorized data extraction.
Sprint 11	Automated Database Backup & System Deployment	F91, F100, F128	- SQL dump routines configured - Storage retention purge set - Staging to production deployment completed	Done	Automated disaster recovery backups active; platform deployed to production.

			- Technical manuscript finalized		
--	--	--	----------------------------------	--	--

Table no. 11 Increment

3.5 Microservices Architecture

The system is designed using a Microservices Architecture that decouples core business functions into independent, specialized services to enhance scalability and simplify maintenance. Central to this architecture is the Authentication Service (Auth.php), a standalone module that handles user identity verification, secure session management, and role-based access control for both Applicants and Admins. By isolating this service, the system ensures that security protocols are applied consistently across all interactions without being tightly bound to the application's spatial logic.

Furthermore, the architecture integrates a Spatial Data Service and an Automated Rule Engine as decoupled logic components. The Spatial Data Service manages the processing of GIS layers, cadastral parcels, and zoning boundaries, while the Rule Engine evaluates development proposals against machine-readable zoning ordinances to determine compliance. These are complemented by the Applicant Self-Service Module, a dedicated service that allows users to manage their permit applications and track statuses through the ApplicantController independently of administrative functions. This separation ensures that high-demand spatial processing does not interfere with the responsiveness of the user dashboard or the LGU's administrative workflows.

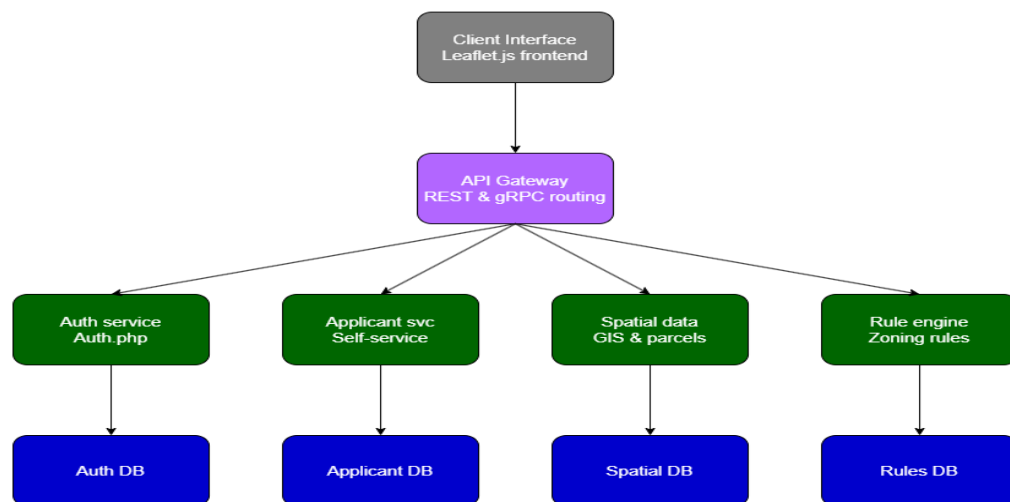


Figure no. 5 Microservices Diagram

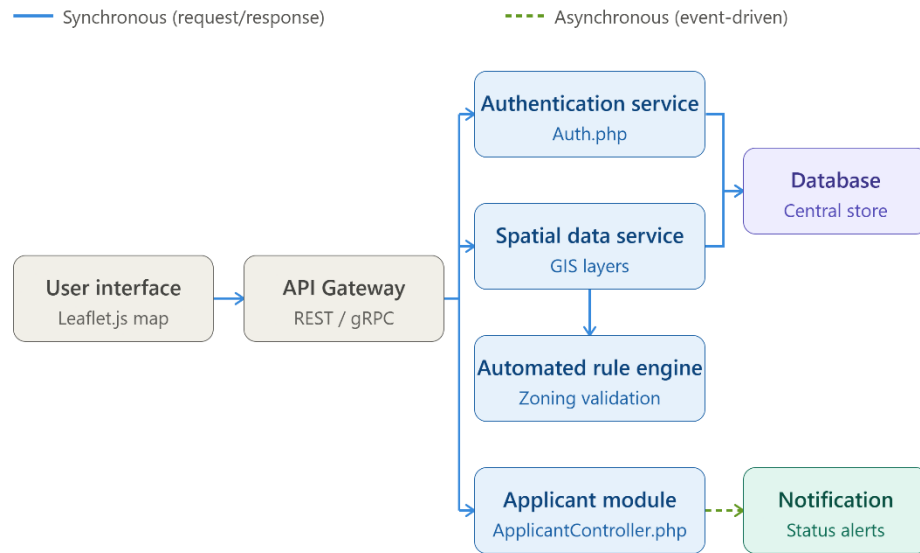


Figure no. 6 Communication Pattern for Microservices

Here's the communication pattern diagram for a web-based GIS system for automated zoning compliance and development permit processing:

- **Blue solid lines → Synchronous** — the UI and API Gateway route requests to the Authentication Service, Spatial Data Service, and Applicant Self-Service Module, which in turn query the central database (Database.php); the Spatial Data Service also feeds the Automated Rule Engine for zoning compliance checks.
- **Green dashed line → Asynchronous** — the Applicant Module triggers the Notification Module for real-time status alerts, consistent

with your document's description of asynchronous processing for time-consuming operations.

This shows the entire web-based GIS system as one process and how external entities interact with it.

- External Entity: Public Applicant / Zoning Officer
- System (Process): Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing
- Data Stores: Property Parcel DB, Zoning Ordinance DB, Permit Application DB, Audit Log DB
- Data Flows: Submit Application, View Map Coordinates, Check Zoning Compliance, Track Status, Receive Permit Notification, Generate Compliance Report

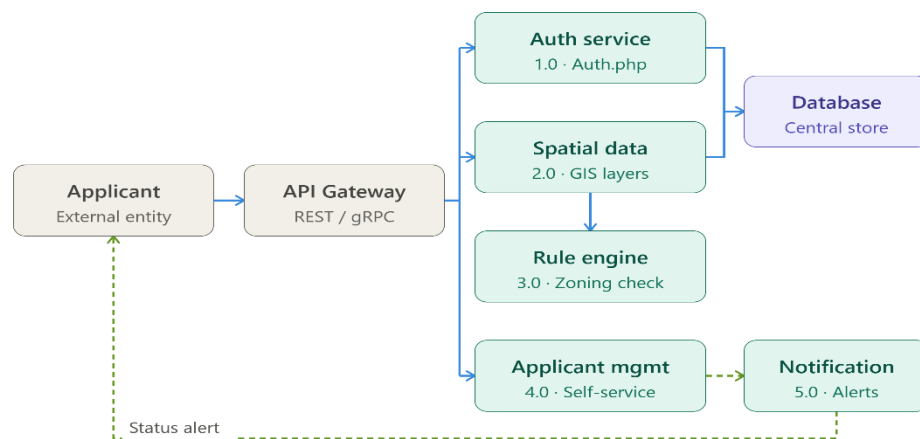


Figure no. 7 Data Flow Diagram for Microservices

3.6 DevOps Implementation

The CI/CD pipeline for the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing automates the software delivery lifecycle from code commitment to production monitoring. The implementation of this pipeline ensures rapid, reliable, and consistent deployments across development and production environments.

The continuous integration and continuous deployment process begins when developers push code changes to the GitHub repository, triggering automated build processes that compile microservices and package spatial processing components. Following this, unit and integration tests are executed automatically to validate backend microservices, database transactions, and GIS spatial compliance logic. Code updates then undergo peer review and administrative sign-off before being merged into the main deployment branch. Approved builds are subsequently deployed automatically to the cloud environment, making updates to the permit processing portal and zoning engine available to users seamlessly. Finally, system performance, application uptime, and deployment status are tracked continuously to ensure reliability and immediate error detection.

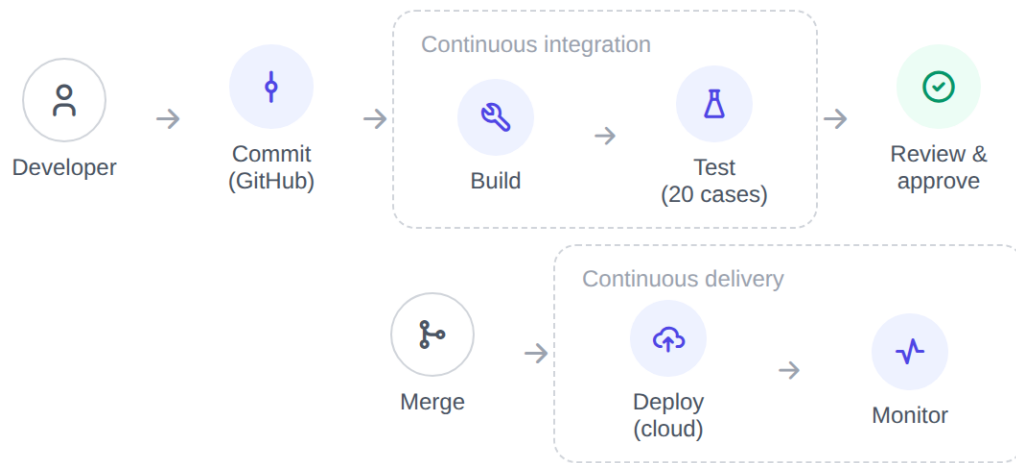


Figure no. 8 CI/CD Pipeline

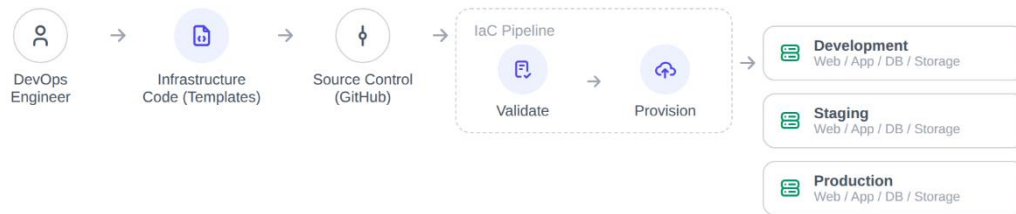


Figure no. 9 Infrastructure as a Code (IaC)

The infrastructure deployment for the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing is managed using Infrastructure as Code (IaC) principles to ensure environment consistency and automated provisioning. The process begins with the DevOps engineer defining the system infrastructure—including web servers, application components, relational databases, and secure storage—using declarative templates. These configuration files are version-controlled and pushed to the GitHub repository. Once committed,

the automated IaC pipeline triggers a validation phase to check syntax and verify resource configurations against cloud provider standards. Following validation, the pipeline provisions and updates the cloud environments systematically, starting with development and staging servers before finally rolling out to production infrastructure.

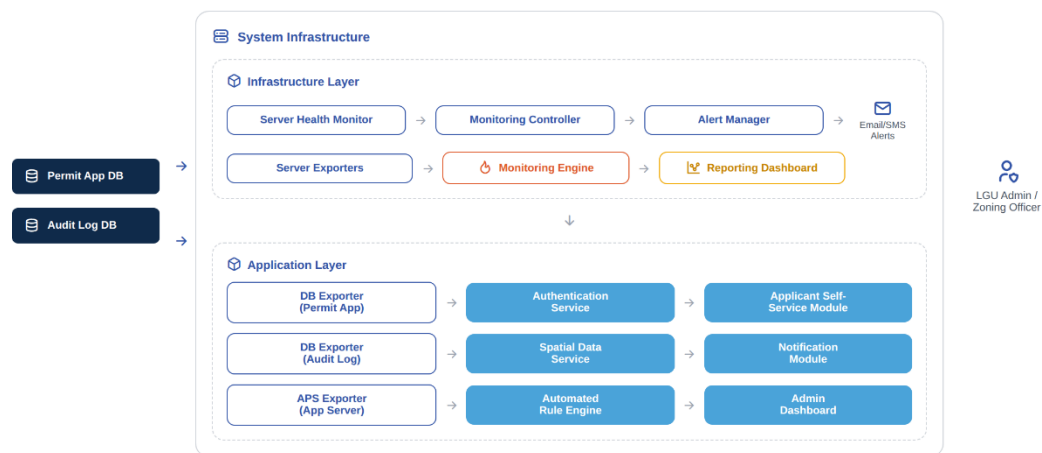


Figure no. 10 Monitoring and Alerting

The monitoring workflow begins at the infrastructure layer, where server health monitors and dedicated server exporters track resource utilization. These metrics are processed by the monitoring controller and evaluated through the monitoring engine to detect performance bottlenecks or system anomalies. When thresholds are breached, the alert manager triggers automated email and SMS notifications to system administrators and zoning officers, while real-time metrics and historical logs are consolidated into the reporting dashboard. Concurrently, the application layer utilizes specific exporters for the permit application database, audit

logs, and application servers to feed data into core services such as authentication, spatial data processing, and the automated rule engine. This integrated monitoring framework ensures high availability, rapid incident response, and reliable tracking of municipal permit operations.

3.7 Integration Approach for Information Systems

The integration strategy for the platform focuses on bridging advanced GIS capabilities with the local government's administrative permitting workflow to create a single, functional system. Central to this approach is Database Integration through the Database.php singleton, which utilizes a centralized spatial database to close the gap between applicant personal information and complex spatial parcel data. This is complemented by API-Driven Connectivity, where frontend interactive maps are integrated with backend PHP controllers to provide users with real-time feedback on zoning compliance as they interact with the map interface.

To ensure practical utility, the system emphasizes LGU Workflow Alignment by providing audit-ready digital workflow and automated report generation specifically designed for urban planners. A critical component of this integration is the Rule Set Conversion strategy, which involves transforming complex, textual zoning ordinances into a structured, machine-readable rule set. This allows the GIS backend to interpret and

execute automated compliance checks such as land-use matches and setbacks effectively digitizing the manual verification process traditionally performed by LGU staff.

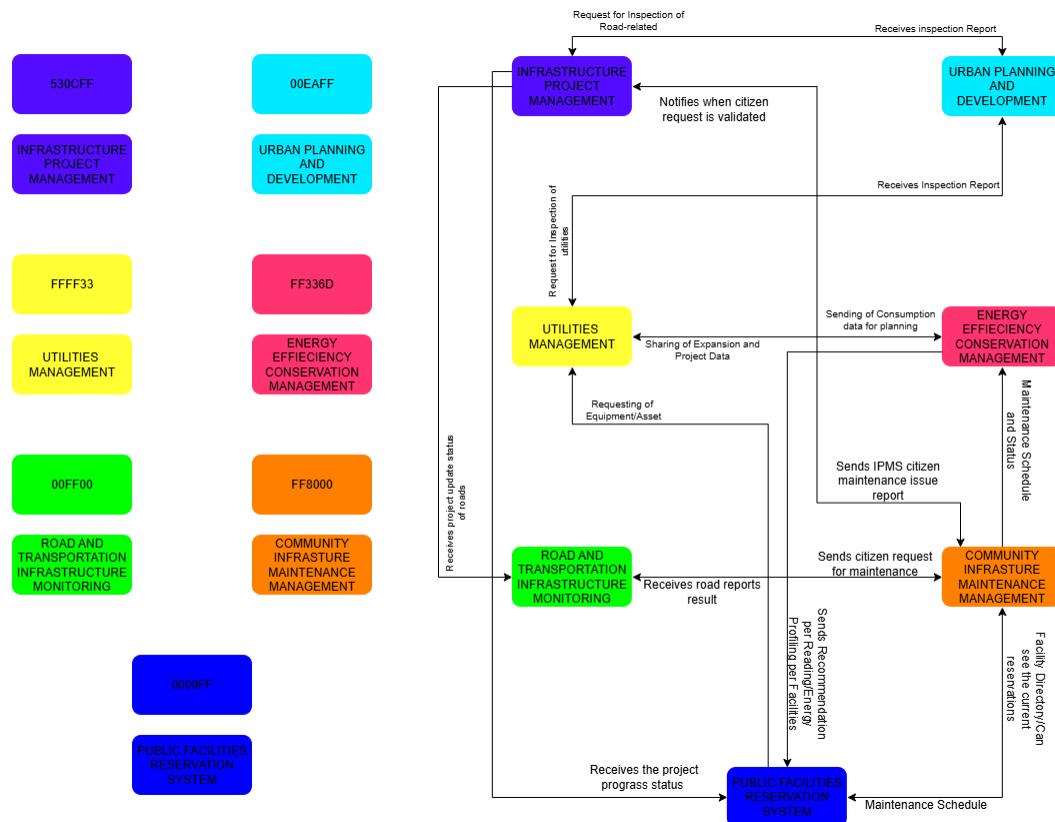


Figure no. 11 Business Process Architecture 1

Business Process Architecture (BPA) for the municipal and departmental system interactions, which maps out how different management divisions coordinate to streamline infrastructure, planning, and community services. The architecture uses a connected flow model to divide responsibilities across various specialized departments, including Infrastructure Project Management, Urban Planning and Development,

Utilities Management, Energy Efficiency Conservation Management, Road and Transportation Infrastructure Monitoring, Community Infrastructure Maintenance Management, and the Public Facilities Reservation System. By integrating automated data exchanges, status tracking, and request handling, this system removes operational bottlenecks and makes inter-departmental workflows much easier to monitor.

The operational network relies heavily on cross-departmental data sharing and reporting. For instance, the Infrastructure Project Management division coordinates with Utilities Management through requests for infrastructure and road-related inspections, while receiving project update statuses from road and facility systems. Meanwhile, Community Infrastructure Maintenance Management acts as a central hub for handling citizen maintenance reports, processing requests for maintenance from the Infrastructure Project Management System (IPMS), and coordinating maintenance schedules and facility directory reservations with the Public Facilities Reservation System.

In the final coordination layer, specialized resource and planning tracking take place between Utilities Management and Energy Efficiency Conservation Management through the sharing of expansion, project, and consumption data. The Public Facilities Reservation System further supports this by profiling data per reading and energy usage to send

recommendations and receive project progress statuses from road and infrastructure monitoring units. This interconnected ecosystem ensures a complete, transparent, and accountable operational history from start to finish across all local government management sectors.

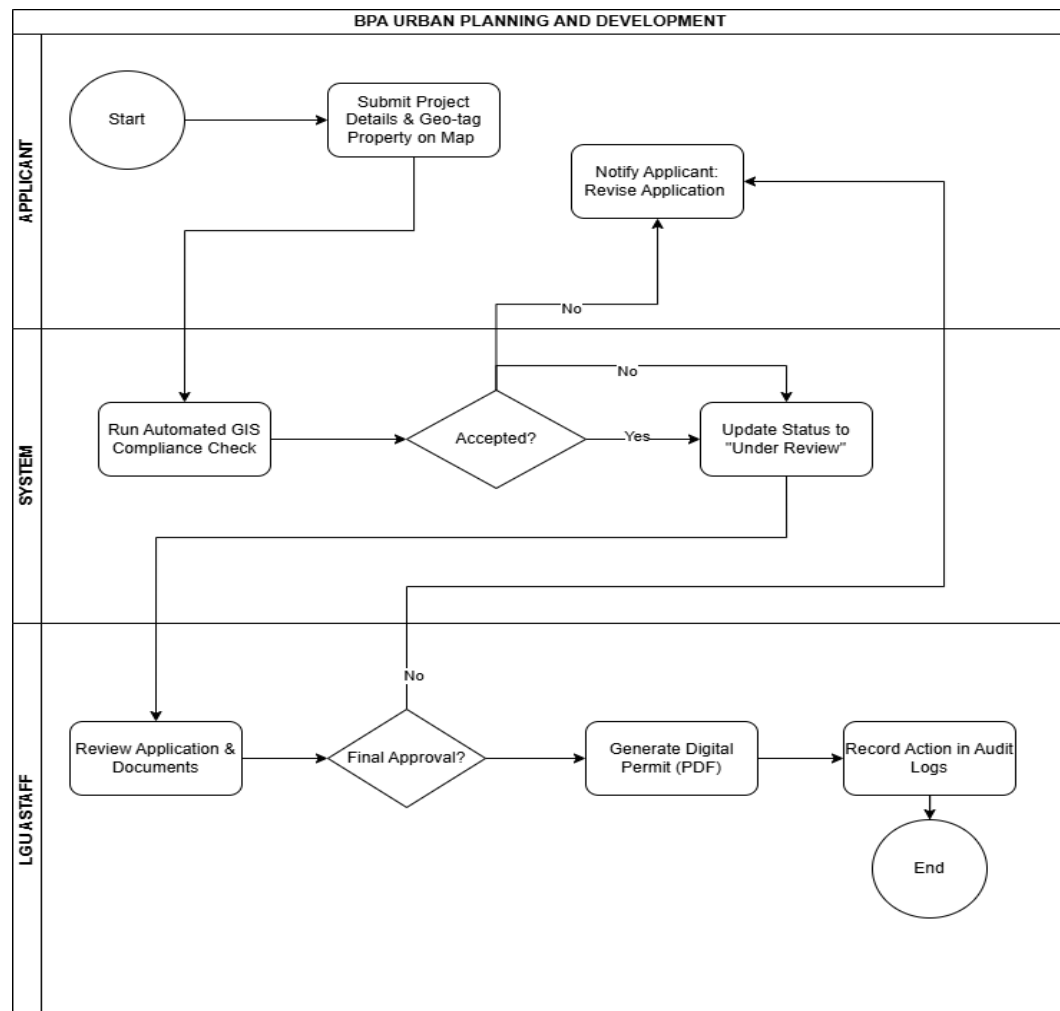


Figure no. 12 Business Process Architecture 2

Business Process Architecture (BPA 2) for the Urban Planning and Development swimlane, which outlines how our system simplifies and

speeds up the local government permitting process. We used a swimlane model to clearly divide the work into three main roles: the Applicant, the System Backend, and the LGU Staff. By combining interactive map tools with automated rule checks and staff approval, this system removes delay points and makes the whole process much easier to track.

The process starts on the Applicant side, where the user enters basic project details and draws their property boundary on an interactive Leaflet map. As soon as they submit, the System Backend takes over to automatically check the map coordinates against current zoning rules. If everything looks good, the system marks the status as "Under Review" and moves it forward, but if the application breaks a rule, it sends a notification asking the applicant to fix their data so bad submissions do not get through.

In the last part, the LGU Staff track handles the actual human review by checking the uploaded documents together with the system-checked map data. Once staff approves it, the system automatically generates an official digital PDF permit for the applicant and records every action, timestamp, and decision into an audit log. This log gives us a complete history of the process, keeping everything clear and accountable from start to finish.

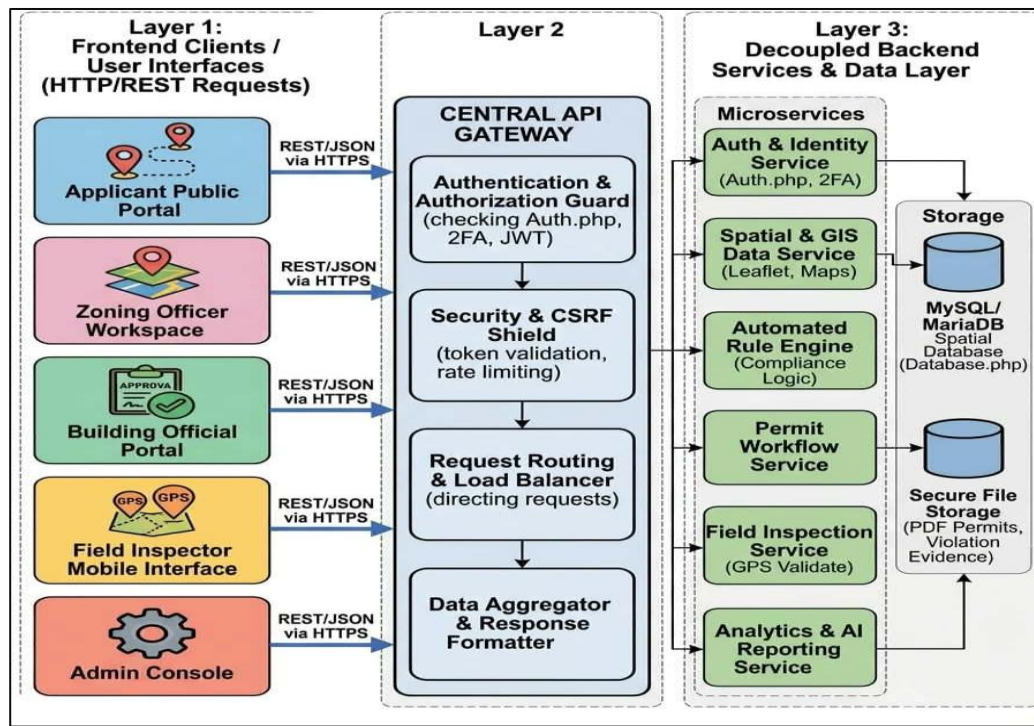


Figure no. 13 API Gateway

API Gateway Integration Architecture, which explains how a central API gateway manages communication between different frontend clients and our separate backend services. The system is split into three main layers, starting with the Frontend Clients used by applicants, officers, and admins. Through these interfaces, users send regular HTTP/REST requests to handle everyday tasks like logging in, mapping property boundaries, and tracking permit statuses.

The Central API Gateway sits in the middle as a security controller that checks every incoming request before it goes any further. It handles essential security checks, such as checking user permissions, blocking

CSRF attacks, and routing traffic to the right place. By keeping these checks at the entrance, the gateway makes sure that only safe and valid requests can actually reach our system's internal services.

Behind the gateway, the third layer holds our backend microservices—like Spatial GIS, Rule Engine, Inspection, and Analytics—which handle the actual business logic. These modular services connect directly to our main database and secure file storage to read and save data when needed. By separating the user interface from the backend using a single gateway, this setup keeps our internal services safe from unauthorized access and keeps our data neat and organized.

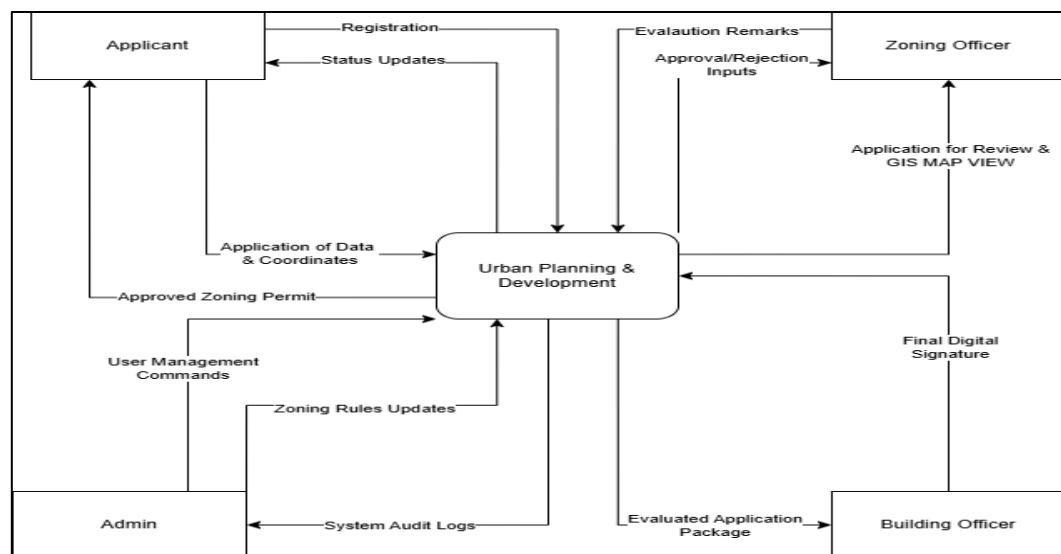


Figure no. 14 Data Flow Diagram Level 0

DFD Level 0 Context Diagram, which sets the overall boundaries for our Urban Planning and Development platform. This high-level view gives

a big picture look at how external users interact with the system without diving into the deep internal coding details. The main users—Applicants, Zoning Officers, Building Officers, and System Administrators—act as the main sources and receivers of the data that keeps the permitting process running.

Throughout the system, information constantly flows back and forth to keep everything moving smoothly. Applicants put in their registration info and map coordinates, and in return, they get status updates and their final permits. At the same time, Zoning and Building Officers input their review notes and digital signatures to move applications along, while Administrators update zoning rules and pull up system audit reports.

By mapping out these high-level data swaps, this context diagram gives us a clean overview of what the system does. It clearly defines the scope of our project and lays out all the main inputs and outputs needed to manage the municipal permitting process.

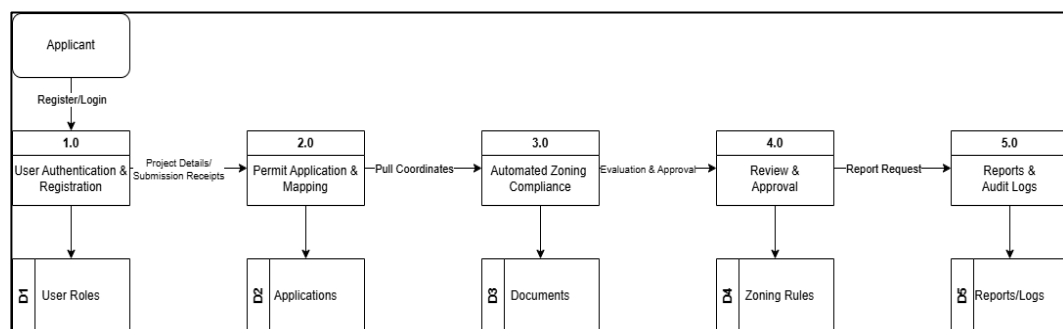


Figure no. 15 Data Flow Diagram Level 1

Breaks down the system into five main internal processes that handle data execution and storage. The flow starts with User Authentication and Registration (1.0), which checks user logins and saves account info into the User Roles data store (*D1*). Next, the Permit Application and Mapping process (2.0) collects basic project details and map coordinates, saving the new submissions right into the Applications data store (*D2*). After that, the Automated Zoning Compliance process (3.0) runs the spatial checks and saves those location results along with uploaded files into the Documents data store (*D3*).

The last two processes cover staff review and overall system logging. In the Review and Approval stage (4.0), city staff pull current policies from the Zoning Rules data store (*D4*) to double-check applications and make their final decisions. Lastly, the Reports and Audit Logs process (5.0) keeps track of every system action, writing event details, created documents, and system stats directly to the Reports/Logs data store (*D5*).

Together, these five connected subsystems create a solid, data-driven pipeline that manages the entire lifecycle of an application, from the moment a user logs in all the way to final audit logging.

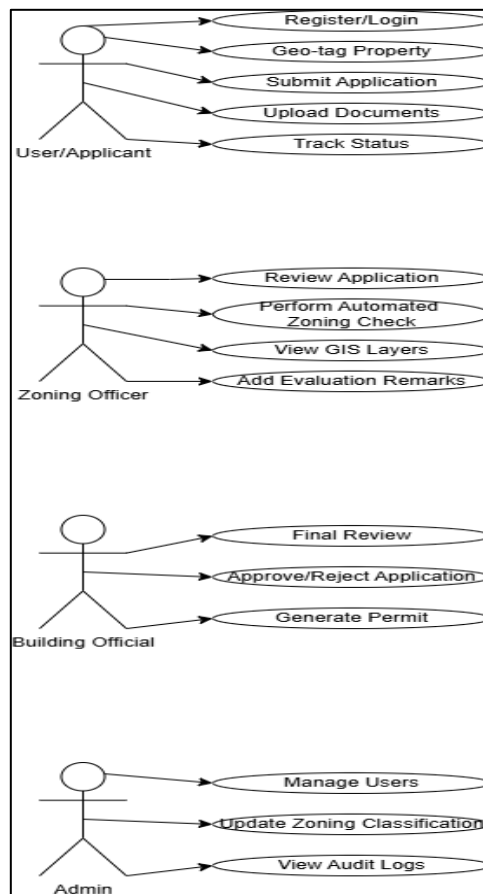


Figure no. 16 Use Case

Which outlines how different users interact with the main features of our system. It splits up the responsibilities among four key roles: Applicants, Zoning Officers, Building Officials, and System Administrators. By connecting each user type to their specific tasks, this diagram clearly sets the boundaries for our system and makes sure everyone only accesses the features they are supposed to.

The main flow starts with Applicants, who do self-service tasks like creating an account, logging in, pinpointing their property on the map,

submitting applications, uploading files, and checking their permit status. After an application is submitted, Zoning Officers step in to review the details, run automated zoning checks, look at the overlapping GIS map layers, and leave evaluation comments.

For the final decision, Building Officials step up to do the high-level administrative review, issue the final approval or rejection, and trigger the creation of the official digital permit. Meanwhile, System Administrators keep the whole platform running smoothly by managing user roles, updating zoning rules, and reviewing audit logs to ensure overall security and compliance.

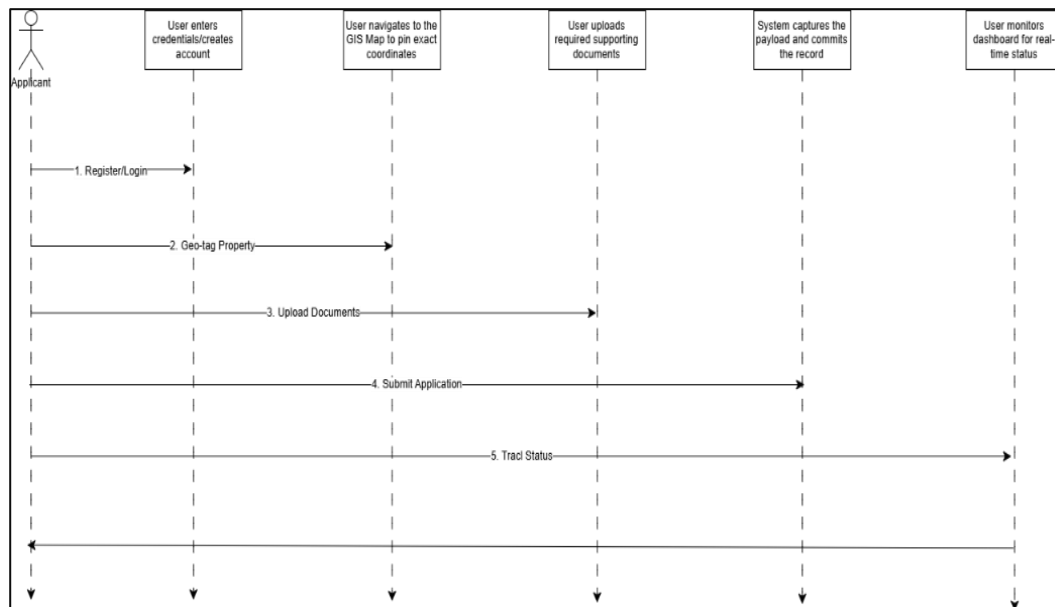


Figure no. 17 Applicant Sequence Diagram

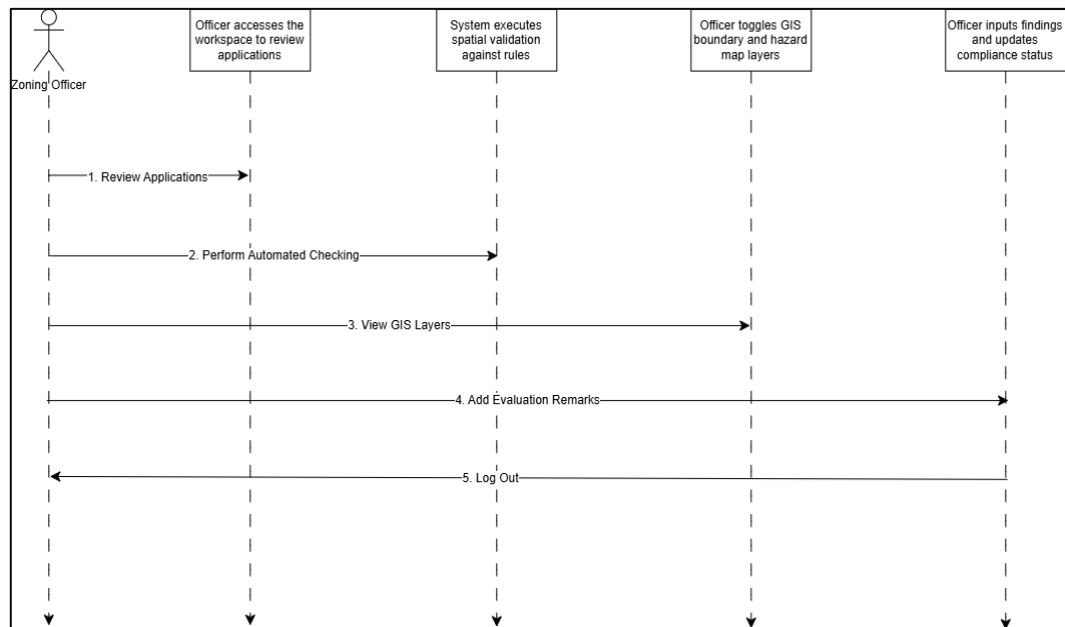


Figure no. 18 Zoning Officer Sequence Diagram

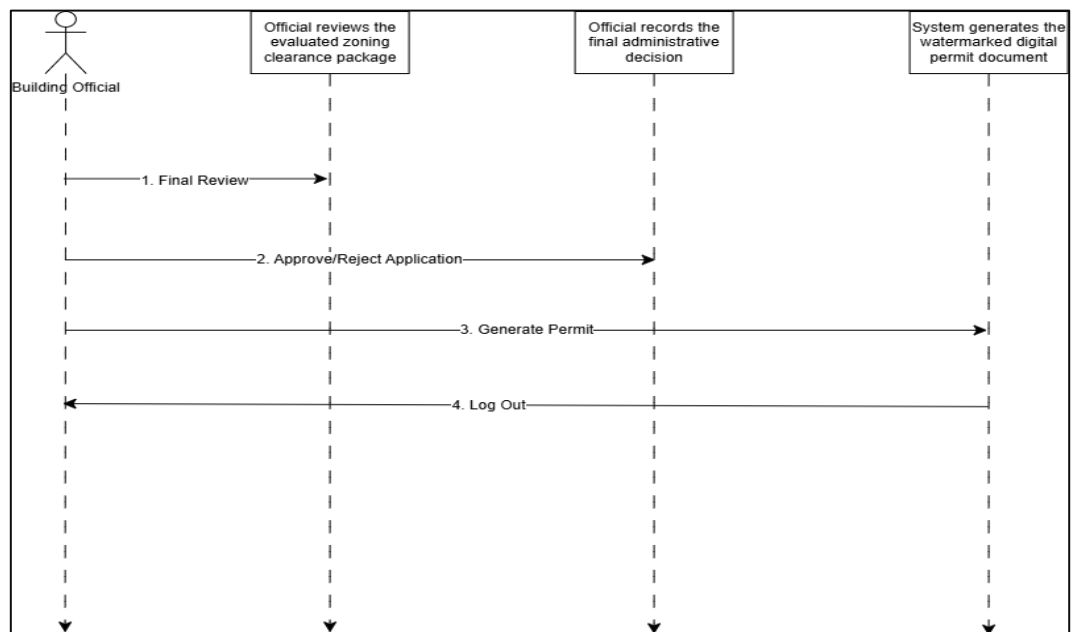


Figure no. 19 Building Officer Sequence Diagram

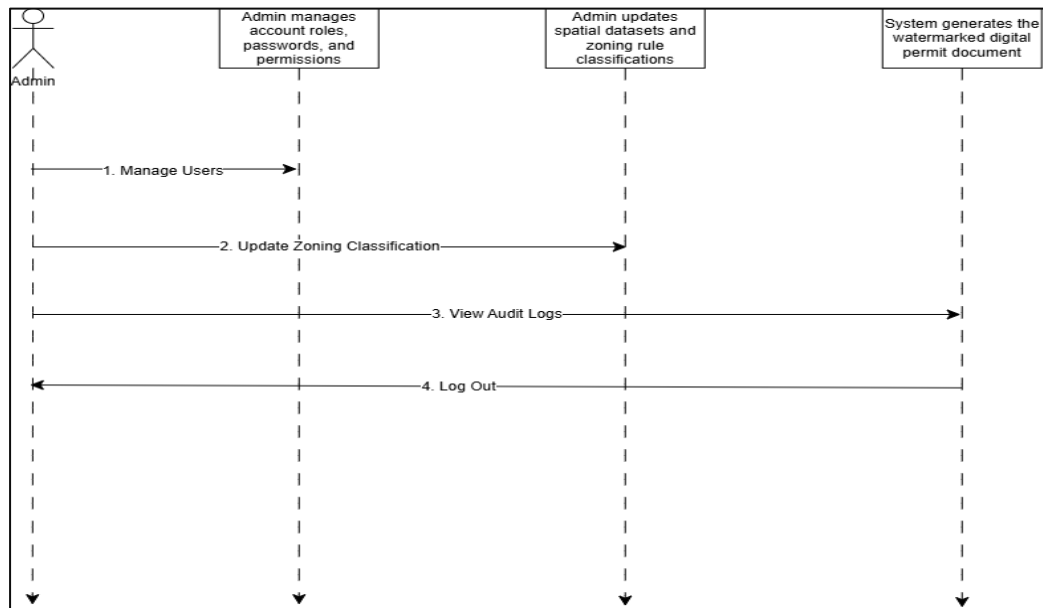


Figure no. 20 Admin Sequence Diagram

The sequence diagrams illustrate the step-by-step interaction flows across the system's primary user roles, detailing how each actor navigates the platform from initial input to system processing. The Applicant workflow opens the sequence as the user logs in, geo-tags property boundaries via an interactive GIS map, attaches required documents, and submits the application payload. Upon submission, the backend processes and stores the transaction data in the database, allowing the applicant to monitor real-time processing updates directly from their tracking dashboard before logging out.

Following initial submission, the Zoning Officer sequence details the technical evaluation and spatial verification phase. The officer accesses assigned queue applications and executed the automated compliance

engine to run point-in-polygon checks against digitized zoning parameters, setbacks, and land-use rules. To perform closer inspection, the officer toggles interactive GIS hazard and boundary layers within the viewport, appends formal assessment remarks, updates the application status in the database, and ends the session.

Once cleared by zoning review, the Building Official sequence maps the final administrative evaluation and permit issuance workflow. The official conducts a final review of the aggregated application package, including previous clearances, structural plans, and departmental notes, before rendering a binding approval or rejection in the system ledger. Upon approval, the system automatically compiles and generates an official, watermarked digital permit embedded with verification details, concluding the official review lifecycle.

Finally, the System Administrator sequence illustrates the continuous platform governance and maintenance framework. The administrator manages system users by configuring role-based permission controls, updating user profiles, and performing bulk dataset updates to reflect evolving municipal land-use ordinances. Additionally, the administrator accesses system audit logs to monitor time-stamped activity

records for compliance oversight, ensuring operational security and data integrity across the platform.

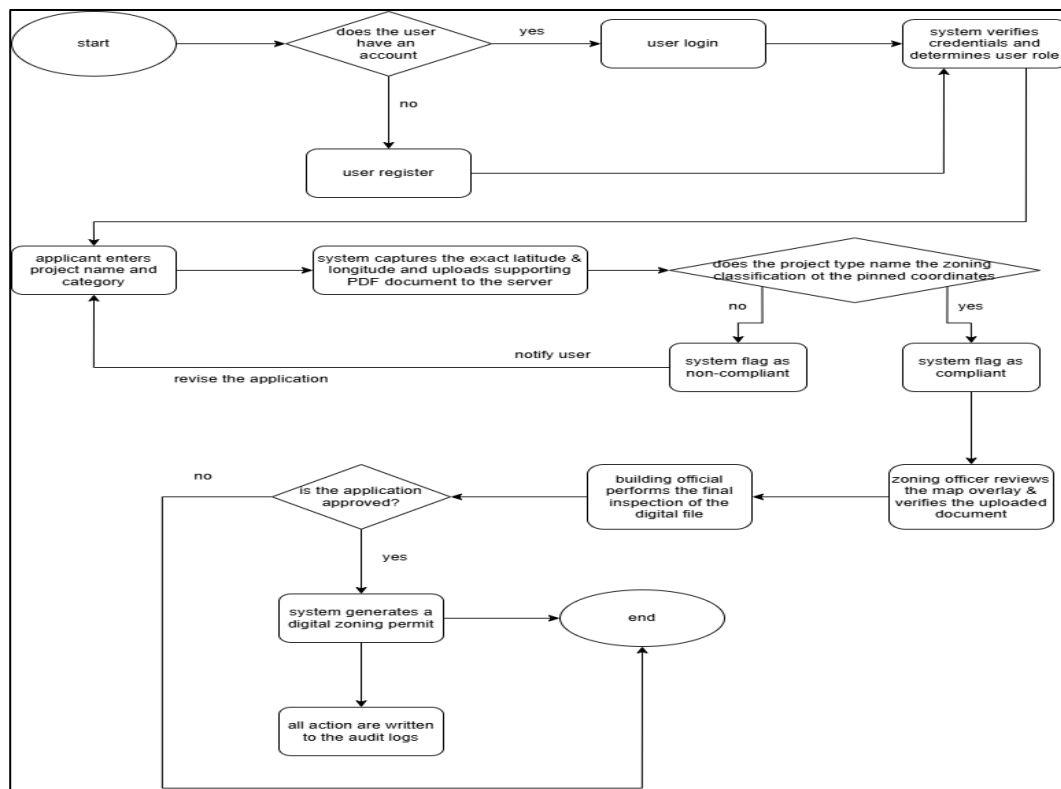


Figure no. 21 Flowchart

Which details the logic and decision paths that control how our system runs. The process starts when a user registers or logs in and plots their property coordinates using the interactive map. Right after they submit, the system automatically runs a geospatial check to see if the project type matches the official zoning rules for that location.

If the spatial check fails, the system marks the application as non-compliant, stops the process right there, and tells the applicant to make the

needed fixes. On the other hand, if the project follows the zoning rules, the system marks it as compliant and sends it down the line for formal review by Zoning and Building Officials.

Once the reviewing officers give their final approval, the system automatically creates the official digital permit and writes the completed transaction into the audit log. This keeps everything transparent and easy to track from start to finish.

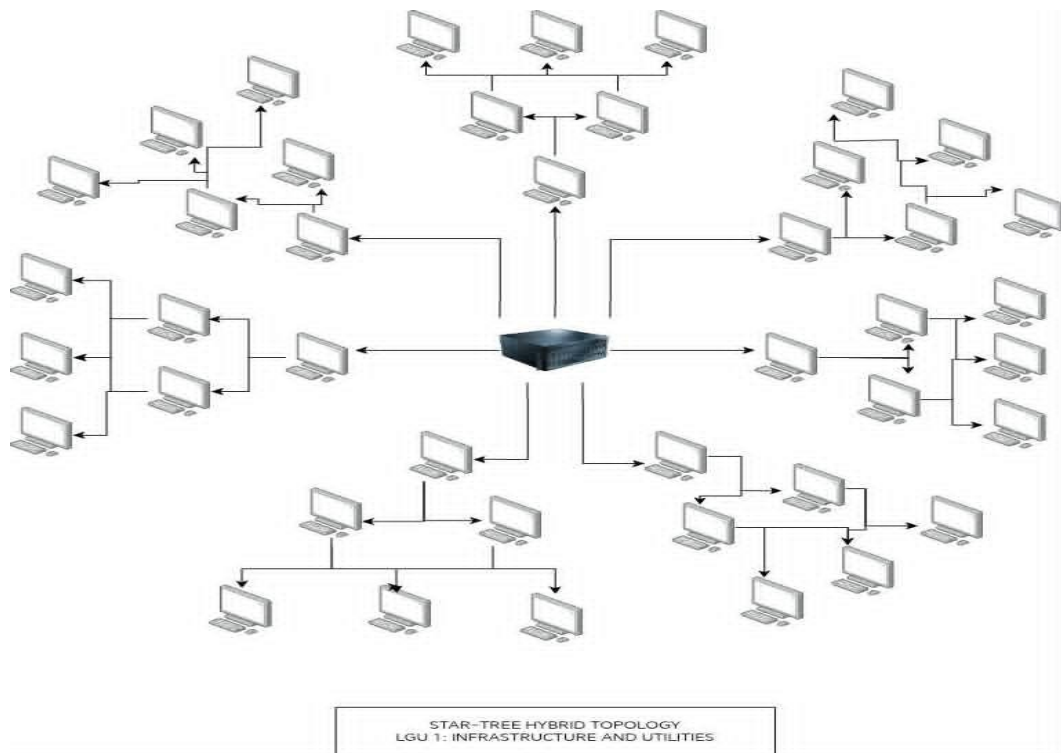


Figure no. 22 Network Topology

The figure details the enterprise-level Star-Tree Hybrid Topology infrastructure deployed across the municipal Local Government Unit (LGU)

framework. This architecture establishes a highly structured network foundation by utilizing a central server and switch hub as the core star node. From this central point, the network branches out into a hierarchical tree arrangement, organizing municipal divisions—including Urban Planning, Zoning, Building Official, Assessor, Inspector, and System Administration—into distinct, high-performance departmental workstation clusters.

By combining the central management of a star topology with the scalable branching of a tree structure, the network effectively distributes traffic across multi-tiered segments. This setup enables disparate LGU departments to concurrently interface with central resources, such as the core database, web-based GIS platform, and automated cloud backup systems. Consequently, the hybrid layout prevents local network bottlenecks, enhances data security across departments, and ensures reliable, high-speed communication throughout the municipal infrastructure.

3.8 Introduction to TOGAF and the Four Architectural Domains

The integration of The Open Group Architecture Framework (TOGAF) provides a structured foundation for the system's development, ensuring the Web-Based GIS System for Automated Zoning and Permits aligns directly with the strategic goals and operational governance of the

Local Government Unit (LGU). By breaking down the municipal enterprise into four architectural domains—Business, Application, Data, and Technology Architecture—TOGAF bridges local urban planning objectives with the underlying digital infrastructure.

1. **Business Architecture**

The Business Architecture domain defines the strategy, governance, and daily operational processes of the municipal permitting workflow. It translates traditional, paper-based LGU tasks—which are inherently prone to administrative delays and human error—into automated, transparent digital procedures. This domain formalizes the roles and interactions of primary actors, including the Applicant, Zoning Officer, Building Official, Assessor, Inspector, Admin, and Super Admin. By standardizing the step-by-step clearance process from coordinate submission to permit release, the Business Architecture ensures the software serves as a formal, legally aligned component of local urban governance.

2. **Data Architecture**

The Data Architecture domain governs the structure, organization, and security of information across the platform. Centralized around Database.php, it manages three core data types: applicant profiles, uploaded verification documents (such as

Titles and Lot Plans), and spatial GIS datasets (cadastral parcels, zoning boundaries, and hazard zones). By enforcing a normalized relational schema and synchronized spatial caching layers, the Data Architecture maintains strict data integrity, eliminates redundancy, and protects sensitive municipal land records against unauthorized modification or data drift.

3. **Application Architecture**

The Application Architecture domain defines the software systems, backend controllers, and their interaction patterns. Adopting a microservices-oriented design, it decouples key functional components—such as the Authentication Service (Auth.php), the Applicant Self-Service Module (ApplicantController.php), the Spatial Data Service, and the Automated Rule Engine—to ensure structural isolation and prevent single points of failure. These independent services communicate through a centralized API Gateway using REST and gRPC protocols, allowing client-side Leaflet.js mapping interfaces to perform asynchronous spatial evaluations without degrading backend performance.

4. **Technology Architecture**

The Technology Architecture domain defines the underlying hardware, software stacks, network topologies, and hosting

platforms required to support high-availability operations. While initial development and prototyping utilize the local XAMPP stack (Apache, PHP, MariaDB/MySQL), the production environment is deployed on scalable, cloud-hosted infrastructure. This ensures that both public applicants and LGU personnel can access real-time spatial analytics, coordinate validation tools, and permit dashboards 24/7 via any standard web browser without requiring client-side installations.

Chapter 4

REQUIREMENTS ANALYSIS

This chapter presents the requirements analysis of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing in Local Government Units (LGUs). It identifies the system stakeholders, the techniques used to gather requirements, and the functional and non-functional requirements needed for the proposed system.

The requirements were analyzed to ensure that the system supports the existing workflows of the Local Government Unit while addressing the limitations of manual permit processing. The succeeding sections discuss the stakeholder identification, requirements gathering techniques, user stories and use cases, and the functional requirements of the system.

4.1 Stakeholder Identification

The table below identifies the key stakeholders involved in the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing in Local Government Units (LGUs). It describes their roles and responsibilities, as well as their level of power and interest in the successful implementation and operation of the proposed system. The stakeholder analysis serves as the basis for determining the project's primary users, decision-makers, and system administrators.

Key	Project Stakeholder	Role / Description	Power Rate (1-5)	Interest Rate (1-5)
A	Super Admin	Has full control over the entire system, including user management, role permissions, system configuration, security settings, and overall platform maintenance.	5	5
B	Building Official	Reviews building plans, verifies compliance	5	5

		with building regulations, approves or rejects permit applications after zoning clearance, and oversees the permit approval process.		
C	Zoning Officer	Evaluates zoning compliance, reviews GIS data, validates land-use classifications, updates zoning layers, and approves	5	5

		or rejects applications based on zoning regulations.		
D	Admin	Manages daily system operations, processes permit applications, generates reports, manages user accounts, and assists in coordinating application workflows.	4	5
E	Inspector	Conducts on-site property	3	5

		inspections, submits inspection reports, and verifies whether construction complies with approved permits.		
F	Assessor	Accesses approved applications to update property records and assess property values for taxation purposes.	3	4

		Does not approve permit applications.		
G	Applicant (Public User)	Submits permit applications, uploads required documents, plots property locations using the GIS map, tracks application status, and receives notifications regarding the application.	2	5

Table no. 12 Stakeholder Identification

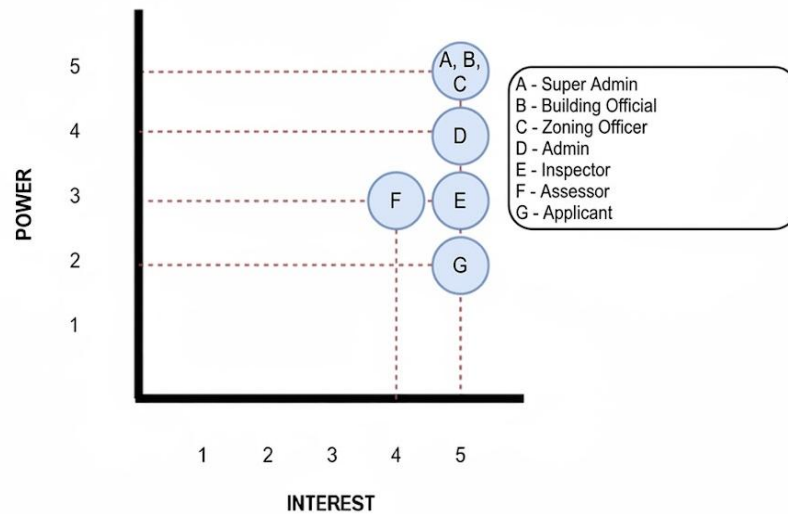


Figure no. 23 Stakeholder Analysis

4.2 Requirements Gathering Techniques

The requirements for the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing in Local Government Units (LGUs) were gathered through observation and interviews. These techniques enabled the researchers to understand the existing permit processing workflow, identify operational challenges, and determine the functional and non-functional requirements of the proposed system.

- **Interview** — An interview was conducted with Mr. Jorge A. Serrana Jr., Information System Analyst III, to gather detailed information regarding the existing zoning compliance and development permit processing procedures of the Local Government Unit (LGU). The

interview focused on understanding the current workflow, identifying the challenges encountered in manual permit processing, and determining the system requirements for the proposed GIS-based system. Mr. Serrana Jr. provided valuable insights into the permit application process, user roles and responsibilities, document requirements, approval workflow, and the desired system features. The information gathered served as the primary basis for identifying the functional and non-functional requirements and ensuring that the proposed system addresses the operational needs of the LGU.

- **Direct Observation** — The researchers conducted direct observation of the existing zoning compliance and development permit processing procedures to understand how applications are received, reviewed, verified, approved, and monitored within the Local Government Unit (LGU). Observing the actual workflow allowed the researchers to identify operational challenges that were not fully discussed during the interview. The observation revealed that the permit processing relied heavily on manual and paper-based procedures, resulting in longer processing times, repetitive data encoding, and difficulties in tracking application status. It also highlighted the absence of a centralized GIS-based system for managing spatial information and verifying zoning compliance,

which served as the basis for the proposed system's features and functionalities.

- **Document Analysis** — The researchers reviewed existing documents related to zoning compliance and development permit processing, including permit application forms, zoning ordinances, workflow documents, and other relevant records. The analysis helped the researchers understand the required information, existing procedures, and regulatory requirements that served as the basis for defining the system's functional and non-functional requirements.
- **Brainstorming** — The researchers conducted brainstorming sessions to discuss and analyze the information gathered from the interviews and direct observations. These sessions enabled the research team to identify possible system features, prioritize user requirements, and develop appropriate solutions to the challenges encountered in the existing permit processing workflow. Through collaborative discussions, the researchers refined the system requirements and ensured that the proposed Web-Based GIS System effectively addresses the operational needs of the Local Government Unit (LGU).

4.3 User Stories and Use Cases

User Stories

The user stories describe the main features of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing in Local Government Units (LGUs) based on the needs of the Applicant, Zoning Officer, Building Official, Inspector, Assessor, Admin, and Super Admin. These user stories are grouped into key system modules that support permit application, zoning compliance verification, GIS mapping, inspection, approval, reporting, and user management.

ID	Module	Role (I want to...)	Goal	Benefits
PA01	Permit Application	As an Applicant, I want to submit my permit application online.	To apply without visiting the LGU office.	Faster and more convenient application process.
PA02	Permit Application	As an Applicant, I want to upload the required documents.	To complete my application digitally.	Reduces paperwork and incomplete submissions.
PA03	Permit Application	As an Applicant, I want to track the status of my application.	To monitor the progress of my permit.	Improves transparency and minimizes follow-up visits.

GIS01	GIS Mapping	As a Zoning Officer, I want to view property locations on the GIS map.	To verify zoning classifications accurately.	Improves zoning validation and decision-making.
GIS02	GIS Mapping	As a Zoning Officer, I want the system to validate zoning compliance automatically.	To reduce manual verification.	Faster and more accurate compliance checking.
IN01	Inspection	As an Inspector, I want to submit inspection reports online.	To record inspection findings immediately.	Improves monitoring and documentation.
AP01	Permit Approval	As a Building Official, I want to review and approve permit applications.	To ensure compliance before approval.	Streamlines the approval process.
RP01	Reports	As an Admin, I want to generate permit and application reports.	To monitor system transactions and statistics.	Supports informed decision-making.
UM01	User Management	As a Super Admin, I want to manage user accounts and permissions.	To secure and control system access.	Enhances system security and administration.
DB01	Property Records	As an Assessor, I want to access	To update property assessment information.	Maintains accurate property records.

		approved permit records.		
--	--	--------------------------	--	--

Table no. 13 User Stories

Use Cases

ID	Module	Use Case Name	Actor(s)	Description
1	Permit Application	Submit Permit Application	Applicant	Allows applicants to submit development permit applications through the online portal.
2	Permit Application	Upload Supporting Documents	Applicant	Enables applicants to upload the required documents for permit processing.
3	Permit Application	Track Application Status	Applicant	Allows applicants to monitor the status and progress of their submitted applications in real time.
4	GIS Mapping	Plot Property Location	Applicant	Enables applicants to identify and plot the location of the property using the interactive GIS map.
5	GIS Mapping	Verify Zoning Classification	Zoning Officer	Displays the property's zoning classification to determine compliance with zoning ordinances.
6	Zoning Compliance	Validate Zoning Compliance	System / Zoning Officer	Automatically evaluates the proposed development against existing zoning

				regulations and spatial data.
7	Inspection	Submit Inspection Report	Inspector	Allows inspectors to record and submit inspection findings through the system.
8	Permit Approval	Review Permit Application	Building Official	Reviews submitted applications, zoning results, and inspection reports before making a decision.
9	Permit Approval	Approve or Reject Permit	Building Official	Approves or rejects permit applications based on compliance with applicable requirements.
10	Online Payment	Process Permit Payment	Applicant / System	Allows applicants to pay permit fees electronically and records successful transactions.
11	Reports	Generate Reports	Admin	Generates reports on permit applications, approvals, inspections, and system activities.
12	Notifications	Send Application Notifications	System	Sends notifications regarding application status, approvals, payment confirmation, and other updates.
13	User Management	Manage User Accounts	Super Admin	Creates, updates, disables, and manages user accounts and role-based access permissions.
14	Property Records	Update Property Records	Assessor	Updates property assessment records

				based on approved development permits.
15	Audit Trail	Record System Activities	System	Maintains logs of user activities and transactions to support accountability and security.

Table no. 14 Use Cases

4.4 Functional Requirements for Integration

The functional requirements define the essential features and capabilities of the proposed Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing in Local Government Units (LGUs). These requirements specify the functions that the system must perform to support permit application processing, zoning compliance verification, GIS mapping, inspection, approval, payment processing, report generation, notifications, user management, and audit logging. Collectively, these requirements ensure that the system effectively supports the operational processes of the Local Government Unit.

1. **User Management** – The system should allow the Super Admin to create, update, deactivate, and manage user accounts while implementing role-based access control for Applicants, Zoning Officers, Building Officials, Inspectors, Assessors, Admins, and Super Admins.

2. **Permit Application Management** – The system should allow applicants to submit development permit applications online, upload the required documents, and securely store application records in the centralized database for further processing.
3. **GIS Mapping and Spatial Visualization** – The system should provide an interactive GIS map that enables users to visualize property locations, zoning boundaries, and land-use classifications for accurate spatial analysis.
4. **Automated Zoning Compliance Validation** – The system should automatically evaluate submitted applications against existing zoning ordinances and spatial data to determine whether the proposed development complies with applicable regulations.
5. **Inspection Management** – The system should allow inspectors to record site inspection findings, upload supporting evidence, and submit inspection reports for review by authorized personnel.
6. **Permit Approval Workflow** – The system should enable Zoning Officers and Building Officials to review permit applications, verify compliance results, approve or reject applications, and record corresponding decisions.
7. **Online Payment Integration** – The system should support electronic payment of permit fees through an integrated online

payment gateway and record payment transactions before the issuance of final approval.

8. **Notification and Status Tracking** – The system should provide real-time application status tracking and automatically notify applicants regarding application updates, approvals, rejections, payment confirmations, and other system-generated events.
9. **Reports Generation** – The system should allow authorized users to generate, filter, print, and export reports related to permit applications, zoning compliance, inspections, approvals, payments, and other administrative records.
10. **Audit Trail and Activity Logs** – The system should record all significant user activities, including logins, application submissions, approvals, updates, and administrative actions, to support accountability, security, and regulatory compliance.

Non-Functional Requirements

The non-functional requirements describe the quality attributes and performance standards that the proposed system must satisfy. These requirements ensure that the system is secure, reliable, efficient, scalable, maintainable, and user-friendly while providing consistent performance and supporting the operational needs of the Local Government Unit.

1. **Performance** – The system should process user requests, map rendering, application submissions, and report generation efficiently under normal operating conditions.
2. **Security** – The system should implement secure authentication, role-based access control, password encryption, and data protection mechanisms in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173).
3. **Scalability** – The system should support future expansion in the number of users, permit applications, spatial datasets, and system modules without significant architectural changes.
4. **Availability and Reliability** – The system should maintain high availability during operational hours and perform regular database backups to minimize data loss and ensure business continuity.
5. **Usability** – The system interface should be intuitive, responsive, and user-friendly across desktop and mobile devices, allowing users to complete transactions with minimal training.
6. **Maintainability** – The system should follow modular software architecture and coding standards to facilitate maintenance, debugging, updates, and future feature enhancements.
7. **Auditability and Compliance** – The system should maintain secure and tamper-resistant audit logs with timestamps and user

information, ensuring transparency and compliance with applicable government regulations.

5 Business Process Architecture

5.1 Identification of Business Processes

The identification of business processes provides a structured understanding of the operational activities supported by the proposed Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing in Local Government Units (LGUs). It describes the sequence of activities performed by the different stakeholders throughout the permit processing lifecycle, from application submission to approval and monitoring. Identifying these business processes serves as the basis for designing an integrated system that streamlines workflows, enhances coordination among users, and improves the efficiency and transparency of zoning compliance and development permit processing.

5.1.1 Recruitment Management Process

The Permit Application Process is the initial stage of the proposed system, enabling public applicants to create an account, authenticate credentials, and submit development permit applications through an online portal. Applicants input required project specifications, upload necessary supporting verification documents (such as Transfer Certificates of Title),

and geo-tag their exact property location using an interactive Leaflet.js GIS map canvas.

The submitted data is automatically validated for completeness and securely stored in the centralized database. This digitized intake replaces traditional manual paperwork, eliminates physical office visits, and ensures that structured, accurate records are immediately available for subsequent administrative review.

5.1.2 Applicants Management Process

The GIS-Based Zoning Compliance Process enables the system to evaluate whether a proposed development strictly adheres to municipal land-use classifications and zoning ordinances. Utilizing the geographic coordinates pinned by the applicant, the automated rule engine performs spatial validation against digitized Comprehensive Land Use Plan (CLUP) boundaries, hazard buffer bands, and setback rules.

The system automatically generates real-time compliance flags (such as "Compliant" or "Non-Compliant") to assist the Zoning Officer during evaluation. This automated spatial verification eliminates the need to manually cross-reference paper maps, drastically reducing processing delays, preventing human error, and ensuring transparent, standardized zoning decisions.

5.1.3 New Hire Onboarding Process

The Permit Approval and Issuance Process supports the final review, clearance, and post-approval documentation within the Local Government Unit (LGU). The system automates administrative routing to ensure speed, transparency, and consistency in granting development permits to public applicants.

Once an application successfully clears initial document review and automated GIS zoning checks, it is routed to the Building Official and relevant LGU personnel for final evaluation. Upon final sign-off, the system automatically generates an official digital Zoning Certificate complete with a unique tracking code and system watermark stamp to prevent fraudulent duplicates. The applicant settles the corresponding permit fees through an integrated online payment gateway, after which the approved permit is released. All final actions, timestamps, and compliance evaluations are saved directly to an unalterable audit log, ensuring strict legal adherence and record integrity.

5.1.4 Recruitment Process Overall

The Business Process Architecture in the Web-Based GIS System manages the end-to-end workflow of zoning compliance verification and development permit processing for the Local Government Unit (LGU).

Previously handled through manual, paper-based methods that caused severe administrative delays and human error, the entire process is now digitized to improve operational efficiency, data accuracy, and inter-departmental coordination.

The overall process begins with the public applicant registering an account and submitting a project application along with required legal documents through the online portal. The applicant uses the integrated Leaflet.js interactive map to geo-tag the precise coordinates of the property parcel. The system securely stores these files in a centralized database while running automated duplicate and error checks.

Next, the automated rule engine performs a spatial intersection check, cross-referencing the submitted coordinates against digitized Comprehensive Land Use Plan (CLUP) boundaries, hazard buffer zones, and setback rules to determine immediate compliance. Zoning Officers and Building Officials review the spatial compliance output alongside submitted site plans within a single evaluation workspace. Field inspectors conduct required on-site reviews and log real-time mobile reports backed by geo-fenced location verification. Once all compliance checks and permit fee payments are completed, the system drafts the official digital permit, notifies the applicant in real time, and logs the complete transaction history for executive reporting and auditing.

5.2 Business Process Diagrams

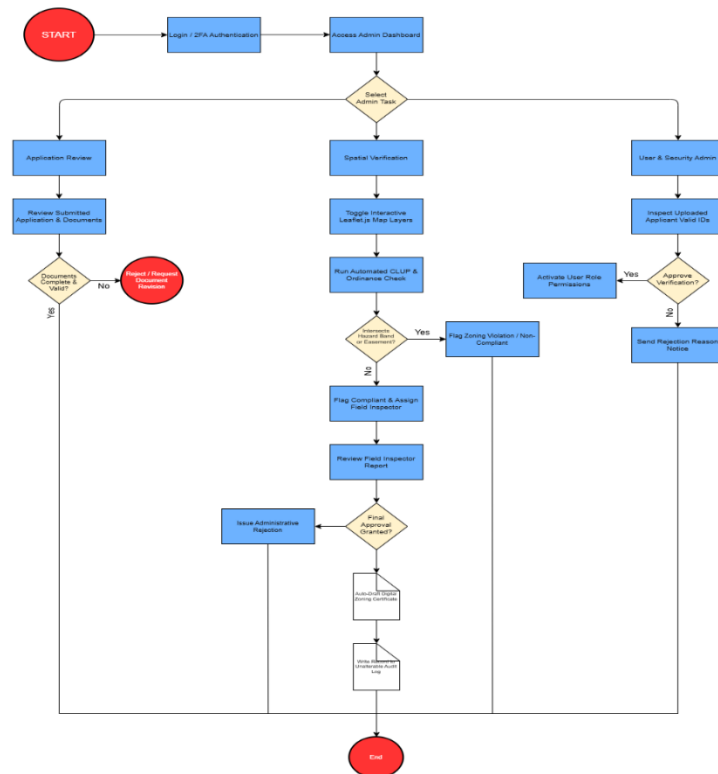


Figure no. 24 Admin/Zoning Officer BPA Diagram

The diagram outlines how administrative users and zoning officers manage permit applications, perform spatial compliance verification, and oversee site inspection processes. It includes application intake review, coordinate verification against Comprehensive Land Use Plan (CLUP) boundaries, and automated hazard buffer checks. The diagram also presents the assignment of field inspectors, review of technical inspection reports, and the generation of administrative analytics.

Additionally, the diagram illustrates the permit approval and issuance workflow, where cleared applications transition through final sign-off, digital Zoning Certificate auto-drafting with embedded security seals, and automated audit logging. The system security management process demonstrates how role-based permissions, two-factor authentication (2FA), and cryptographic token validations are applied to safeguard municipal spatial and administrative records.

Lastly, the diagram highlights the user management process, where applicant identity verification documents are reviewed, approved, or rejected with contextual feedback notes. Overall, the Admin/Zoning Officer BPA diagram demonstrates how the system integrates multiple LGU administrative functions into a centralized, automated, and efficient workflow.

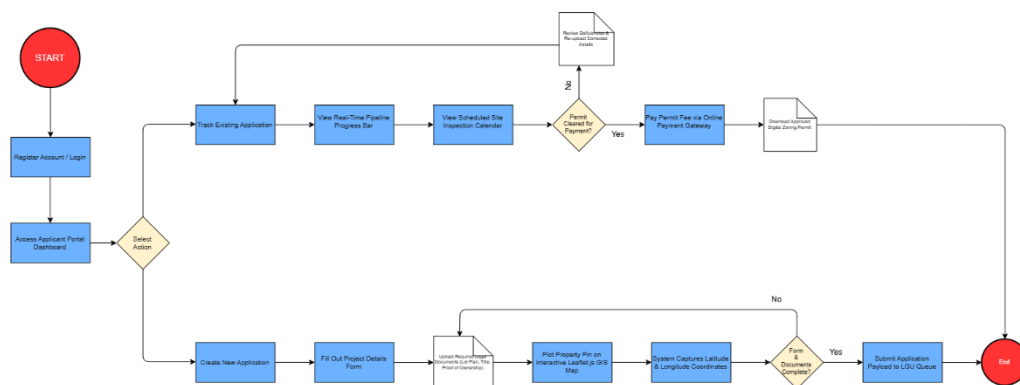


Figure no. 25 Public Applicant BPA for Web-Based GIS System

The process begins with applicants registering an account or logging into the online portal to gain authenticated access to their

dashboard workspace. Once authenticated, applicants submit development permit applications by filling out structured project profile details and uploading required legal documents (such as Lot Plans and Proof of Ownership). The system validates the completeness of submitted files before allowing the application payload to be finalized.

Applicants plot their property location using an interactive Leaflet.js GIS map canvas, which automatically populates high-precision latitude and longitude coordinates into the submission form. Through the user dashboard, applicants track real-time application status updates along a linear color-coded pipeline, view inspection schedules, and settle required permit processing fees through an integrated online payment gateway. If document deficiencies or zoning non-compliance flags are raised, applicants receive detailed revision notes allowing them to rectify their submission.

Overall, the Applicant BPA Diagram demonstrates how the system provides a structured, transparent, and user-friendly permitting process

that ensures proper data validation, real-time tracking, and streamlined interaction with local government offices.

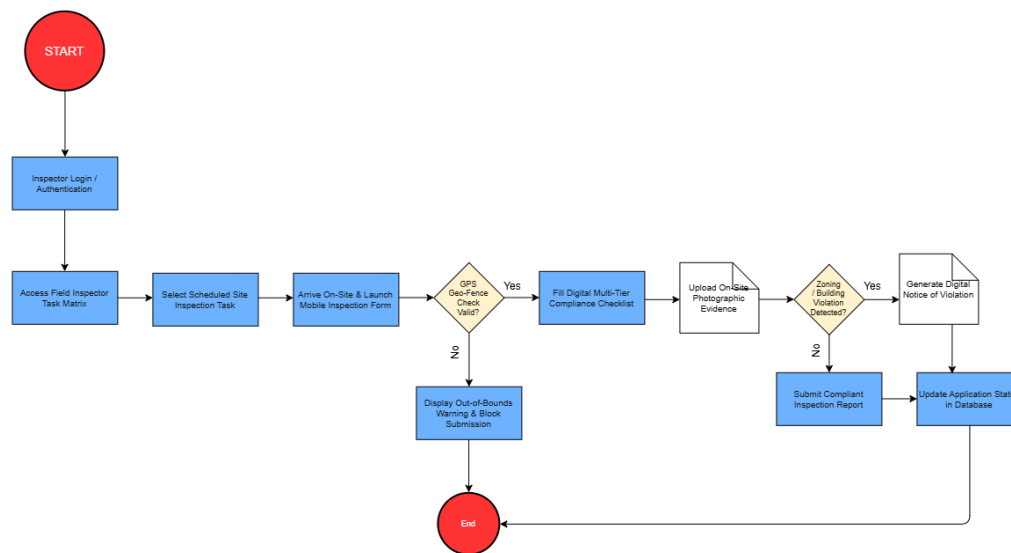


Figure no. 26 Inspector BPA for Web-Based GIS System

The process begins with field inspector login and authentication, granting access to a specialized inspection task matrix dashboard. Inspectors view their chronologically assigned properties, review submitted site plans, and navigate directly to scheduled property inspection tasks.

The diagram presents the field inspection workflow, where inspectors conduct on-site evaluations using a mobile-optimized interface. The mobile module performs real-time geo-fenced GPS verification to validate proximity before allowing the inspector to complete the digital compliance checklist, log field observations, and upload multi-format photographic evidence.

Additionally, if structural violations or unauthorized land expansion are detected during site checks, the workflow enables inspectors to rapidly generate a standardized digital Notice of Violation. Overall, the Inspector BPA Diagram demonstrates how the platform supports field mobility, data accuracy, and enforcement transparency across LGU zoning operations.

5.3 Alignment of Integrated System with Business Processes

This section explains how the Local Government Unit 1: Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing aligns with and enhances existing municipal urban planning and permitting processes. Through automation and spatial system integration, the platform improves operational productivity by reducing reliance on manual map verification, fragmented paper records, and inconsistent decision-making practices.

By working closely with the Zoning and Planning Office, the system transforms traditional permitting activities into efficient, data-driven workflows. The platform streamlines permit application submission, coordinate-based zoning validation, field inspection routing, and post-approval monitoring, enabling faster clearance times, improved data accuracy, and better coordination across municipal departments.

5.3.1 Correlation with Permit Application and GIS Zoning Verification Processes

Permit application intake and zoning verification in local government units are typically managed using analogue procedures, relying heavily on physical paper forms, manual map cross-referencing, and fragmented records, which leads to operational slowdowns and evaluation overlaps. The Web-Based GIS System complements and modernizes the current municipal permitting framework by digitalizing permit submission forms, applicant profile intake, and initial spatial screenings.

The system's coordinate-based automated compliance engine synchronizes applicant-pinned geographic locations with digitized Comprehensive Land Use Plan (CLUP) boundaries and hazard zone layers, streamlining spatial evaluation efforts, resolving data inaccuracies, and expediting overall permit processing.

5.3.2 Alignment with Application Tracking and Communication Processes

The current permit tracking and communication workflows in local government units are typically fragmented, requiring information regarding application status to be pieced together from different paper ledgers, physical office logs, or separate departmental spreadsheets. This division

sacrifices visibility, creates a transparency gap, and causes severe administrative delays. The Web-Based GIS System consolidates all applicant details, land parcel coordinates, uploaded legal assets, and inspection histories into a single centralized platform to satisfy the municipality's need for up-to-date, accurate tracking. The level of transparency between LGU zoning personnel and public applicants is significantly enhanced through real-time pipeline status updates and automated notifications. This modernizes inter-departmental communication, keeps applicants continuously informed throughout the evaluation lifecycle, and ultimately improves public trust and institutional credibility.

5.3.3 Consonance with Permit Approval and Clearance Issuance Processes

Most local government units do not have a standardized digital clearance process; permit evaluations are often inconsistent, and approvals do not strictly adhere to established zoning policies. Relatedly, the Web-Based GIS System modernizes the permit clearance pipeline through task automation surrounding applicant records, spatial verification of documents, and automated officer assignments. The system improves the existing approval process by providing standardized evaluation workflows, centralized administrative dashboards, and instant access to

machine-readable zoning rules and CLUP ordinances. It is an alignment that streamlines inter-departmental routing, cuts down on the time spent on manual administration, and ramps up the speed and accuracy of official permit issuance.

5.3.4 Performance Monitoring and Operational Analytics Alignment

Traditional local government permitting systems typically provide subjective and ad-hoc performance evaluations of administrative workflows. The Web-Based GIS System meets modern performance-management standards, since inspection task tracking, SLA overdue monitoring, and processing productivity digitization are its core features. The performance data will be used by LGU management for making better decisions, through dashboard analytics and reports (analytical), providing transparency and fairness. This congruence informs ongoing operational monitoring and enables the municipal leadership to perform evidence-based evaluations that align with organizational objectives.

5.3.5 Consonance with Administrative Accountability and Staff Engagement Processes

Evaluation and task distribution among local government personnel are often ad-hoc or inconsistent. Consolidating operational metrics,

merging daily activities together to generate a precise and clear description of officer contributions is accommodated within the Web-Based GIS System by formalizing performance tracking as part of an automated workflow involving workload balancing. The system layers operational metrics over regular LGU processes on a daily basis, displaying top-performing barangays, active caseloads, and completed inspection tasks directly within administrative dashboards. This identification will lead to higher departmental engagement, staff morale, and efficiency, fostering an accountable and positive public service workplace culture.

5.3.6 Alignment by System Integration and Centralization of Data

The Web-Based GIS System improves business process alignment by centralizing all municipal zoning and permit processing activities. Centralizing permitting activities in the system eliminates data entry redundancy and errors associated with duplicate entries. Centralizing zoning activities also encourages smooth exchanges of information across application intake, automated GIS compliance checking, field inspections, and permit approvals.

Role-based access control (RBAC) enables public applicants, zoning officers, building officials, inspectors, assessors, and administrators to access the system strictly as per their defined role and responsibilities

towards the platform. The integrated model of the system promotes compliance with both LGU policies and regulatory requirements in all areas of municipal urban planning operations.

5.4 Business Process Improvements

In this section, the researchers elaborate on how the efficiency, accuracy, and level of municipal services rendered by the Local Government Unit (LGU) have been significantly enhanced through the introduction of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. By comparing the traditional AS-IS manual permitting processes of the municipality with the improved TO-BE digital processes enabled by the platform, the advantages brought by integrating GIS mapping, automated coordinate validation, and centralized data management become clearly evident.

5.4.1 Permit Application and Inspection Scheduling Upgrades

Improvements to Permit Application and Inspection Scheduling. AS-IS Evolution: The permitting process was largely manual, relying on paper-based application forms, physical folders, and manual spreadsheets. Setting up site inspections required repeated coordination between public applicants, zoning officers, and field inspectors, usually taking too long and

creating scheduling bottlenecks and communication noise. TO-BE Process: The Web-Based GIS System automates the applicant intake and verification process alongside initial spatial screening. The Automated Inspection Scheduler digitizes calendar booking and cross-references inspector availability with scheduled site visits, taking into account official LGU office hours and holiday constraints. This is achieved through real-time updates and notifications that are automatically sent to both applicants and inspectors.

Benefits Gained:

- Reduced time taken for inspection scheduling.
- Elimination of scheduling hassles and double-booking conflicts.
- Quicker permit evaluation process—both in site scheduling and clearance issuance.
- Superior user experience for applicants through real-time status tracking and updated data.

5.4.2 Spatial Data Management and Application Tracking Enhancements

AS-IS Evolution: Applicant records and legal documents were scattered across various physical filing cabinets and un-synchronized spreadsheets; hence, some files were misplaced, record redundancies

were common, visibility into application status was poor or absent, and communication with applicants was sluggish and reliant on physical follow-up visits.

TO-BE Process: All applicant data, spatial coordinates, and uploaded verification assets are consolidated in a centralized database. The Web-Based GIS System provides real-time pipeline status tracking and automated system alerts, notifying applicants of their submission progress and providing contextual instructions at every evaluation stage.

Improvements Achieved:

- Accurate and centralized applicant and land parcel records.
- Elimination of duplicate data entries and loss of physical records.
- Significantly better status visibility and real-time public communication.
- Shortened processing cycle time to transition an application from initial filing to approved zoning clearance.

5.4.3 Clearance Workflow and Verification Improvements

AS-IS Evolution: The permit intake and verification process was unorganized and inconsistent; manual document verification and land title cross-referencing used to take many days or weeks, leaving applicants confused about regulatory demands and LGU zoning requirements.

TO-BE Process: The Web-Based GIS System fully automates application record logging, document upload checks, and task assignments across reviewing officers. The platform features a centralized administrative dashboard where official zoning guidelines, municipal announcements, and spatial requirements are directly accessible online.

Improvements Achieved:

- Consistent and standardized application verification workflow.
- Drastic reduction in processing and evaluation time.
- Accelerated administrative readiness and improved staff work efficiency.
- Proper, precise, and accurate document compliance and spatial records management.

5.4.4 Performance Monitoring and Operational Evaluation

AS-IS Evolution: Performance evaluation of zoning administrative operations was reliant on hard-copy written logs and subjective ratings. Non-continuous and unorganized data made it difficult for LGU leadership to track processing bottlenecks.

TO-BE Process: The Web-Based GIS System audits staff task completion, inspection field reports, and overall processing productivity in

real time. It provides analytical dashboards and automated reports that deliver operational performance data dynamically.

Improvements Achieved:

- Evidence-based and objective administrative assessments.
- Enhanced operational accountability and organizational transparency.
- Immediate, real-time feedback on system processing speeds and SLA compliance.
- Better data-driven decision-making by LGU management and urban planners.

5.4.5 Administrative Accountability and Staff Engagement Enhancements

AS-IS Evolution: Staff recognition and productivity tracking were extremely informal, irregular, not well documented, and demotivating.

TO-BE Process: The Web-Based GIS System integrates staff productivity tracking into the platform through active caseload monitoring, top-performing barangay analytics, and clear performance logging.

Improvements Achieved:

- An open, transparent, and frequent operational tracking system.

- Higher job satisfaction and motivation among LGU personnel.
- A stronger culture of accountability across municipal departments.
- Improved staff retention and workplace efficiency.

5.4.6 General Operational Efficiency Across the Organization

AS-IS Evolution: Legacy tools and manual procedures were fragmented across different offices, leading to widespread operational inefficiencies, permit evaluation delays, and an overwhelming administrative workload.

TO-BE Process: The Web-Based GIS System provides a single unified platform with integrated role-based access control, encompassing applicant intake, GIS spatial validation, field inspection management, and municipal reporting analytics.

Improvements Achieved:

- Lower overall administrative workload for LGU staff.
- Superior data accuracy and systemic record integrity.
- Faster decision-making and approval turnaround times.
- Seamless organizational alignment across municipal departments.

6 Application Architecture

6.1 Components of Application Architecture

The Application Architecture defines the main components and modules of the Local Government Unit (LGU) Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. It illustrates how these components interact to deliver system functionality and describes the overall structure, layers, and responsibilities of each module.

The architecture is designed to be modular, scalable, and easy to maintain, ensuring that it can support the evolving urban planning and permitting needs of local government units. The system follows a layered architectural model, where each layer serves a specific function and communicates with other layers through clearly defined interfaces. This layered approach improves system reliability, flexibility, and long-term maintainability.

The Presentation Layer

The Presentation Layer acts as the interface between end users and the application. It provides internet-based access to municipal administrative staff, zoning officers, field inspectors, local planners, and public permit applicants, facilitating easy interaction across different operational roles. This layer handles:

- Presentation of interactive map interfaces, application forms, administrative dashboards, and evaluation reports.
- User input validation and spatial coordinate pin entries.
- Inter-module navigation within the system.

The interface is responsive and user-friendly to ensure compatibility with various web browsers and mobile devices, making it accessible to both technical GIS specialists and non-technical applicants.

Business Logic Layer

The Business Logic Layer encompasses the core processing rules of the system. It is responsible for enforcing municipal zoning policies and automated compliance workflows. This layer manages:

- Screening of permit applications and initial document verification.
- Coordinate-based GIS spatial overlays and automated CLUP rule validation.
- Management of zoning clearance and development permit applications.
- Field inspection scheduling and routing workflows.

- Compliance evaluation rationale and decision engine enforcement.
- System notification rules and automated status alerts.

This layer ensures all zoning processes follow standardized municipal ordinances and enables seamless automation across modules.

Data Layer

Data storage, retrieval, and spatial data management are handled by the Data Layer. It employs a centralized relational database integrated with spatial extension engines (e.g., PostGIS) to maintain data integrity and spatial consistency. This layer stores:

- Applicant profiles, land ownership records, and submitted legal assets.
- Geographic coordinates, digitized CLUP zoning maps, and hazard layer boundaries.
- Inspection schedules and field evaluation logs.
- SLA processing times and operational performance metrics.
- System audit trails and decision logs.

Access to data is strictly regulated through role-based permissions to guarantee security, data authenticity, and compliance with data privacy standards.

Core Application Modules

The following core modules make up the Application Architecture, functioning as independent yet interconnected components:

- **Permit Intake Module:** Processes public application submissions, initial document uploads, duplicate record verification, and initial spatial validation.
- **Application Tracking Module:** Monitors permit progress across review stages and sends automated status updates to applicants.
- **Automated GIS Compliance Engine:** Cross-references submitted parcel coordinates against digitized land-use maps and spatial restriction layers to assess zoning compliance automatically.
- **Inspection Management Module:** Automates field inspection assignments, verifies site findings, and manages inspector schedules.

- **Clearance & Approval Module:** Handles official evaluations, administrative sign-offs, and digital zoning clearance generation.
- **Analytics & Performance Module:** Monitors LGU processing speeds, active caseloads, officer productivity, and generates operational reports.

These modules communicate via secure internal interfaces to maintain smooth, uninterrupted data flow across the entire platform.

Overview Application Architecture

The Application Architecture offers a systematic and organized framework that enables efficient, data-driven municipal processes. The system achieves long-term maintainability, scalability, and flexibility by decoupling presentation, business logic, and spatial data management layers. Furthermore, this modular architecture facilitates future integration with external government platforms, such as property tax assessment systems, building permit engines, and advanced spatial analytics tools.

6.2 Application Architecture Diagrams

The section below displays the Application Architecture Diagrams of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. These diagrams

graphically represent the logical and physical design of the application, illustrating how system components interact as well as how the application is deployed within its operational environment.

The diagrams assist in providing better insight into the overall system structure, the relationships between core components, and the flow of spatial and administrative data across system layers and modules.

6.2.1 Logical Application Architecture Diagram

The Logical Application Architecture Diagram represents the functional structure of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing and details the communication between application layers and core modules. It emphasizes what the system is doing rather than where the components are physically deployed.

At the top layer, users including Public Applicants, Zoning Officers, Field Inspectors, Municipal Planning Administrators, and Local Executives interact with the system via a responsive web-based interface. The Presentation Layer receives user requests and forwards them to the Business Logic Layer where core municipal evaluation workflows and GIS validation rules are executed. The

Business Logic Layer interacts directly with the Data Layer to securely store, query, and retrieve spatial boundaries and application records.

Key Components Shown in the Logical Diagram:

- Presentation Layer (Web Interface / Interactive Dashboards / Spatial Forms)
- Business Logic Layer
 - Permit Intake Module
 - Application Tracking Module
 - Automated GIS Compliance Engine
 - Field Inspection Management Module
 - Clearance & Approval Module
 - Analytics & Performance Module
- Data Layer (Centralized Database & Spatial Data Engine)

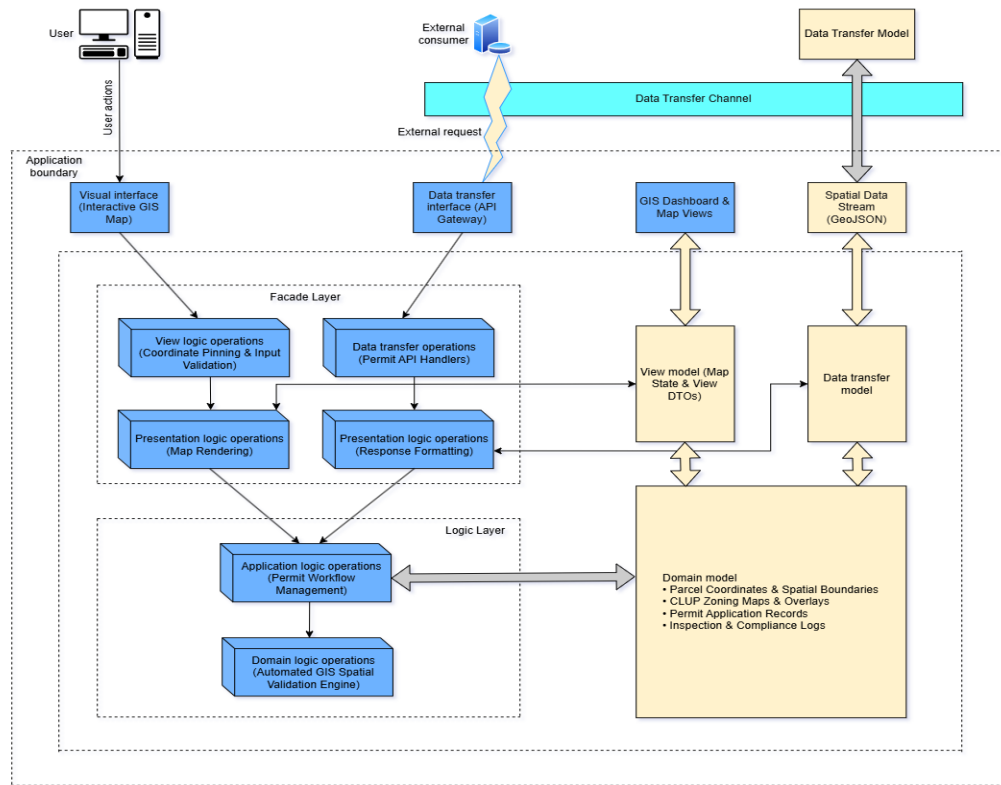


Figure no. 27 Application Architecture

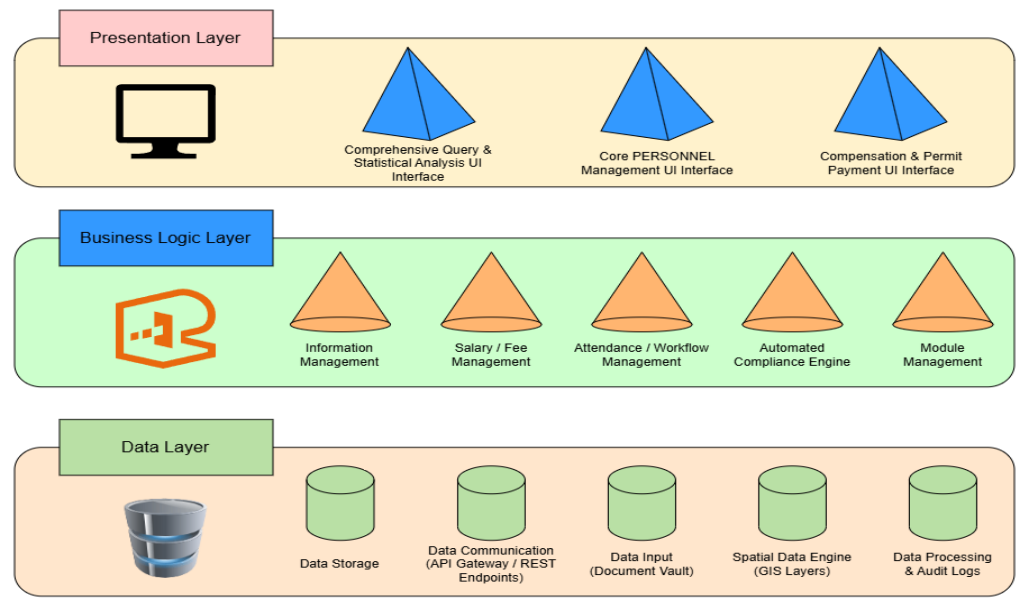


Figure no. 28 Architecture Diagram

6.2.2 Physical Application Architecture Diagram

The Physical Application Architecture Diagram demonstrates the physical deployment structure, infrastructure components, and network associations used in hosting and accessing the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing.

This architecture allows stakeholders (such as public applicants, zoning officers, building officials, and administrators) to access the system via standard web browsers over the internet. User requests are directed to a web server hosting the presentation layer and interactive Leaflet.js mapping interface, which connects to the application server. Core business logic—including the automated GIS compliance engine and workflow routing—is processed by the application server, which communicates securely with the centralized database server containing applicant profiles, parcel coordinates, and zoning ordinance records.

Key Components Shown in the Physical Diagram:

- **Client Devices** (Desktops, Laptops, Mobile Devices used by applicants and LGU personnel)

- **Internet / Network Connection** (Secure communication channel)
- **Web Server** (Hosting the presentation layer and map interface components)
- **Application Server** (Processing business logic, user authentication, and the automated GIS compliance rule engine)
- **Database Server** (Centralized relational and spatial database management system)

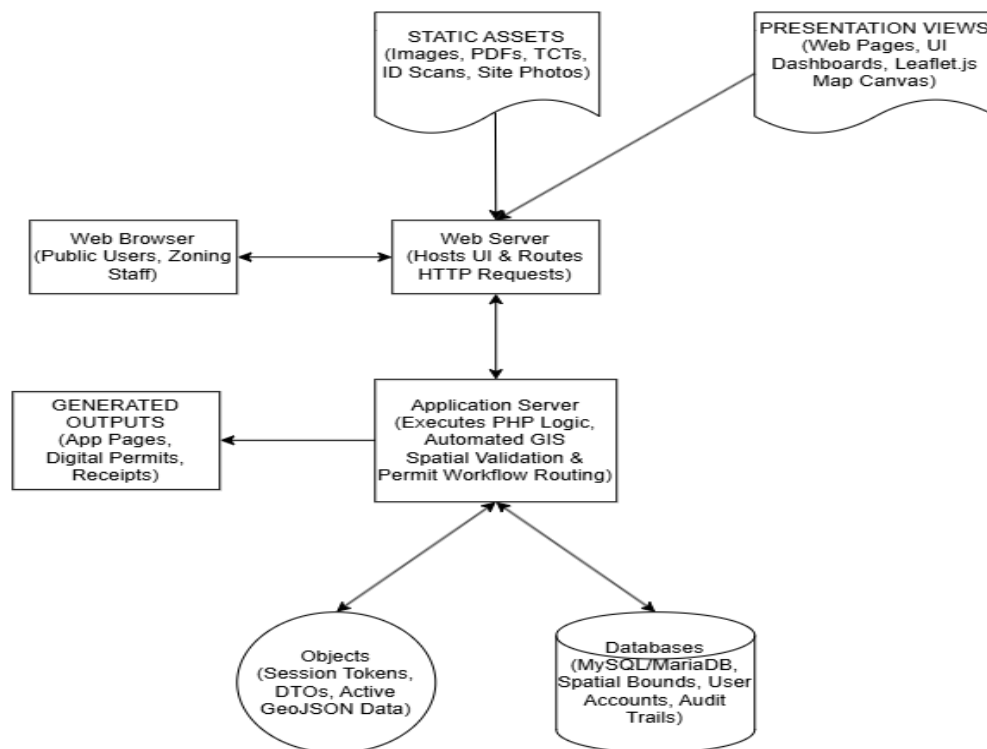


Figure no. 29 Physical View of the Architecture of a Web Application

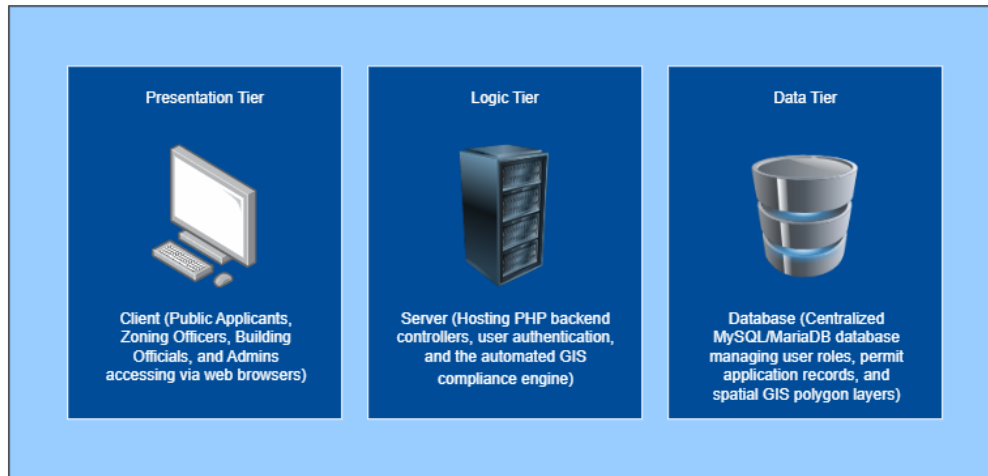


Figure no. 30 Understanding the Architecture 3-tier

6.2.3. Relations between Logical and Physical Architecture

The physical architecture and logical architecture are complementary to one another. Although the logical architecture establishes system functionality, user roles, and data interactions, the physical architecture establishes deployment infrastructure, servers, and network connections. The combination of both makes it possible to guarantee that the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing is not only functionally sound but also technically capable of operating reliably within the actual operational environment of a local government unit.

6.2.4 Summary Application Architecture Diagram

As the Application Architecture Diagrams show, the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing features a structured, layered, and modular design. The logical diagram provides a clear distinction of concerns between system layers and modules, whereas the physical diagram outlines effective deployment, server infrastructure, and remote access. Together, these figures confirm the scalability of the implementation and support the future growth and digital transformation of the municipal platform.

6.3 Integration of Software Modules

6.3.1 Integration Architecture Overview

The Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing is developed on the basis of a layered and modular architecture, with modules interconnected through the business logic layer and a centralized database. Each of the modules performs a specific task, yet all modules share common identifiers such as User ID, Application Number, and Parcel Identification Number (PIN) that make data transitions and tracking of the permit processing lifecycle seamless.

Integration is primarily achieved through:

- Shared data repositories
- API-based internal communication
- Event-driven triggers
- Automated notification services

6.3.2 Permit Application and Zoning Compliance Integration

The integration between the Permit Application Module and the Automated Zoning Compliance Module occurs immediately after an applicant submits their property location and development proposal.

- The Permit Application Module captures applicant data, parcel details, and uploaded document attachments.
- Validated spatial data is stored in the centralized municipal database.
- The Automated Zoning Compliance Module accesses this data to perform point-in-polygon checks against the Comprehensive Land Use Plan (CLUP).
- Status updates (e.g., Pending, Under Review, Compliant, Non-Compliant, Approved) are synchronized in real time.

- Automated notifications are triggered when application status or compliance results change.

This integration eliminates duplicated records and ensures transparency for both local government unit (LGU) officials and applicants.

6.3.3 Integration of Automated GIS Compliance and Inspection Scheduling

The automated GIS compliance engine is tightly integrated with the Permit Application and Inspection Scheduling modules.

- Applicant property coordinates and boundaries are analyzed against zoning rules using spatial processing.
- The scheduler communicates with zoning officer calendars and field availability records.
- Inspection schedules are generated and validated against local holidays and officer workloads.
- Once confirmed, inspection details are saved in the database and shared with the Permit Application module.
- Notification services inform both applicants and zoning inspectors of the scheduled site visits.

This integration significantly reduces manual coordination while maintaining strict compliance oversight.

6.3.4 Integration with Approved Permit Issuance and Clearance Module

Once a permit application is marked as Approved, the system automatically initiates the Clearance and Certificate Issuance Module.

- Approved application records are converted into official municipal permit and zoning certificate profiles.
- A unique Permit Tracking Number is generated and shared across all relevant modules.
- Digital certificate documents and compliance clearance tasks are assigned automatically.
- Issuance dashboards access data from both the Permit Application and Spatial Zoning Compliance modules.

This automated handoff ensures continuity and minimizes delays in releasing official LGU documents to applicants.

6.3.5 Integration of Zoning Officer Performance and Task Tracking Module

The Zoning Officer Performance and Task Tracking Module integrates with the permit review and site inspection systems to begin monitoring staff productivity and task completion after activation.

- Inspection logs, application review turnaround times, and field schedules are linked to staff IDs.
- Productivity and compliance processing metrics are stored in the municipal database.
- Performance summaries and workload analytics are generated using shared user data.

This integration allows department heads and administrators to access comprehensive operational performance information in one platform.

6.3.6 Integration of Public Notice and Transparency Module

The Public Notice and Transparency Module uses data from the Permit Application and Spatial Zoning Compliance modules to publish official advisories and public disclosures.

- Application status and land-use decisions are evaluated against public disclosure criteria.
- Approved zoning notices and public clearances are recorded in the transparency database.
- Compliance history and permit logs are linked back to parcel and user profiles.
- Notifications and public advisories are automatically generated and displayed on user and public dashboards.

This integration reinforces a data-driven and transparent LGU governance system.

6.4 Communication and Integration Patterns

6.4.1 Communication Protocols

The Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing uses standard web communication protocols to enable interaction between clients, application services, and the centralized database.

- HTTP/HTTPS — All client–server communication is conducted using HTTP, with HTTPS recommended to secure data transmission. This protocol enables users to access the system

via standard web browsers without additional software installation.

- RESTful APIs — Internal communication between software modules is implemented using REST (Representational State Transfer) APIs. Each module exposes well-defined endpoints for creating, retrieving, updating, and deleting data (CRUD operations).

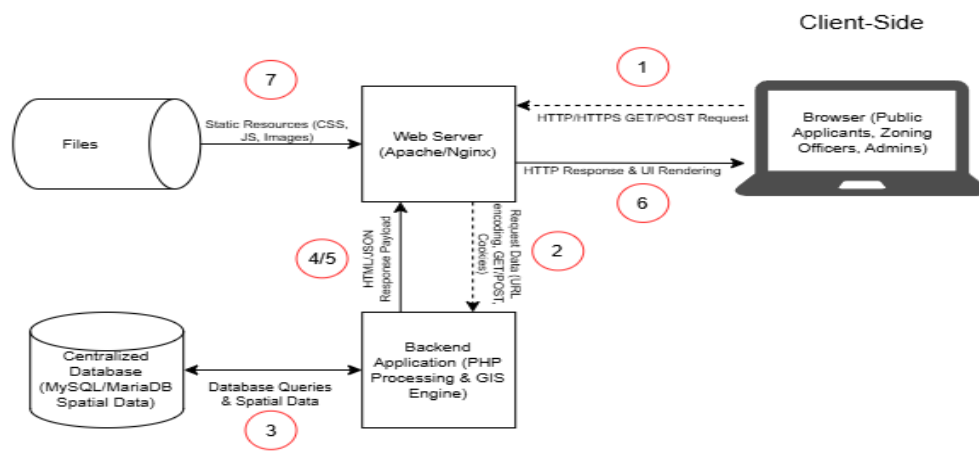


Figure no. 31 Web application data flow

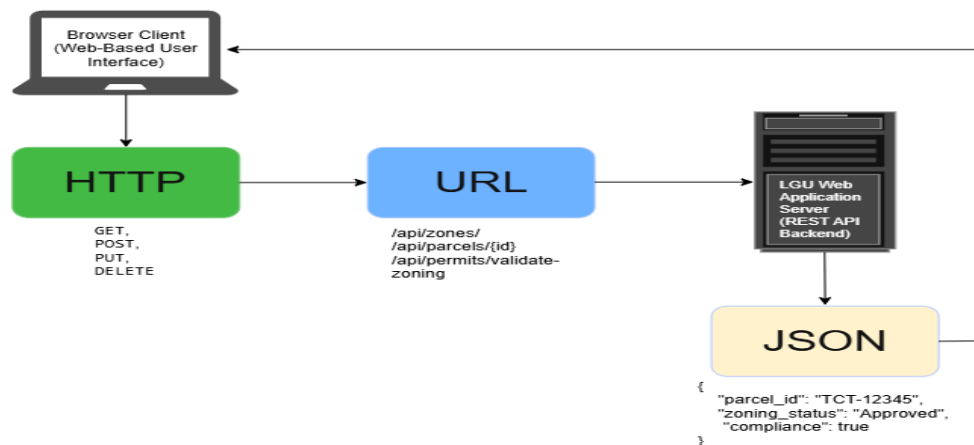


Figure no. 32 REST API architecture

7 Data Architecture

7.1 Data Sources and Types

This section describes the data sources and data types used in the LGU Web-Based GIS System for Zoning Compliance. The data architecture is designed to support accurate spatial data collection, secure cloud storage, and efficient processing of information across all local government unit (LGU) land-use evaluation functions.

The system handles both structured and unstructured data originating from multiple internal and external sources.

7.1.1 Data Sources

The primary data sources of the LGU Web-Based GIS System for Zoning Compliance are generated through system interactions and user inputs. These sources include:

1. User-Generated Data - Data entered by public applicants, zoning officers, and system administrators through web forms. This includes applicant details, parcel boundary descriptions, business application specifics, and zoning compliance logs.
2. Applicant-Submitted Documents - Files uploaded by applicants such as Transfer Certificates of Title (TCT) scans, site

development plans, architectural drawings, and supporting legal documents during the clearance process.

3. System-Generated Data - Automatically created data such as transaction IDs, audit logs, system performance metrics, notification records, and spatial calculation logs.
4. GIS and Spatial Data - Geospatial datasets derived from mapping layers, coordinate overlays, polygon boundaries, and spatial zoning databases analyzed by the system's mapping engine.

7.1.2 Types of Data

The Web-Based GIS System for Automated Zoning Compliance Development Permit Processing processes different types of data based on its purpose and format. These data types enable efficient permit processing, spatial analysis, secure record management, and seamless integration between the system's modules.

Structured Data

Structured data is stored in a relational database (SQL-based) and follows a predefined schema. This type of data

supports efficient querying, reporting, and integration across different system modules.

Examples of structured data include:

- Applicant personal information and account credentials
- Permit application records and application status
- Property and parcel information
- Zoning classifications and land-use categories
- Inspection schedules and assigned personnel
- Payment transaction records
- User roles and access permissions
- Audit logs and system activity records

This structured data ensures data consistency, accuracy, and integrity across the entire system.

Unstructured Data

Unstructured data consists of information that does not follow a rigid schema and is typically stored as files or text-based content.

Examples of unstructured data include:

- Uploaded land titles and supporting documents (PDF, DOCX)
- Building plans, site development plans, and technical drawings

- Valid identification documents
- Inspection photos and supporting images
- Applicant remarks and additional comments
- System-generated notification and email content

These data types are stored in file repositories and are linked to structured database records for easy retrieval and verification.

Semi-Structured Data

Semi-structured data contains elements of structure while allowing flexibility in content. It is commonly used for communication between system components and external services.

Examples of semi-structured data include:

- JSON-formatted API requests and responses
- GeoJSON files containing spatial features and map layers
- System configuration files
- Authentication tokens and session data
- GIS coordinate and metadata exchanged between mapping services

This type of data enables efficient interoperability between the GIS engine, database, and other integrated modules while maintaining flexibility for data exchange.

7.1.3 Data Usage Across Modules

Different modules of the Web-Based GIS System for Automated Zoning Compliance Development Permit Processing consume, process, and generate data to support the permit application workflow and zoning compliance verification. The data usage across modules is described as follows:

- Applicant Management Module uses applicant personal information, account credentials, uploaded identification documents, and permit application records.
- GIS Map and Spatial Validation Module processes parcel information, zoning classifications, geographic coordinates, and land-use boundaries to verify zoning compliance.
- Permit Processing Module manages permit applications, application status, assigned personnel, approval history, and workflow records.
- Inspection Management Module utilizes inspection schedules, inspection reports, photographs, and compliance findings submitted by inspectors.
- Payment Management Module records payment transactions, payment confirmations, official receipts, and billing information for permit processing.

- Notification Module generates email notifications, application updates, approval alerts, rejection notices, and system messages for applicants and LGU personnel.
- Reporting and Analytics Module consolidates application statistics, permit processing reports, zoning compliance summaries, audit logs, and system performance metrics to support administrative decision-making.
- Administration and Security Module manages user accounts, role-based access permissions, activity logs, authentication records, and system configuration data to ensure secure and reliable system operation.

7.2 Data Flow Diagrams

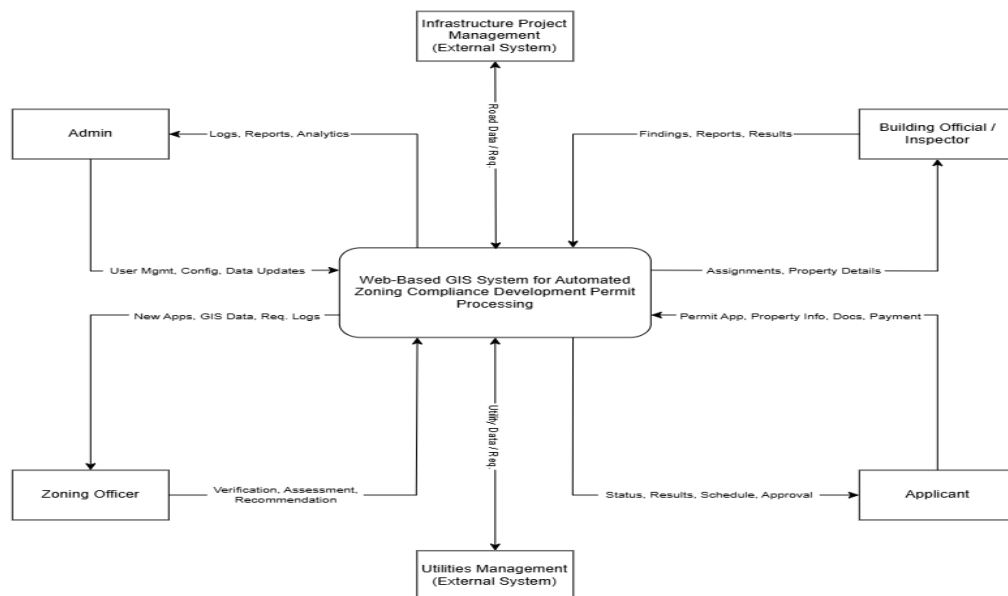


Figure no. 33 DFD lvl 0

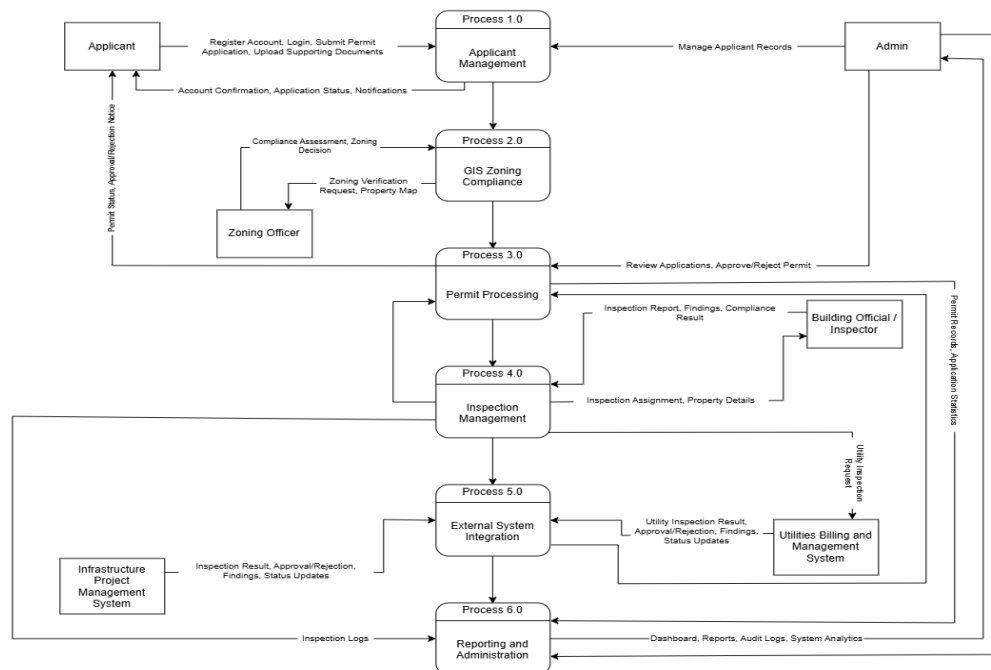


Figure no. 34 DFD IV 1

7.3 Data Storage and Management

This section describes the data storage mechanisms, database systems, and data management strategies used in the Web-Based GIS System for Automated Zoning Compliance Development Permit Processing. The system's data architecture is designed to ensure data integrity, security, scalability, and efficient access to spatial and non-spatial information throughout the permit application and zoning compliance process. By utilizing a centralized database integrated with GIS technologies, the system enables reliable storage, retrieval, and management of applicant records, property information, zoning data,

inspection results, permit records, and other related documents while maintaining consistency and supporting real-time decision-making.

7.3.1 Database System

The Web-Based GIS System for Automated Zoning Compliance Development Permit Processing uses a relational database management system (RDBMS), specifically a SQL-based database, to store and manage structured data related to permit applications, applicant information, property records, zoning classifications, inspection reports, payment transactions, and system user accounts. The relational database model was selected because of its reliability, support for data relationships, efficient transaction processing, and ability to maintain data integrity across interconnected system modules.

The database serves as a centralized repository, ensuring that all modules access a single source of truth for permit records, GIS spatial references, zoning information, inspection findings, and administrative data. By organizing data into related tables with primary and foreign key relationships, the system minimizes data redundancy, improves consistency, and enables efficient retrieval of information for permit processing, zoning compliance verification, reporting, and decision-making.

The database is integrated with the GIS component of the system to associate spatial information with non-spatial records. This integration allows authorized users to visualize property locations, validate zoning classifications, and retrieve geographic information alongside permit application data, supporting accurate and efficient processing of development permits.

7.3.2 Data Storage Methods

The system employs multiple data storage methods based on the type of data being processed to ensure efficient storage, retrieval, security, and data integrity.

1. **Structured Data Storage** – Structured data such as applicant information, permit application records, property details, zoning classifications, inspection schedules, payment transactions, and user accounts are stored in normalized relational database tables. This ensures:
 - Data consistency
 - Reduced data redundancy
 - Efficient querying and reporting
 - Improved data integrity through relational constraints
2. **Unstructured Data Storage** – Files such as land titles, building plans, site development plans, supporting documents, inspection

photographs, valid identification cards, and other uploaded attachments are stored in a secure file storage system. Metadata, including file name, file type, upload date, storage location, and associated application ID, are maintained in the database to facilitate efficient retrieval and management.

3. Semi-Structured Data Storage – Semi-structured data, including JSON-formatted API requests and responses, GeoJSON files, GIS metadata, and system configuration data, are stored either as JSON fields within the database or processed through the application's service layer before being converted into structured records. This storage method supports interoperability between the GIS engine, external integrated systems, and internal application modules while maintaining flexibility in data exchange.

7.3.3 Database Organization

The database is logically organized into functional domains that correspond to the major modules of the Web-Based GIS System for Automated Zoning Compliance Development Permit Processing. This organization promotes efficient data management, minimizes redundancy, and supports seamless data sharing across the system.

- Applicant Management Tables – Store applicant account information, personal details, contact information, uploaded identification documents, and permit application records.
- Property and GIS Tables – Maintain property information, parcel records, geographic coordinates, zoning classifications, land-use categories, and other spatial data required for GIS-based zoning compliance verification.
- Permit Processing Tables – Store permit application details, application status, document verification records, approval history, assigned personnel, and permit issuance information.
- Inspection Management Tables – Capture inspection requests, inspection schedules, assigned inspectors, inspection findings, compliance reports, and supporting photographs.
- Payment Management Tables – Store payment transactions, official receipt information, payment status, billing records, and transaction history associated with permit applications.
- Administration and Audit Tables – Maintain user accounts, role-based access permissions, system configuration settings, activity logs, audit trails, notifications, and report generation records.

Each table utilizes primary keys and foreign keys to establish relationships between related records, ensuring referential integrity, reducing data redundancy, and supporting accurate retrieval of information throughout the permit processing workflow.

7.3.4 Data Access and Retrieval

Data access in the Web-Based GIS System for Automated Zoning Compliance Development Permit Processing is managed through the application's business logic layer, which serves as an intermediary between the user interface, system modules, GIS services, external integrated systems, and the database. This architecture ensures secure, reliable, and efficient retrieval and manipulation of data throughout the permit processing workflow.

- Role-based access control (RBAC) ensures that applicants, zoning officers, inspectors, administrators, and other authorized users can only access data and system functions appropriate to their assigned roles.
- CRUD (Create, Read, Update, Delete) operations are performed using secure SQL queries and validated through the business logic layer to maintain data integrity and prevent unauthorized modifications.

- Database indexing and query optimization are implemented for frequently accessed records, such as permit applications, property information, zoning classifications, inspection reports, and payment transactions, to improve system performance and reduce response time.
- API-based data access enables seamless communication between the GIS engine, external systems — IPMS (Infrastructure Project Management System) and UMAN (Utilities Billing and Management System) — and the application's internal modules. Integration with these external systems is secured through API key authentication and HMAC-signed webhook callbacks, ensuring authenticated and tamper-proof data exchange and synchronization for outgoing inspection requests and incoming inspection results.

This approach ensures secure, accurate, and efficient data retrieval while preventing unauthorized access and maintaining the confidentiality, integrity, and availability of information stored within the system.

7.3.5 Data Management and Maintenance

To maintain data quality, system reliability, and the accuracy of permit processing and zoning compliance verification, the following data management practices are implemented:

- Input validation to prevent incomplete, inaccurate, or invalid applicant information, permit application details, property records, and uploaded documents from being stored in the system.
- Duplicate record detection to identify and prevent duplicate applicant accounts, permit applications, and property records, ensuring data consistency and reducing redundancy.
- Regular database backups to safeguard critical system data, including permit records, GIS information, inspection reports, payment transactions, and user accounts against accidental loss or system failure.
- Audit logging of significant system activities, such as application submissions, permit approvals, inspection updates, user account modifications, and administrative actions, to ensure accountability and support system monitoring.
- Error handling and transaction management to maintain database consistency by preventing incomplete transactions,

preserving data integrity, and ensuring reliable processing of permit applications and system operations.

These data management and maintenance practices help ensure the confidentiality, integrity, and availability of information while supporting the secure, efficient, and continuous operation of the Web-Based GIS System for Automated Zoning Compliance Development Permit Processing.

7.3.6 Data Storage and Management Summary

The data storage and management design of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing provides a secure, structured, and scalable foundation for municipal operations. By combining relational databases, secure file storage repositories for titles and plans, and controlled access mechanisms, the system ensures reliable data handling while supporting intelligent features such as automated coordinate validation and real-time spatial analytics. This architecture enables efficient data retrieval, long-term maintainability, and future system expansion.

7.4 Data Synchronization Across Systems

This section explains how the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing ensures data consistency, accuracy, and real-time synchronization across all system modules. Effective data synchronization is essential to maintain a single, reliable source of information throughout the municipal permit processing lifecycle.

7.4.1 Centralized Data Repository

The system uses a centralized relational database as the primary source of truth. All modules—including Applicant Management, GIS Mapping, Permit Processing, Inspection Management, and Reporting—read from and write to this shared database.

By centralizing data storage:

- Duplicate and conflicting records are minimized.
- Updates made by one module are immediately visible to others.
- Data integrity is preserved across the system.

7.4.2 Shared Unique Identifiers

Synchronization across modules is achieved using unique identifiers, such as:

- User ID – used during registration, authentication, and user management.
- Application Number – used during permit intake, GIS mapping, and zoning compliance tracking.
- Parcel Identification Number (PIN) – used during spatial coordinate validation and property record archiving.

These identifiers allow seamless data mapping and traceability as records move across modules.

7.4.3 Real-Time Update Mechanisms

Data changes are synchronized in real time through controlled system transactions:

- When an application status changes, updates are instantly reflected across permit processing and applicant tracking modules.
- Inspection schedule updates are automatically reflected in applicant dashboards and LGU inspector calendars.

- Zoning compliance and evaluation updates are immediately accessible to reviewing officers and administrative modules.

This real-time synchronization ensures that all users view the most current data.

7.4.4 Event-Driven Data Synchronization

The Web-Based GIS System uses an event-driven approach to trigger automated updates between modules.

Examples of synchronization events:

- Permit application submission → applicant record creation and tracking initialization
- Inspection scheduled → notification and calendar update
- Permit approved → digital zoning certificate generation
- Compliance violation detected → audit log generation and notification dispatch

This approach reduces manual intervention and ensures timely data propagation.

7.4.5 Transaction Management and Consistency

The system uses database transactions to ensure consistency during critical operations. If a process fails at any stage,

changes are rolled back to prevent partial updates or data corruption.

This guarantees:

- Atomic updates
- Consistent data states
- Reliable system behavior during high workloads

7.4.6 Synchronization Between Logical Modules

While modules operate independently, synchronization is maintained through:

- Shared database tables
- Controlled API interactions
- Standardized data formats

This loose coupling allows module updates or maintenance without affecting overall system integrity.

8 Technology Architecture

8.1 Technology Stack and Infrastructure

This section presents the technology stack and infrastructure used in the development and deployment of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. The

selected technologies were chosen to ensure system reliability, scalability, security, and ease of maintenance while supporting automated spatial validation, interactive mapping, and municipal reporting.

8.1.1 Software Technologies

Front-End Technologies

The front-end of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing is responsible for user interaction and system presentation.

- HTML5 – Structure of web pages and forms
- CSS3 – Styling and responsive layout design
- Bootstrap – CSS framework utilized for responsive layout design, grid systems, and mobile-friendly UI components
- JavaScript – Client-side logic, form validation, and dynamic content rendering (including Leaflet.js map integration)
- Web Browser – System access via standard browsers (Chrome, Edge, Firefox)

These technologies ensure a responsive and user-friendly interface across devices.

Back-End Technologies

The back-end handles business logic, data processing, and module integration.

- Server-Side Programming Language – PHP
- RESTful API Framework – Enables communication between system modules
- Spatial Processing & GIS Libraries – Used for coordinate validation and automated mapping analysis
- Authentication and Authorization Mechanisms – Role-based access control (RBAC) supporting Applicant, Zoning Officer, Building Official, Inspector, Assessor, Admin, and Super Admin roles

This layer ensures efficient processing and secure system operations.

Database Technologies

Data storage and management are handled using:

- Relational Database Management System (RDBMS) – MySQL / MariaDB
- SQL – Data definition, manipulation, and querying
- Database Management Tools – phpMyAdmin or equivalent

These technologies support structured data storage and ensure data integrity.

8.1.2 Infrastructure Components

Application Hosting Environment

The system is deployed in a web-based environment, allowing remote access.

- Web Server (Apache)
- Application Server – Hosts PHP business logic and automated GIS spatial validation rules
- Localhost/Test Server – Used during development and testing (XAMPP environment)

This setup enables scalability and controlled deployment.

Storage Infrastructure

- Database Server – Centralized storage for structured data (MySQL/MariaDB)
- File Storage System – Secure storage for uploaded verification documents, land titles, and building plans
- Backup Storage – Periodic data backups for recovery and reliability

Network and Connectivity

- Internet Connectivity – Enables browser-based system access
- HTTP/HTTPS Protocols – Secure data transmission
- Firewall and Access Controls – Protect system resources

8.1.3 Development and Management Tools

To support collaboration and quality assurance, the following tools were used:

- Version Control System – Git
- Code Repository – GitHub
- Agile Project Management Tools – Scrum boards, sprint trackers, and Agile Scrum ceremonies
- Testing Tools – Manual testing, integration testing, and debugging utilities
- Documentation Tools – Word processing and diagramming tools (e.g., Draw.io)

8.1.4 Hardware Requirements

The system operates on standard hardware configurations:

- **Server Hardware** – Web and database server for hosting the application backend, spatial datasets, and audit databases (cloud or on-premise)
- **Client Devices** – Desktop, laptop, or mobile devices used by public applicants, zoning officers, building officials, and field inspectors
- **Minimum Requirements** – Stable internet connection and modern web browser to access the interactive Leaflet.js mapping interface and administrative dashboards

8.2 Software Technologies

This section describes the software technologies, including programming languages, libraries, and frameworks, used in the development of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. These technologies were selected to support system functionality, modularity, security, and automated spatial processing.

8.2.1 Programming Languages

The following programming languages and frameworks were used to develop the system:

- HTML5 – Used to structure the web pages, forms, and system layout.
- CSS3 – Responsible for page styling, layout design, and responsive interface implementation.
- Bootstrap – Utilized as the CSS framework for responsive grid architecture and mobile-first user interface styling.
- JavaScript – Handles client-side interactivity, form validation, and dynamic content rendering, including Leaflet.js map integration.
- Server-Side Language (PHP) – Implements core business logic, system workflows, and server-side automated GIS validation processing.

These languages work together to deliver a functional and interactive Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing.

8.2.2 Frameworks

Frameworks were utilized to improve development efficiency, structure the architectural domains, and ensure a maintainable code base:

- TOGAF (The Open Group Architecture Framework) – Used to guide the enterprise architecture by dividing municipal operations into four core domains: Business, Application, Data, and Technology.
- Agile Scrum Framework – The iterative software development methodology structured around time-boxed sprints, product backlogs, and team collaboration ceremonies.
- Web Application Framework – Supports routing, session management, and modular development.
- RESTful API Framework – Enables communication between system modules through standardized endpoints.
- Authentication Framework – Manages user login, role-based access control (RBAC), and session security.
- Bootstrap – Acts as the front-end CSS framework for responsive layout grids, styling, and mobile-first user interface components.

Framework usage ensures consistent coding standards, structured enterprise alignment, and simplifies system expansion.

8.2.3 Libraries

Several libraries were integrated to enhance functionality:

- Leaflet.js Library – Used for client-side geospatial mapping, rendering interactive tiles, and managing draggable property coordinate pins.
- JavaScript UI & Charting Libraries – Support user interface behavior, form handling, asynchronous data requests, and dynamic execution of Chart.js visual graphs (doughnut, pie, bar, and trend line charts).
- Database Access Libraries – Facilitate secure and efficient interaction with the centralized relational and spatial database.
- Composer-Managed Libraries (e.g., PHPMailer / Mail Transport Packages) – Manage the backend generation of secure 6-digit OTP codes and enable automated email notifications, system alerts, and status updates.

These libraries reduce development time while adding advanced features such as interactive GIS spatial validation, automated compliance checking, and real-time dashboard analytics.

8.2.4 Database Technologies

The system uses structured data management technologies to store and retrieve information:

- MySQL / MariaDB – Relational database management system for storing applicant accounts, permit application records, spatial boundaries, and audit logs.
- SQL – Used for querying, inserting, updating, and managing relational database records.
- Database Administration Tools – Used for database configuration, secure backups, and maintenance.

These technologies ensure data consistency and reliability.

8.2.5 Development and Testing Tools

To support development quality and collaboration, the following tools were used:

- Version Control System – Git – Used for tracking code changes, managing branches, and supporting collaborative software development throughout the sprint cycles.
- Code Repositories – GitHub – Utilized for collaborative repository hosting and version control management.

- **Testing Utilities** – Browser developer tools and debugging utilities – Used for inspecting elements, analyzing asynchronous requests, and troubleshooting front-end code.
- **Documentation and Diagram Tools** – Used for technical documentation and architecture diagrams – Supported the creation of system models, database diagrams, and the capstone manuscript.

8.3 Scalability and Performance Considerations

This section discusses how the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing was designed to support system growth while maintaining reliable performance. Scalability and performance were key considerations due to the expanding volume of geospatial datasets, property boundaries, and concurrent public users accessing permit applications and spatial maps.

8.3.1 Modular and Layered Architecture

The system follows a modular and layered architecture, allowing individual modules—such as Permit Intake, GIS Mapping and Spatial Validation, Permit Processing, Inspection Management, Payment Management, and Reporting and Analytics—to scale independently. This separation of concerns ensures that increased

workload in one module does not degrade overall system performance.

8.3.2 Database Scalability

To handle growing data volumes, the system uses a relational database optimized for scalability:

- Normalized database design to reduce redundancy
- Indexing on frequently accessed fields (Applicant ID, User ID, Parcel Identification Number)
- Efficient query design to minimize response time
- Support for database scaling through hardware upgrades or database replication

These measures ensure consistent performance even as data grows.

8.3.3 Efficient Data Processing

The system minimizes performance bottlenecks by:

- Processing only necessary data during transactions
- Using server-side validation to reduce client-side load
- Separating transactional operations from analytical processing

This approach ensures faster response times for end users.

8.3.4 Asynchronous Processing

Time-consuming operations are handled asynchronously to maintain responsiveness:

- Email notifications and OTP dispatches
- Inspection schedule generation and calendar updates
- Performance analytics and chart data rendering
- System announcements and digital certificate generation

Asynchronous execution prevents system slowdowns during peak usage.

8.3.5 Load Management and Concurrent Users

The Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing is designed to support multiple concurrent users by:

- Managing user sessions efficiently
- Optimizing API requests and backend queries
- Using stateless communication where possible

This ensures that LGU staff, zoning officers, inspectors, and public applicants can access the system simultaneously without degradation in performance.

8.3.6 Spatial and GIS Performance Optimization

The automated GIS compliance engine and mapping interface are optimized by:

- Preprocessing spatial coordinates and vector boundaries
- Limiting spatial intersection checks to relevant zoning ordinance layers
- Converting unstructured geographic data into structured GeoJSON formats early in the workflow

This reduces processing overhead, minimizes latency during map rendering, and improves overall spatial validation efficiency.

8.3.7 Infrastructure Scalability

The web-based deployment allows infrastructure scaling through:

- Vertical scaling (upgrading server resources such as RAM, CPU, and storage capacity)
- Horizontal scaling (adding servers or worker nodes if deployed in a distributed environment to handle high volumes of zoning requests)
- Flexible hosting options (on-premise local government server deployment or cloud-based hosting)

This ensures adaptability to future organizational growth and municipal expansion.

8.3.8 Performance Monitoring and Maintenance

System performance is maintained through:

- Logging and monitoring of system and spatial mapping activities
- Regular database maintenance and spatial index optimization
- Error detection and resolution mechanisms for zoning verification failures

These practices help sustain long-term system reliability.

8.3.9 Scalability and Performance Summary

The Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing was designed with scalability and performance in mind by leveraging modular architecture, optimized relational and spatial database design, asynchronous processing, and flexible infrastructure. These considerations ensure that the system can efficiently handle increasing numbers of concurrent users, property parcels, and geographic data while maintaining responsive and reliable municipal operations.

9 Development Process

9.1 Agile Scrum Roles and Responsibilities

The development of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing utilized the Agile Scrum methodology to ensure a collaborative, adaptive, and incremental approach to building the system. The following roles were pivotal:

Team Member	Assigned Role	Responsibilities
Ma. Cristina Buenaflor	Project Manager	Oversees the overall project execution and lifecycle. Manages Agile/Scrum sprints, coordinates team tasks, tracks progress against deadlines, and leads team communication.
Rosary Mae C. Velicaria Ma. Cristina Buenaflor	Quality Assurance	Conducts rigorous system testing (unit, integration, and user acceptance) on the

		GIS mapping and permit routing flows. Identifies bugs and ensures the system meets the defined objectives.
Althaniel O. Abordo Joshua Canceran Justin Michael Espino	Document Specialist	Assists in writing, editing, and formatting the capstone manuscript. Ensures all system documentation, diagrams, and literature reviews meet institutional standards.
Yumie Margareth N. Dalagan	Full-Stack Developer	Responsible for writing the code for both the frontend and backend of the system. Integrates GIS mapping features,

		builds the automated zoning logic, and manages the database architecture.
--	--	---

Table no. 15 Roles and Responsibilities

9.2 Sprint Planning and Backlog Management

Product Backlog Management

- All system tasks, features, and user stories were consolidated into a single product backlog.
- Backlog items were prioritized based on:
 - Business value
 - Urgency
 - Functional dependencies
- Continuous backlog refinement was performed to:
 - Review and clarify requirements
 - Break down complex features into smaller, actionable tasks

Communication and Coordination

- Stakeholders were actively involved to:
 - Validate system requirements
 - Provide feedback and revisions

- Team discussions ensured that:
 - All members clearly understood backlog items
 - Scope and objectives were aligned

Sprint Goal Setting

- A clear objective was defined for each sprint.
- Backlog items were selected based on:
 - Priority
 - Relevance to the sprint goal
- Each sprint was planned to deliver a functional system increment.

Task Breakdown and Estimation

- User stories were decomposed into smaller development tasks.
- Effort estimation was performed for each task to:
 - Manage workload effectively
 - Monitor sprint progress

Capacity Planning

- Team availability and workload capacity were evaluated.
- Sprint commitments were based on:
 - Sprint duration
 - Realistic team capacity
- Overcommitment was avoided to ensure consistent progress.

Timeboxing

- Each sprint followed a fixed time duration, typically:
 - 1 to 4 weeks
- Agile ceremonies were scheduled, including:
 - Daily stand-up meetings
 - Sprint reviews
 - Sprint retrospectives

9.3 Sprint Execution and Deliverables

During sprint execution, the development team collaboratively worked to achieve the defined sprint goals for the Local Government Unit 1: Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. Sprint activities were carried out based on the approved sprint backlog, with tasks executed, monitored, and validated within each sprint iteration. Coordination among team members ensured that development activities were aligned with the system objectives and requirements defined in earlier chapters. Continuous review and testing were performed to confirm that completed features met functional and usability standards.

Key Deliverables

- User Interface and User Experience (UI/UX) Design – A user-friendly and responsive interface was developed to allow public applicants, zoning officers, and LGU administrators to efficiently interact with the system.
- Permit Intake and Tracking Module – Modules were implemented to manage permit applications, document submissions, status tracking, and automated notifications throughout the zoning approval process.
- Automated GIS Zoning Compliance Engine – An intelligent spatial validation feature was developed to analyze coordinate inputs, check land-use zoning layers, and automatically verify setbacks, height restrictions, and floor area ratios.
- Post-Approval Monitoring Module – A module was created to track construction adherence and manage municipal inspection requests and schedules.
- Reporting and Analytics Module – Features for generating summary reports, visualizing permit metrics, and providing real-time operational insights for LGU officials were implemented.
- System Integration and Testing – All modules were integrated into a unified system, followed by unit testing, integration testing, and

user acceptance testing to ensure system reliability and correctness.

9.4 Challenges Faced in the Development Process

The development of the Local Government Unit 1: Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing presented several technical, operational, and coordination-related challenges. These challenges emerged due to the complexity of integrating spatial mapping technologies, automated land-use compliance logic, and municipal permit workflows into a single unified system.

Key Challenges

- **Module Integration and Data Synchronization** – Ensuring seamless communication and consistent data flow among modules such as Permit Intake, GIS Mapping, Automated Compliance Validation, and Post-Approval Monitoring required careful design of shared spatial data structures and system workflows. Maintaining data consistency across modules during early development iterations posed significant challenges.
- **Processing Unstructured Spatial and Zoning Data** – Handling complex geographic inputs, particularly varying coordinate formats, vector boundary layers, and municipal zoning ordinance

rules used for automated compliance checks, introduced complexity in spatial data interpretation and transformation. Converting this data into structured and reliable zoning verification reports required multiple refinements of the processing logic.

- **Role-Based Access and User Management** – Implementing secure role-based access control for public applicants, zoning officers, municipal inspectors, and LGU administrators required extensive validation. Ensuring that users could access appropriate system features without compromising security or usability was a key challenge during development.
- **System Integration and Workflow Alignment** – Aligning online permit submissions with automated zoning validation, inspection scheduling, and approval processes required careful coordination to ensure continuity across the municipal permit lifecycle. Integrating these workflows without disrupting existing data relationships added complexity to system design.
- **Time Constraints and Resource Limitations** – Balancing system development with academic schedules and limited development resources affected sprint timelines. This required careful task

prioritization and realistic sprint commitments to maintain steady progress.

- Testing and Validation Constraints – User acceptance testing with actual municipal zoning officers and LGU personnel was limited due to availability constraints. As a result, testing cycles relied heavily on simulated scenarios and iterative internal reviews to validate system functionality and usability.

SUMMARY

Despite these challenges, each obstacle provided valuable insights that contributed to the refinement and robustness of the system. Through continuous iteration, careful planning, and collaborative problem-solving, the development team successfully delivered a functional, scalable, and user-centered Web-Based GIS system tailored to the needs of local government units in managing zoning compliance and permit processing.

10 Implementation

10.1 Technical Implementation Details

The implementation of the Local Government Unit 1: Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing follows a layered and modular architectural approach to ensure

system maintainability, scalability, and efficient integration of municipal zoning and permit workflows.

Presentation Layer

The Presentation Layer provides the user interface of the system and is responsible for user interaction. It was developed using standard web technologies to deliver a responsive and user-friendly experience for different system users, including public applicants, zoning officers, municipal inspectors, and LGU administrators.

This layer handles:

- User dashboards and navigation
- Data input through web forms and interactive map interfaces
- Display of spatial maps, zoning reports, schedules, and notifications

The design emphasizes usability and accessibility to support both technical and non-technical users.

Application Logic Layer

The Application Logic Layer encapsulates the core business processes of the system. It manages the execution of municipal workflows and enforces business rules across modules such as permit intake, spatial

validation, zoning compliance checks, inspection management, and reporting.

Key responsibilities of this layer include:

- Processing permit applications and review workflows
- Implementing the automated GIS spatial validation logic for zoning compliance
- Managing role-based authentication and authorization
- Coordinating communication between system modules
- Handling notifications and system events

This layer ensures that all municipal operations follow standardized and consistent logic.

Data Management Layer

The Data Management Layer is responsible for storing, retrieving, and maintaining system data. Structured data such as applicant records, property parcels, zoning classifications, inspection schedules, and performance metrics are stored in a relational database management system (MySQL) equipped with spatial extensions.

Unstructured and semi-structured data, including land titles, building plans, architectural drawings, and uploaded documents, are stored in a secure file storage system and linked to structured records through unique

identifiers. This separation ensures efficient data organization and access control.

Integration and Communication Layer

An internal integration layer facilitates communication between system modules through standardized service calls and shared data services. This design supports seamless data synchronization and allows the system to be extended with additional modules or external municipal services in the future.

Intelligent Processing Layer (Spatial and GIS Integration)

The system integrates an Intelligent Processing Layer to support the automated zoning compliance engine. Spatial data processing and GIS techniques are used to analyze applicant-provided coordinate boundaries against municipal zoning ordinances, automatically verifying setbacks, height limits, and floor area ratios. This layer enhances automation while reducing manual map verification by zoning officers.

10.2. Tools and Technologies Used

This section presents the tools and technologies utilized in the development, deployment, and collaboration processes of the Local Government Unit 1: Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. These tools supported

efficient system development, testing, deployment, and team coordination throughout the project lifecycle.

Development Tools and Technologies

The following tools and technologies were used to design and implement the system:

Tool/Technology	Purpose
HTML5, CSS3, JavaScript	Used for developing the system's user interface, ensuring responsiveness and interactive functionality, including Leaflet.js map integration for the interactive zoning maps.
Bootstrap	CSS framework used for responsive layout design, grid systems, and mobile-friendly UI components across devices.
Server-Side Programming Language (PHP)	Implemented core business logic, system workflows, automated GIS spatial validation, and server-side processing.

Leaflet.js Library	Used for client-side geospatial mapping, rendering interactive tiles, and managing draggable property coordinate pins for zoning and permit verification.
Spatial Processing & GIS Libraries	Used for coordinate validation and automated mapping/compliance analysis.
RESTful API Architecture	Enabled communication between system modules through standardized, well-defined endpoints supporting CRUD (Create, Retrieve, Update, Delete) operations.
MySQL / MariaDB	Relational database management system used for storing structured HR-independent data such as applicant accounts, permit application records, spatial boundaries, and audit logs.

SQL	Used for querying, inserting, updating, and managing relational database records.
Composer-Managed Libraries (e.g., PHPMailer)	Used for generating secure OTP codes and enabling automated email notifications and status alerts.

Table no. 16 Development Tools and Technologies

Deployment and Infrastructure Tools

To support system hosting and deployment, the following tools were utilized:

Tool/Technology	Purpose
Web Server (Apache)	Hosted the application and handled client-server communication.
Application Server	Hosted PHP business logic and automated GIS spatial validation rules.
Localhost/Test Server (XAMPP Environment)	Used during development and testing phases.

Database Server	Centralized storage for structured data (MySQL/MariaDB).
File Storage System	Managed secure storage of uploaded verification documents, land titles, and building plans.
Backup Storage	Ensured periodic data backups for recovery and system reliability.
HTTP/HTTPS Protocols	Enabled secure data transmission across the network.
Firewall and Access Controls	Protected system resources and restricted unauthorized access.

Table no. 17 Deployment and Infrastructure Tools

Testing and Quality Assurance Tools

Testing tools were used to validate system functionality and performance:

Tools/Technology	Purpose
Browser Developer Tools	Used for inspecting elements, analyzing asynchronous requests, debugging, and troubleshooting

	front-end code and client-side behavior.
Manual Testing	Applied for functional testing, integration testing, and debugging throughout the sprint cycles.
Version Control System (Git)	Used for tracking code changes, managing branches, and supporting collaborative testing and development.
Code Repository (GitHub)	Utilized for collaborative repository hosting, version control, and coordinating testing efforts across the development team.
Documentation and Diagram Tools (e.g., Draw.io)	Supported technical documentation, architecture diagrams, and tracking of system models during quality assurance.

Table no. 18 Testing and Quality Assurance Tools

Collaboration and Project Management Tools

The following tools supported team collaboration and Agile Scrum implementation:

Tools/Technology	Purpose
Version Control System (Git)	Managed source code changes, branching, and collaboration throughout the sprint cycles.
Code Repository (GitHub)	Hosted the project code and facilitated collaborative version control among the development team.
Agile Project Management Tools	Used for sprint planning, task tracking, and progress monitoring through Scrum boards, sprint trackers, and Agile Scrum ceremonies.
Documentation and Diagramming Tools (e.g., Draw.io)	Used for preparing technical documentation, system

	diagrams, database diagrams, and the capstone manuscript.
--	---

Table no. 19 Collaboration and Project Management Tools

These technologies provide a reliable, scalable, and user-friendly web application for zoning compliance and permit management.

10.3. Code Integration and Interoperability

The system integrates multiple software components to ensure smooth communication between application modules, as well as with two external systems developed by partner groups: IPMS (Infrastructure Project Management System) and UMAN (Utilities Billing and Management System).

The frontend communicates with the backend through HTTP requests. PHP processes user inputs and interacts with the MySQL database to retrieve and store information. The GIS module uses Leaflet.js together with OpenStreetMap to display interactive maps and zoning boundaries.

When an applicant selects a property location, the GIS module retrieves the corresponding zoning information. The backend compares the selected location with zoning regulations stored in the database. If the proposed development complies with zoning requirements, the application

proceeds to the review stage. Otherwise, the system automatically notifies the applicant of the zoning conflict.

Connection with IPMS (Infrastructure Project Management System)

- Outbound (To IPMS): The system sends inspection requests for road-related concerns identified during zoning or permit evaluation.
- Inbound (From IPMS): The system receives inspection results, including whether the request is approved or rejected, along with details on identified problems, findings, and status updates.

Connection with UMAN (Utilities Billing and Management System)

- Outbound (To UMAN): The system sends inspection requests for utility-related concerns, such as water and electricity services.
- Inbound (From UMAN): The system receives inspection results, including approval or rejection status, along with identified issues, findings, and status updates.

Both external integrations are secured through API key authentication and HMAC-signed webhook callbacks. Outbound requests include a shared API key that must match the corresponding key configured on each partner system, ensuring only authorized systems can exchange data. For asynchronous updates — such as inspection results — each external system sends data back to a dedicated webhook callback

endpoint hosted by this system. Every webhook payload is signed using a shared HMAC secret, which the receiving endpoint verifies before accepting the data, protecting against spoofed or tampered requests. Environment-specific configuration (production versus local development) is managed through environment variables, keeping credentials out of version control.

The interoperability of these technologies enables real-time validation, secure data exchange, and accurate mapping while maintaining consistency across all internal system modules and connected external systems.

10.4. Integration Testing and Debugging

To ensure system reliability and seamless interaction among modules, multiple testing activities were conducted throughout the development of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing, including its integration with the external IPMS and UMAN systems. These testing processes verified that all integrated components functioned correctly as a unified system.

Testing Activities Conducted

- **Unit Testing** – Individual system functions and core logic were tested to ensure correct behavior of each module before integration.
- **Module Testing** – Each system module, including Applicant Management, GIS Map and Spatial Validation, Permit Processing, Inspection Management, Payment Management, Notification, and Reporting and Analytics, was tested independently to verify functional completeness.
- **Integration Testing** – Tests were performed to validate communication and data flow between modules using RESTful API calls and shared database access. This ensured that updates in one module, such as a permit status change, were accurately reflected across the system.
- **GIS Spatial Validation Testing** – The automated zoning compliance engine was tested using various coordinate and property boundary inputs to verify the accuracy of spatial (point-in-polygon) validation and rule-based compliance checks.
- **External Integration Testing** – The API key authentication and HMAC webhook signing mechanisms used for communication with IPMS and UMAN were tested to verify secure request handling and accurate receipt of inspection results.

- **End-to-End Testing** – Manual walkthroughs were conducted to simulate real user scenarios, covering the complete permit workflow from application submission to permit issuance.

The following issues were identified and resolved during debugging:

- Incorrect map coordinates
- Database connection errors
- Form validation issues
- Duplicate application submissions
- Slow loading of GIS layers
- Session timeout handling
- Responsive design compatibility

SUMMARY

During debugging, identified issues such as incorrect map coordinates, database connection errors, form validation issues, duplicate submissions, and responsive design inconsistencies were resolved through iterative code refinement and configuration adjustments. Error handling and logging mechanisms were used to trace and fix issues, ensuring system stability prior to final deployment.

11 Testing and Quality Assurance

11.1. Testing Strategies and Methodologies

A combination of manual and feature-based testing approaches was applied to validate the implementation of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing. Each core system module was tested independently and later validated as part of the integrated system to ensure correctness and reliability. The system underwent Unit Testing, Integration Testing, System Testing, User Acceptance Testing (UAT), and Security Testing to verify functionality, reliability, and security.

Testing Approaches Applied

- **Backend Functional Testing** – Core system logic and data processing workflows were tested to validate permit intake, applicant tracking, automated zoning compliance evaluation, inspection scheduling, payment processing, and permit issuance.
- **Frontend and User Interface Testing** – Browser-based testing was conducted to ensure correct user interface behavior, form validation, navigation flow, and accurate rendering of the Leaflet.js mapping interface across different user roles.

- **Integration Testing** – Communication and data synchronization between modules were tested to confirm that updates in one module, such as a zoning compliance result or payment confirmation, were properly reflected across the system using RESTful API calls and a centralized database.
- **External System Integration Testing** – API key authentication and HMAC webhook signing mechanisms were tested to validate secure request handling and accurate receipt of inspection results from IPMS and UMAN.
- **Black Box Testing** – System functionality was evaluated from the user's perspective by validating expected outputs based on user inputs, without examining internal code structure.
- **End-to-End Testing** – Complete permit workflows were simulated, from applicant coordinate submission and automated zoning validation to inspection scheduling, permit fee payment, and final permit issuance.
- **Security Testing** – Role-based access control, session management, and authentication mechanisms were tested to ensure that users could only access data and functions appropriate to their assigned roles.
- **Final System-Wide Testing** – A comprehensive test was performed after full module integration to ensure the system

operated correctly as a unified zoning and permit processing platform.

Testing Strategy Summary

The layered testing strategies ensured that the Web-Based GIS System met functional requirements, maintained system stability, and delivered reliable performance across all zoning compliance and permit processing operations prior to deployment.

11.2. Test Cases and Test Data

Test cases were derived from the system requirements, use case scenarios, and user stories defined in earlier chapters. Each test case was designed to validate both normal system behavior and exception handling across the modules of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing.

Test Case ID	Module	Test Description	Expected Result	Status
TC-01	Applicant Manageme nt	Test applicant registration and account creation	Applicants can successfully	Passed

			create and verify an account	
TC-02	Applicant Management	Test document upload for land titles and supporting files	Applicants can upload required documents in supported formats	Passed
TC-03	Applicant Management	Test file validation for invalid or incomplete uploads	System rejects invalid or incomplete files with an error message	Passed
TC-04	Applicant Management	Test duplicate application checker using applicant details	System prevents duplicate application submissions	On-going
TC-05	GIS Map and Spatial Validation	Test property coordinate plotting on the Leaflet.js map	System accurately captures latitude and longitude coordinates	Passed

TC-06	GIS Map and Spatial Validation	Test automated zoning compliance engine (point-in- polygon check)	System correctly flags application as Compliant or Non-Compliant	Passed
TC-07	GIS Map and Spatial Validation	Test setback and floor area ratio validation	System accurately validates against CLUP-derived setback rules	On- going
TC-08	GIS Map and Spatial Validation	Test hazard buffer zone detection	System correctly identifies properties within hazard buffer zones	Not Tested
TC-09	Permit Processing	Test permit application intake and status tracking	Applicants can track real-time application status	Passed
TC-10	Permit Processing	Test Zoning Officer	Application status updates correctly	Passed

		approval/rejection workflow	based on officer decision	
TC-11	Permit Processing	Test Building Official final review and permit issuance	Digital permit is generated upon final sign-off	On-going
TC-12	Permit Processing	Test digital Zoning Certificate generation with tracking code	System generates certificate with unique tracking code and watermark	On-going
TC-13	Inspection Management	Test inspection scheduling and assignment to field inspectors	Inspection schedules sync correctly with inspector calendars	Passed
TC-14	Inspection Management	Test geo-fenced GPS verification for on-site inspections	System validates inspector proximity before	On-going

			allowing checklist access	
TC-15	Inspection Management	Test inspection report submission with photographic evidence	Inspectors can submit findings with attached photos	Passed
TC-16	Inspection Management	Test Notice of Violation generation	System generates a standardized digital violation notice	Not Tested
TC-17	Payment Management	Test permit fee payment through integrated payment gateway	Payment transaction is recorded and confirmed before approval release	On-going
TC-18	Notification Module	Test automated notifications for application status updates	Applicants and inspectors receive real-time notifications	Passed

TC-19	Reporting and Analytics	Test generation of administrative and compliance reports	System generates accurate summary reports and analytics	On-going
TC-20	Administration and Security	Test role-based access control across all user roles	Users can only access data and functions per assigned role	Passed
TC-21	Administration and Security	Test two-factor authentication (2FA) during login	System requires valid OTP before granting access	Passed
TC-22	Administration and Security	Test session timeout and re-authentication	Inactive sessions expire and require re-login	Passed
TC-23	Administration and Security	Test audit log recording for critical actions	All approvals, rejections, and edits are logged with timestamps	Passed

TC-24	External Integration (IPMS)	Test outbound inspection request for road- related concerns	IPMS receives a properly formatted, authenticated request	On- going
TC-25	External Integration (IPMS)	Test inbound webhook for IPMS inspection results	System correctly records approval/rejection and findings from IPMS	On- going
TC-26	External Integration (UMAN)	Test outbound inspection request for utility- related concerns	UMAN receives a properly formatted, authenticated request	On- going
TC-27	External Integration (UMAN)	Test inbound webhook for UMAN inspection results	System correctly records approval/rejection and findings from UMAN	On- going

TC-28	External Integration (IPMS/UM AN)	Test API key mismatch handling	System rejects requests with invalid or missing API key	Passed
TC-29	External Integration (IPMS/UM AN)	Test HMAC webhook signature verification	System rejects webhook payloads with invalid or missing signature	Passed

Table no. 20 Sample Test Cases and Test Data

11.3. Test Results and Bug Reports

All major modules of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing underwent integration and system testing based on the test cases presented in Section 11.2. Most test cases were successfully executed, while minor issues were identified during early testing phases. These issues were logged, analyzed, and resolved before final validation.

Test Results and Bug Reports

Bug ID	Related Test Case ID	Description	Resolution Applied	Severity	Status
BUG-GIS-01	TC-05	Incorrect map coordinates captured during property plotting	Recalibrated Leaflet.js coordinate handling and validated against OpenStreetMap reference points	Medium	Resolved
BUG-DB-02	TC-01	Database connection errors during peak testing	Optimized database connection handling and added	High	Resolved

			retry/error logging		
BUG-FORM-03	TC-02	Form validation issues on document upload fields	Strengthened client-side and server-side input validation rules	Medium	Resolved
BUG-APP-04	TC-04	Duplicate application submissions accepted by the system	Implemented duplicate-record checker using applicant details	Medium	Resolved
BUG-GIS-05	TC-06	Slow loading of GIS zoning layers on the map interface	Optimized spatial query performance and layer rendering	Low	Resolved
BUG-SEC-06	TC-22	Session timeout not properly	Corrected session expiration logic and re-	Medium	Resolved

		enforced on inactivity	authentication flow		
BUG- UI-07		Responsive design compatibility issues across device sizes	Adjusted CSS/Bootstrap responsive breakpoints and layout grid	Low	Resolved
BUG- SYNC- 08	TC-11	Module integration and data synchronizati on delays between Permit Processing and Inspection Management	Refined internal service calls and shared database access to ensure real- time consistency across modules	High	Resolved
BUG- SPAT- 09	TC-07, TC-08	Difficulty processing unstructured	Refined spatial validation logic and improved	High	Resolved

		spatial and zoning data (irregular parcel boundaries, hazard buffer overlaps)	parsing of geographic input data		
BUG-RBAC-10	TC-20	Role-based access control misconfiguration allowed restricted features to be accessed by unauthorized roles	Reviewed and corrected role-permission mappings across all seven user roles	High	Resolved
BUG-WF-11	TC-09, TC-10	Workflow alignment issue between	Adjusted workflow sequencing logic to	Medium	Resolved

		online permit submission and automated zoning/inspection processes	correctly route applications between zoning, building, and inspection stages		
BUG-EXT-12	TC-24, TC-26	API key mismatch caused outbound inspection requests to IPMS and UMAN to be rejected	Corrected environment variable configuration to ensure API keys matched on both systems	High	Resolved
BUG-EXT-13	TC-25, TC-27	HMAC webhook signature verification failures when	Fixed webhook secret mismatch and corrected signature-	High	Resolved

		receiving inspection results from IPMS/UMAN	generation logic on the callback endpoint		
BUG- EXT- 14	TC- 24– TC-27	Incorrect webhook callback URL configuration caused inspection results from IPMS/UMAN to fail delivery	Corrected callback URL to point to the system's actual physical file path	Mediu m	Resolved

Table no. 21 Test Results and Bug Reports

Test Results Summary

After applying the necessary fixes, all affected test cases were re-executed and marked as Passed. The resolved issues did not impact the overall system architecture and were successfully addressed through debugging and refinement. This confirms that the Web-Based GIS System meets its functional requirements and operates reliably as an integrated

zoning compliance and permit processing platform, including its integrations with the external IPMS and UMAN systems.

11.4 Quality Assurance Measures

To ensure that the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing met the required quality, performance, and reliability standards, several quality assurance measures were consistently applied throughout the development process.

- **Code Reviews** were regularly conducted during sprint cycles to ensure code maintainability, correctness, and adherence to agreed coding standards.
- **Peer Testing** was applied to cross-check module behavior, identify inconsistencies, and validate functionality across the seven user roles (Applicant, Zoning Officer, Building Official, Assessor, Inspector, Admin, Super Admin).
- **Version Control (Git and GitHub)** was strictly followed to manage code changes, handle branching, and prevent integration conflicts across sprint cycles.
- **Agile Scrum Ceremonies** — including daily stand-ups, sprint reviews, and sprint retrospectives — were used to continuously monitor progress, surface defects early, and adjust priorities based on testing outcomes.

- **Role-Based Access Control (RBAC) Validation** was performed to ensure users could only access data and system functions appropriate to their assigned roles, preventing unauthorized access.
- **Security Quality Checks** were applied to verify session regeneration on login, two-factor authentication (2FA) through OTPs, cryptographic token validation on administrative writes, and audit logging of critical actions.
- **External Integration Validation** was performed to confirm that API key authentication and HMAC webhook signature verification correctly secured data exchange with the IPMS and UMAN systems.
- **Final Quality Assurance Validation** was performed to confirm that all functional and non-functional requirements — including reliability, scalability, and security — were met prior to system completion.

The Web-Based GIS System underwent comprehensive testing and validation using a combination of manual testing and iterative review. All critical functionalities, including automated zoning compliance verification, permit processing, inspection scheduling, and external system integration, were verified through defined test cases, and all identified issues were resolved before final validation. Through consistent application of code reviews, peer testing, version control practices, and security validation, the system achieved a stable, reliable, and scalable solution suitable for

automating zoning compliance and development permit processing for Local Government Units.

12 Results and Evaluation

12.1 Project Outcomes and Deliverables

The implementation of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing produced significant outcomes and deliverables that directly addressed the project objectives of automating zoning verification, improving permit processing efficiency, and enhancing transparency for Local Government Units (LGUs). The key outcomes and deliverables are outlined below:

Automated Zoning Compliance Verification

- Coordinate-based, rule-based compliance engine cross-referencing applicant-plotted coordinates against digitized zoning boundaries
- Automatic flagging of properties intersecting protected hazard bands or restricted easements
- Reduced manual, paper-based cross-referencing previously performed by zoning officers
- Improved consistency and accuracy in zoning decisions

Interactive GIS Mapping Module

- Fully responsive Leaflet.js-based mapping interface with a coordinate picker for applicants
- Toggleable zoning and buffer-zone overlays for reviewing officers
- Geofenced viewport restricting navigation to the pilot city/municipality
- Switchable road, satellite, hybrid, and terrain map layers

Real-Time Application Status Tracking

- Localized application registry with a color-coded progress pipeline
- Interactive inspection calendar allowing applicants to monitor and request rescheduled inspections
- Reduced need for physical visits to the LGU office

Secure, Centralized Data Management

- Centralized relational database housing all permit applications, zoning records, and associated documents
- Session regeneration on login, mandatory two-factor authentication (2FA) via OTPs, and cryptographic token validation on administrative writes
- Multi-table transaction rollback safeguards and an unalterable audit log
- Role-based access control (RBAC) across all seven user roles

External System Integration

- Secure, API key-authenticated communication with the IPMS (Infrastructure Project Management System) and UMAN (Utilities Billing and Management System)
- HMAC-signed webhook callbacks for receiving inspection results, approval/rejection status, and identified findings

Digital Audit Trail and Reporting

- Automated digital Zoning Certificate drafting with embedded security seals and unique tracking codes
- Secure, password-verified database backups
- Multi-parameter reporting exports for executive and compliance reporting

Integrated and Scalable Platform

- Fully integrated system modules (Applicant Management, GIS/Spatial Validation, Permit Processing, Inspection Management, Payment, Notification, Reporting, Administration/Security)
- TOGAF-aligned enterprise architecture supporting future scalability
- Successful system testing and validation across 120+ completed user stories

- Completion of all planned project deliverables by the conclusion of the final sprint

12.2 Alignment with Project Objectives

The Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing has successfully aligned with the project's original objectives. The evaluation of system outcomes demonstrates that each objective was addressed through implemented features and validated through testing and deployment.

Objective 1: Automated, Coordinate-Based Compliance Validation

The system achieved this objective by developing a rule-based validation module that automatically compares proposed land-use applications against existing zoning ordinances through coordinate-based checks — covering setbacks, height limits, and floor area ratios. This automation replaced manual, subjective cross-referencing of paper-based CLUP maps and significantly improved consistency in zoning decisions.

Objective 2: Secure, Centralized Database

This objective was achieved through the implementation of a secure, centralized relational database that manages and protects sensitive applicant records and spatial layers from unauthorized access or data corruption. Session regeneration, two-factor authentication (2FA),

cryptographic token validation, and role-based access control (RBAC) were implemented to safeguard data integrity across all seven user roles.

Objective 3: Interactive GIS Mapping Interface

The system fulfilled this objective by delivering a high-precision, Leaflet.js-based interactive mapping interface that provides planners and the public with a visual "ground truth" for spatial analysis. This includes a coordinate picker, toggleable zoning and buffer-zone overlays, and switchable map layers, replacing static paper maps.

Objective 4: Real-Time Application Status Tracking

This objective was achieved through a real-time status-tracking dashboard that allows applicants to monitor their submissions from initial filing through final approval without physical follow-up visits, including an interactive inspection calendar for scheduling and rescheduling site visits.

Objective 5: Reliable Digital Audit Trail

The system addressed this objective by establishing a digital audit trail complete with time-stamped logs, automated Zoning Certificate drafting with embedded security seals, and PDF summary receipts that archive every step of the permitting process — minimizing human error and administrative discretion in zoning enforcement.

Overall Evaluation

Based on the implemented features and testing results, the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing successfully met all five defined project objectives. The system demonstrates strong alignment with its intended goals and provides an effective, automated solution for zoning compliance verification and development permit processing within Local Government Units.

12.3 Stakeholder and User Feedback

This section presents an evaluation of stakeholder engagement and feedback gathered during the development and testing of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing.

Throughout the Agile Scrum development process, stakeholders — including the project adviser and panel — were consulted at key milestones to provide feedback and revisions on system design, workflow accuracy, and alignment with LGU zoning and permitting processes. This iterative feedback loop informed refinements to the GIS mapping interface, the automated compliance engine, and the overall application workflow throughout the sprint cycles.

However, formal User Acceptance Testing (UAT) with actual municipal zoning officers and LGU personnel was limited due to availability constraints, as noted in the challenges encountered during development. As a result, the system's validation relied primarily on internal testing, adviser and panel review, and simulated user scenarios based on the documented business processes of the Zoning and Planning Office, rather than extensive direct feedback from LGU end-users in a live operational setting.

This limitation is acknowledged as an area for future work: full-scale UAT involving actual LGU personnel — Zoning Officers, Building Officials, Assessors, and Inspectors — would be necessary to validate the system's usability, workflow accuracy, and operational fit under real municipal conditions before any production deployment. Conducting this structured stakeholder validation is recommended as a priority step in future enhancement phases of the project.

12.4 Lessons Learned

The development of the Web-Based GIS System for Automated Zoning Compliance and Development Permit Processing provided the research team with valuable insights that extended beyond the technical implementation of the platform.

Value of Iterative, Agile Development

The adoption of the Agile Scrum methodology proved instrumental in translating the study's objectives into a working system. The disciplined rhythm of Sprint Planning, Daily Standups, Sprint Reviews, and Retrospectives allowed the development team to remain responsive to emerging technical challenges, particularly in reconciling spatial data accuracy with the practical realities of zoning ordinance enforcement.

Complexity of Module Integration and Data Synchronization

Ensuring seamless communication and consistent data synchronization across interdependent modules — such as Permit Processing, Inspection Management, and the external IPMS and UMAN integrations — proved more complex than initially anticipated. This reinforced the importance of standardized API contracts, consistent data structures, and thorough integration testing early in the development cycle rather than only at the final stages.

Challenges of Processing Unstructured Spatial Data

Handling complex geographic inputs, irregular parcel boundaries, and hazard buffer overlaps required more careful validation logic than standard structured data processing. This highlighted the importance of

allocating sufficient time for spatial-data-specific testing when working with GIS-based systems.

Security Cannot Be an Afterthought

Implementing secure role-based access control across seven distinct user roles, alongside session management, two-factor authentication, and cryptographic token validation, required extensive validation effort. This reinforced that access control and information-security design decisions are most effective when addressed early in the architecture phase rather than layered on afterward.

The Limits of Simulated Testing

Limited availability of actual municipal zoning officers and LGU personnel for User Acceptance Testing constrained how thoroughly the system's real-world usability could be validated prior to completion. This experience underscored that early and sustained engagement with actual end-users — not just technical stakeholders — is critical for producing a system that fits real institutional workflows, and should be planned for from the earliest possible stage of future similar projects.

Balancing Automation with Human Oversight

The project affirmed that technology, when carefully designed around genuine institutional needs, can meaningfully reduce administrative

burden and restore public trust through greater transparency. At the same time, the team recognized that the system is not intended to replace human judgment or on-site verification, but to serve as a reliable digital foundation upon which zoning officers and building officials can exercise their expertise with greater speed and confidence.

Chapter 5

CONCLUSION

13.1 Key Takeaways and Summary

1. Increased Productivity

The suggested web-based system reduces manual paperwork and processing time by automating the verification of zoning compliance and the processing of development permits.

2. Improved Precision

Applicants and evaluators can reduce human error and guarantee that permit applications adhere to local development policies by incorporating zoning regulations into the system.

3. Increased Openness

The system allows applicants to monitor the status of their applications in real time, promoting accountability and improving communication between the local government and stakeholders.

4. Management of Centralized Data

A centralized database houses all zoning records, permit applications, and associated paperwork, which improves the organization and security of data retrieval, reporting, and record management.

5. Encouragement of Sustainable Local Development

The system helps local government units make informed decisions by ensuring that development projects conform to approved land use plans and zoning ordinances.

Summary

The Web-Based System for Automated Zoning Compliance Development Permit Processing is designed to modernize the way local government units manage zoning compliance and permit applications. By replacing manual procedures with an automated digital platform, the system streamlines application processing, reduces delays, and improves the overall quality of public service.

The system promotes transparency, accuracy, and accountability through automated validation, centralized record management, and real-time application tracking. Ultimately, it supports efficient governance and sustainable local development by ensuring that construction and development projects comply with existing zoning regulations while providing a more convenient experience for both applicants and government personnel. Additionally, by encouraging the use of contemporary information technologies in public administration, this system's implementation supports local government entities' ongoing digital transformation initiatives. It lessens reliance on manual procedures, which are frequently prone to delays, lost paperwork, and irregularities. Both government employees and the general public gain from the system's

digitization of zoning compliance verification and permit processing, which produces a more responsive, dependable, and citizen-centered service.

Lastly, by guaranteeing that development projects adhere to current zoning laws, thorough land use plans, and regulatory requirements prior to approval, the suggested system promotes sustainable local development. The deployment of this web-based system offers a workable and scalable way to boost service quality, support long-term community development, and improve regulatory compliance as local governments continue to adopt digital governance.

13.2 Project Achievements and Contributions

Project Achievements

Over the course of its development, the Web-Based GIS System for Automated Zoning and Permits successfully translated each of the objectives outlined in Chapter 1 into a working, tested feature of the platform. The project's Product Backlog, which enumerated over 120 user stories spanning registration and identity verification, authentication, dashboard analytics, application intake, GIS spatial verification, inspection scheduling, administrative reporting, and audit logging, was completed in full, with every backlog item marked "Done" by the conclusion of the final sprint.

Specifically, the automated, coordinate-based compliance engine envisioned in the study's first objective was realized through a rule-based validation module that cross-references applicant-plotted coordinates against digitized zoning boundaries, automatically flagging properties that intersect protected hazard bands or restricted easements before an officer signs off. This automation directly replaced the manual, paper-based cross-referencing that previously slowed permit evaluation and introduced inconsistency into zoning decisions.

The secure, centralized database called for in the second objective was implemented alongside a dedicated information-security backlog: session regeneration upon login, mandatory two-factor authentication through short-lived one-time passwords, cryptographic token validation on every administrative write, multi-table transaction rollback safeguards, and an unalterable audit log that records every re-routing, parameter change, and final evaluation decision. Role-based access control further ensures that the Applicant, Zoning Officer, Building Official, Assessor, Inspector, Admin, and Super Admin each interact only with the modules and data appropriate to their function.

The interactive, Leaflet.js-based mapping interface described in the third objective was delivered as a fully responsive GIS module, complete with a coordinate picker for applicants, toggleable zoning and buffer-zone overlays for reviewing officers, a geofenced viewport restricting navigation

to the pilot city, and a floating layer-control button for switching between road, satellite, hybrid, and terrain tiles - replacing static paper maps with a dynamic digital "ground truth."

The real-time status-tracking dashboard called for in the fourth objective was achieved through a localized application registry, a color-coded progress pipeline, and an interactive inspection calendar that lets applicants monitor their submissions, and even request a rescheduled inspection, without a physical visit to the LGU office.

Finally, the reliable digital audit trail called for in the fifth objective is supported by automated certificate drafting with embedded digital seals, secure and password-verified database backups, and multi-parameter reporting exports, giving the LGU a dependable, time-stamped record of every permitting decision. Beyond this functional backlog, the project also delivered on its non-functional commitments, verifying the platform against defined mobile-responsiveness breakpoints, bilingual English and Filipino localization, and accessible, color-coded status indicators, ensuring the system is usable by both LGU personnel and the general public it serves.

Contributions

Beyond its technical deliverables, the project contributes concrete value to each of the stakeholders identified in Chapter 1's discussion of significance. For the Local Government Unit, the system offers a durable digital infrastructure that enforces the Comprehensive Land Use Plan

consistently and reduces the institution's reliance on fragmented paper records. For Zoning Officers, the automated compliance engine and structured evaluation workspace remove the burden of manually cross-referencing maps and titles, freeing them to focus on higher-level urban-planning judgment. For Building Officials, consolidated compliance reports and a single-view evaluation workspace allow final permit decisions to be rendered more quickly without sacrificing regulatory accuracy.

For Inspectors, the project contributes a dedicated field-reporting workflow, complete with scheduling, geo-fenced location validation, and violation-notice generation, that did not exist under the previous manual process. For the general public and permit applicants, the project's most tangible contribution is transparency: a citizen no longer needs to make repeated in-person visits to know where an application stands, what documents remain deficient, or when a site inspection is scheduled. Finally, for future researchers and IT practitioners, the completed system, together with its documented Agile Scrum process, TOGAF-aligned enterprise architecture, and microservices-oriented technical design, serves as a scalable and replicable reference for applying GIS and modern software-engineering practices to other local-governance challenges, providing a concrete foundation upon which the enhancements discussed in the following section can be built.

13.3 Future Work and Enhancements

Although the Web-Based GIS System for Automated Zoning Compliance Development Permit Processing in Local Development successfully achieved its primary objectives of automating zoning compliance verification, streamlining permit processing, and improving transparency within the Local Government Unit (LGU), there remain several opportunities for future enhancements that can further improve the system's efficiency, scalability, and functionality.

First, future researchers may integrate the system with other government information systems such as the Building Permit System, Business Permit and Licensing System (BPLS), Tax Assessment System, and the Land Registration Authority (LRA). Such integration would eliminate duplicate data entry, improve data consistency across government agencies, and provide a unified digital platform for processing development-related applications.

Second, the system can be enhanced by incorporating mobile application support for Android and iOS devices. A mobile version would enable applicants to submit applications, monitor permit status, receive notifications, and access zoning information anytime and anywhere. Likewise, LGU personnel and field inspectors could perform on-site

inspections using mobile GIS tools with GPS capabilities, allowing real-time validation and faster reporting.

Another significant enhancement is the implementation of advanced GIS capabilities, including high-resolution satellite imagery, drone mapping integration, three-dimensional (3D) visualization, and cadastral parcel management. These features would provide more accurate spatial analysis and improve decision-making during zoning evaluation and urban planning activities.

Future development may also incorporate Artificial Intelligence (AI) and Machine Learning (ML) technologies to assist zoning officers in identifying potential zoning conflicts, predicting future land-use trends, detecting irregular development patterns, and recommending appropriate zoning classifications based on historical permit records and spatial data.

To further improve public service delivery, the system may include an online payment gateway integrated with accredited financial institutions and electronic payment platforms. This enhancement would allow applicants to pay permit processing fees electronically, reducing physical transactions and supporting a more convenient end-to-end digital permitting process.

The researchers also recommend enhancing the notification system by integrating SMS, email, and push notifications to provide applicants with real-time updates regarding application status, document deficiencies,

inspection schedules, and permit approvals. This feature would further strengthen transparency and reduce unnecessary follow-up inquiries at the LGU offices.

From an information security perspective, future versions of the system should implement additional security measures such as multi-factor authentication (MFA), stronger encryption techniques, automated backup and disaster recovery mechanisms, intrusion detection systems, and continuous security monitoring to ensure the confidentiality, integrity, and availability of sensitive government and applicant information.

Finally, as municipalities continue to expand, the system should be designed for greater scalability through cloud-based deployment and microservices optimization. This would allow multiple municipalities or cities to adopt the platform while supporting larger GIS datasets, increased numbers of concurrent users, and future government digital transformation initiatives.

Overall, these recommended enhancements will further strengthen the capability of the proposed Web-Based GIS System, making it more intelligent, secure, interoperable, and sustainable. Future implementations can build upon the current system to support comprehensive e-governance initiatives, improve public service delivery, and contribute to more efficient and data-driven urban planning and development management within Local Government Units.

13.4 Closing Remarks

The completion of this capstone project marks the culmination of a journey that began with the recognition of a fundamental inefficiency within local governance: the reliance on manual, paper-based procedures for zoning compliance and permit processing. What started as an examination of the operational bottlenecks faced by the Local Government Unit's Zoning and Planning Office has evolved, through structured research, iterative development, and rigorous requirements analysis, into a functional Web-Based GIS System capable of transforming how zoning compliance is verified and how development permits are processed.

Throughout this undertaking, the adoption of the Agile Scrum methodology proved instrumental in translating the study's objectives into a working system. The disciplined rhythm of Sprint Planning, Daily Standups, Sprint Reviews, and Retrospectives allowed the development team to remain responsive to emerging technical challenges, particularly in reconciling spatial data accuracy with the practical realities of zoning ordinance enforcement. This iterative approach ensured that each component of the system, from the interactive GIS mapping module to the automated compliance engine, was continuously refined and validated against the actual needs of the Applicant, Zoning Officer, Building Official, and System Administrator.

More than a technical accomplishment, this project represents a deliberate step toward the broader digital transformation of local governance. It affirms the researchers' conviction that technology, when carefully designed around genuine institutional needs, can meaningfully reduce administrative burden, minimize human error, and restore public trust through greater transparency and accountability. At the same time, the researchers recognize that the system is not intended to replace human judgment or on-site verification, but rather to serve as a reliable digital foundation upon which zoning officers and building officials can exercise their expertise with greater speed and confidence.

As this study closes, the researchers hope that the Web-Based GIS System for Automated Zoning and Permits will serve not only as a practical tool for the pilot LGU but as a demonstrable model for other municipalities seeking to modernize their own permitting processes. It is the researchers' aspiration that this work, and the recommendations offered for its continued enhancement, will inspire further innovation at the intersection of geographic information systems and local governance, ultimately contributing to more efficient, transparent, and sustainable urban development for communities in the years to come.

REFERENCES

- Abin, S. M., Shaju, A. E., Priya, J. K. C., Abhijith, V., & Jose, T. (2021). Role of GIS & analytics in transforming municipal administration. *International Journal of Engineering Research & Technology (IJERT)*, NCIIE 2021, 9(6). <https://doi.org/10.17577/IJERTCONV9IS06016>
- Carrido, A. J. DC., Felix, P. A. G., Soriano, F. A., Esmeralda, A. K. B., Petacio, A. F., Aminulla, M. H. K., Roxas, R. R., Magayanes, L. M., Padoa, I. C., & Rivera, M. A. (2026). A geospatial analytical framework for assessing land use and zoning compliance: The case of Zamboanga City. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, X-5/W4-2025, 135–142. <https://doi.org/10.5194/isprs-annals-X-5-W4-2025-135-2026>
- Cina, E., Elbasi, E., Elmazi, G., & AlArnaout, Z. (2025). The role of AI in predictive modelling for sustainable urban development: Challenges and opportunities. *Sustainability*, 17(11), 5148. <https://doi.org/10.3390/su17115148>
- De Villa, C. D. V., Florencondia, N. T., & Aduna, J. R. (2025). Digitalization and innovation in building permit systems in the Philippines: Enhancing efficiency, compliance, and safety. *International Journal of Research in Engineering, Science and Management*, 8(6), 112–119. <https://journal.ijresm.com/index.php/ijresm/article/view/3308>
- Felix, P. A. G., Carrido, A. J. DC., Soriano, F. A., Esmeralda, A. K. B., Petacio, A. F., Aminulla, M. H. K., Roxas, R. R., Magayanes, L. M., Padoa, I. C., & Rivera, M. A. (2026). Assessing institutional readiness and willingness of Philippine LGUs to adopt geospatial land use monitoring tools. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-5/W4-2025, 95–102. <https://doi.org/10.5194/isprs-archives-XLVIII-5-W4-2025-95-2026>
- Khanna, A., Singh, P., & Angra, P. K. (2026). Blockchain-enabled e-governance framework for secure and transparent land record management. *International Research Journal of Multidisciplinary Technovation*, 8(4), 73–90. <https://doi.org/10.54392/irjmt2645>

Osakwe, J., Akongne, N. N., Osakwe, N. C., & Gondo, D. (2025). Integrating data analytics and GIS for smart city development: A systematic literature review. In *Proceedings of the 2025 IST-Africa Conference (IST-Africa)* (pp. 1–11). IEEE.
https://www.researchgate.net/publication/392865907_Integrating_Data_Analytics_and_GIS_for_Smart_City_Development_A_Systematic_Literature_Review

Rivas, R. M. (2026). Barangay information management system with automated mobile messaging for improved community engagement. *International Journal on Science and Technology (IJSAT)*, 17(1).
<https://www.ijstat.org/papers/2026/1/10412.pdf>

Santillan, J. R., & Heipke, C. (2024). Visualizing the Philippines' SDG 11.3.1 status and progress using open-source geospatial software and earth observation datasets. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, X-5-2024, 173–180.
<https://doi.org/10.5194/isprs-annals-X-5-2024-173-2024>

Sassa, A. C., de Almeida, I. A., Pereira, T. N. F., & de Oliveira, M. S. (2023). Scrum: A systematic literature review. *International Journal of Advanced Computer Science and Applications*, 14(4), 173–181.
https://thesai.org/Downloads/Volume14No4/Paper_20-Scrum_A_Systematic_Literature_Review.pdf

Schwaber, K., & Sutherland, J. (2020). *The Scrum guide: The definitive guide to Scrum — The rules of the game*. Scrum.org.
<https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>

Shahi, K., McCabe, B. Y., & Shahi, A. (2019). Framework for automated model-based e-permitting system for municipal jurisdictions. *Journal of Management in Engineering*, 35(6).
[https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000712](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000712)

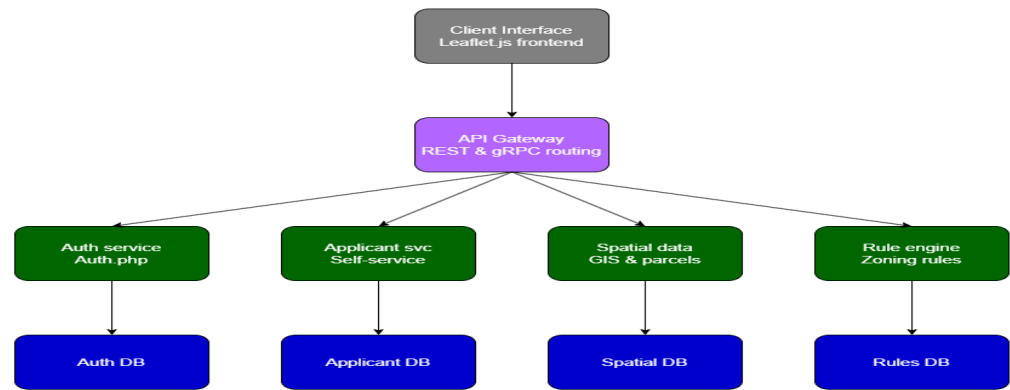
Wonohardjo, E. P., Sunaryo, R. F., & Sudiyono, Y. (2019). A systematic review of SCRUM in software development. *JOIV: International Journal on Informatics Visualization*, 3(2), 108–112.
<https://doi.org/10.30630/joiv.3.2.167>

APPENDICES

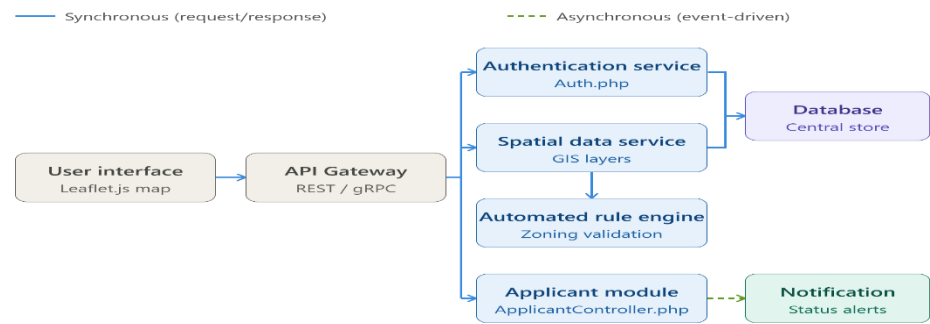
Appendix A: Detailed Technical Documentation

1. System Architecture

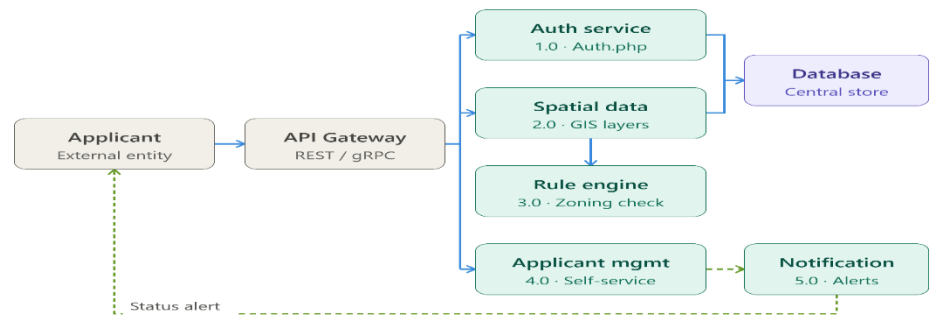
Microservices Diagram



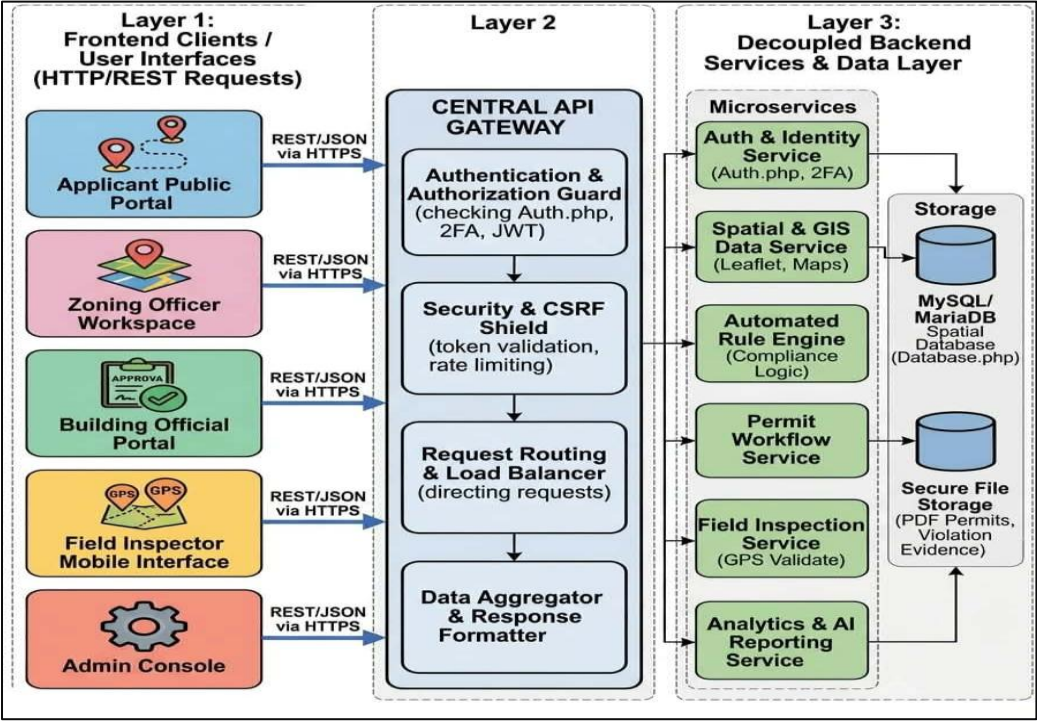
Communication Pattern Diagram



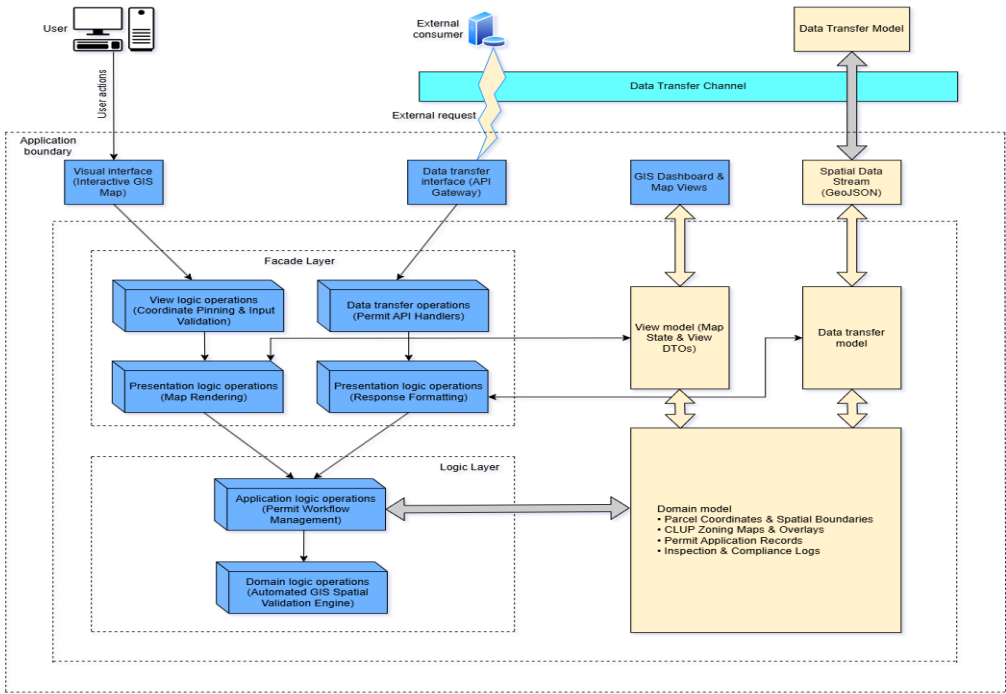
Data Flow Diagram for Microservices



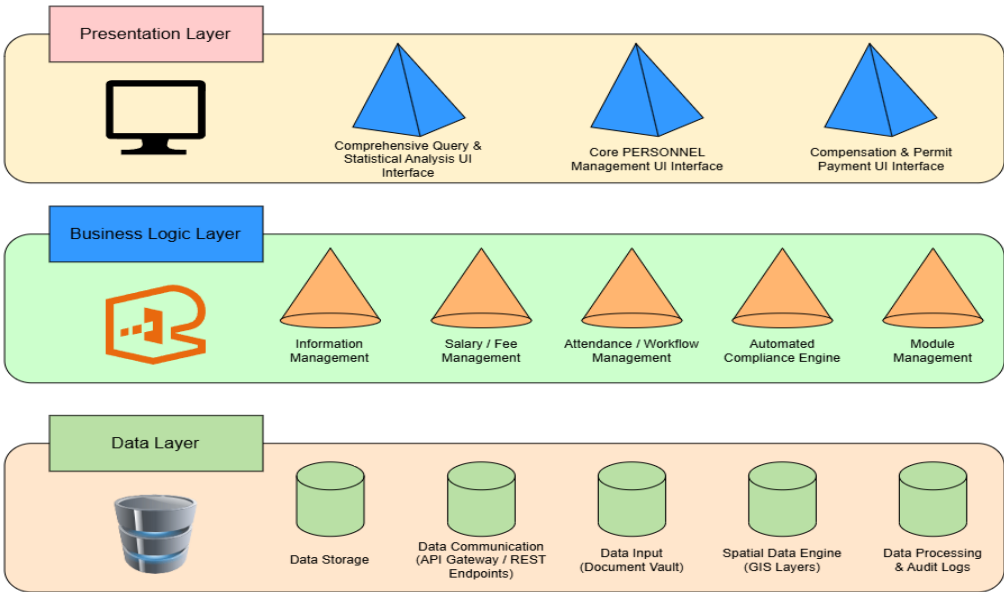
API Gateway



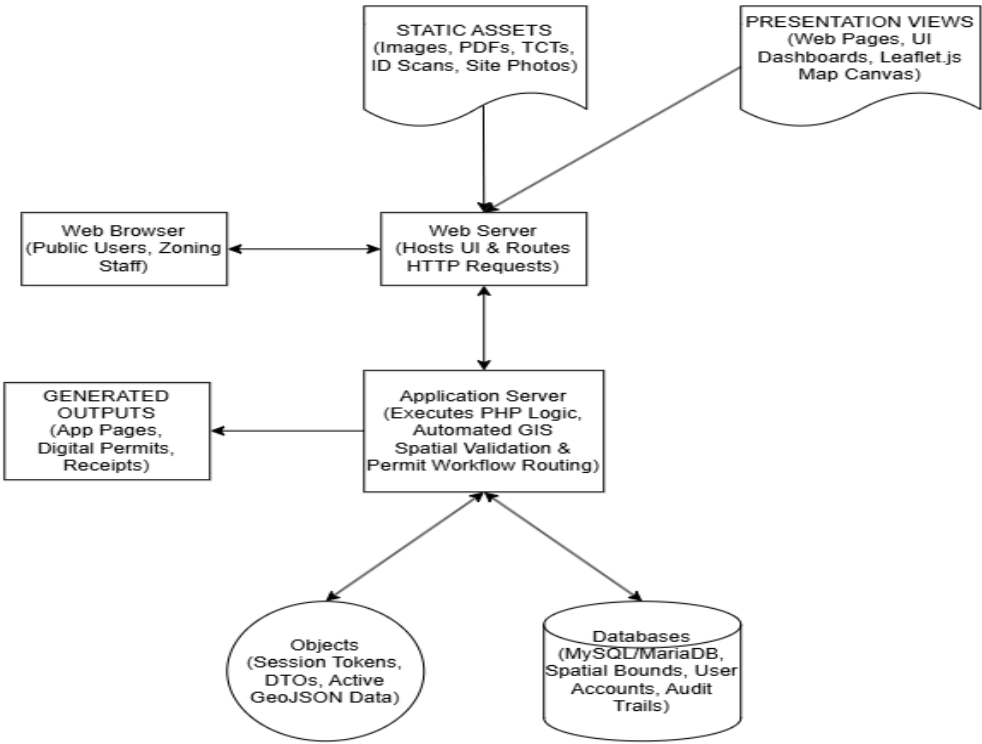
Application Architecture



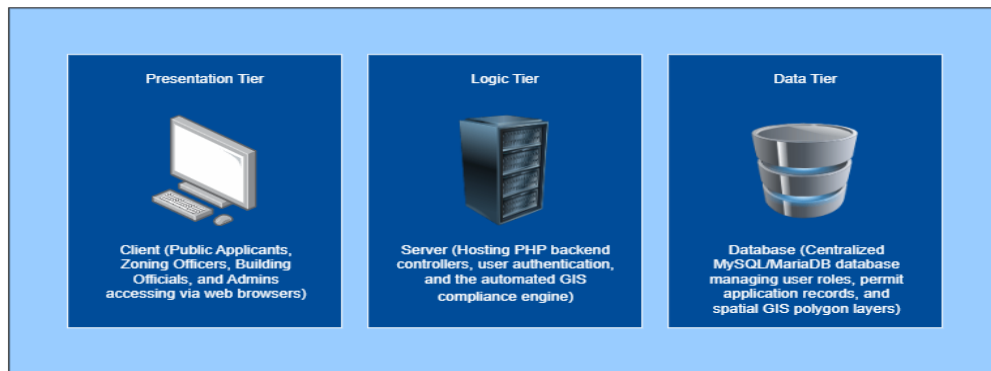
Architecture Diagram



Physical View of the Architecture of a Web Application

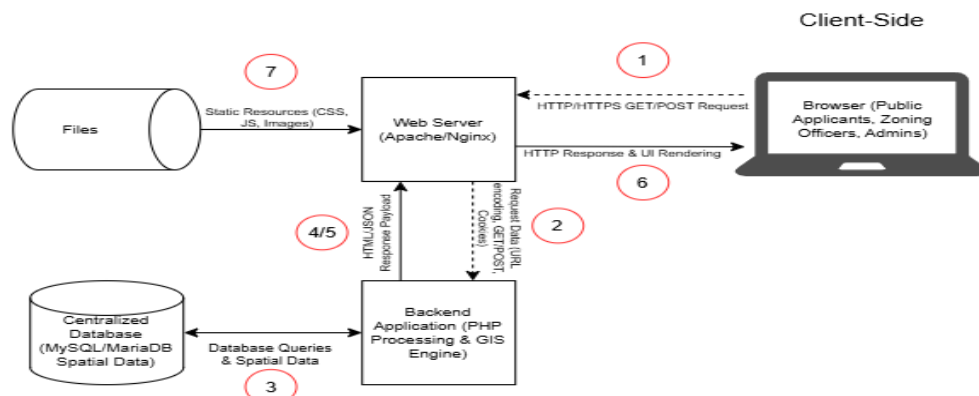


Understanding the Architecture 3-tier

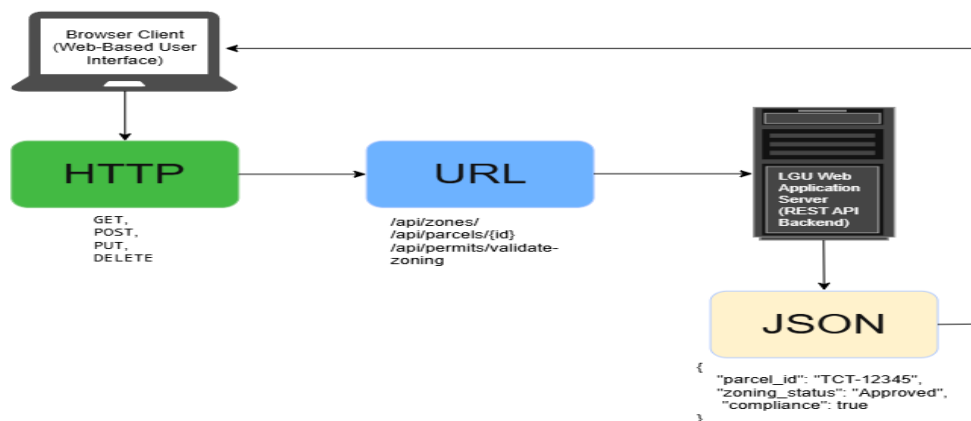


2. Information Systems Integration

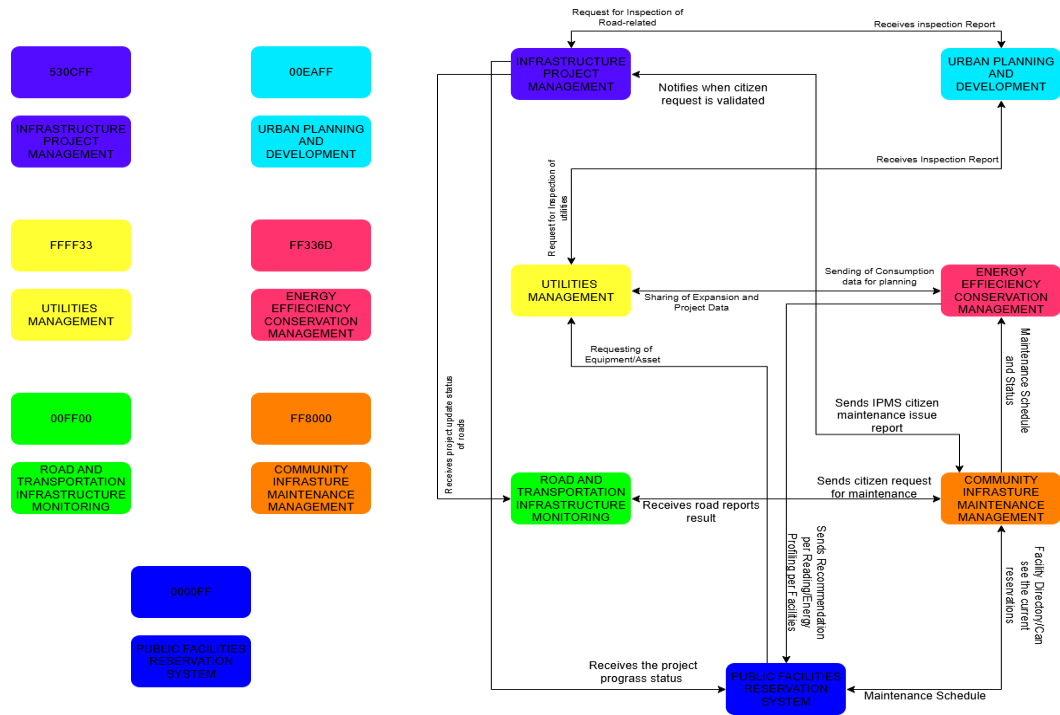
Web Application Data Flow



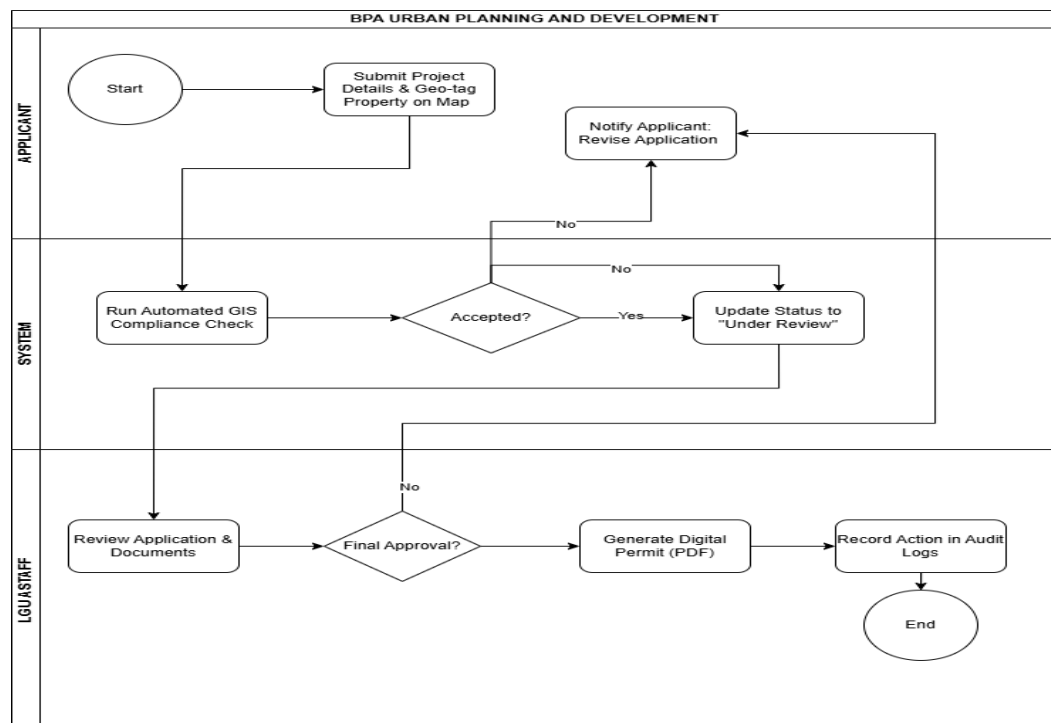
REST API architecture



Business Process Architecture 1

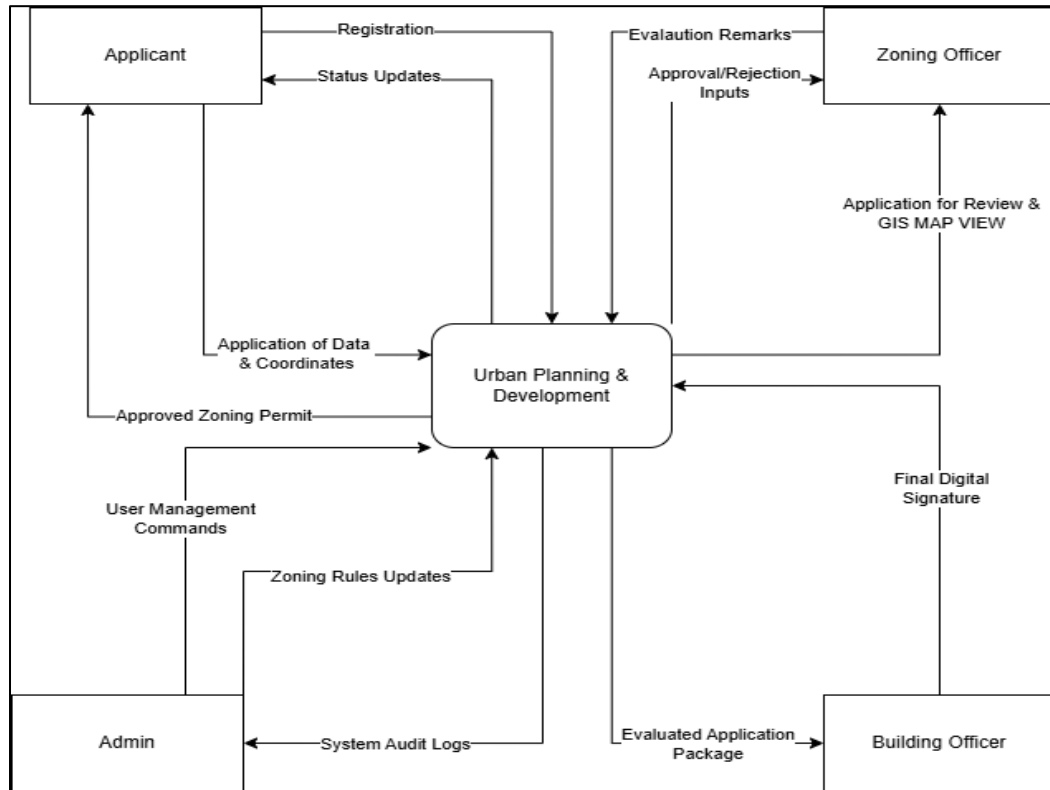


Business Process Architecture 2

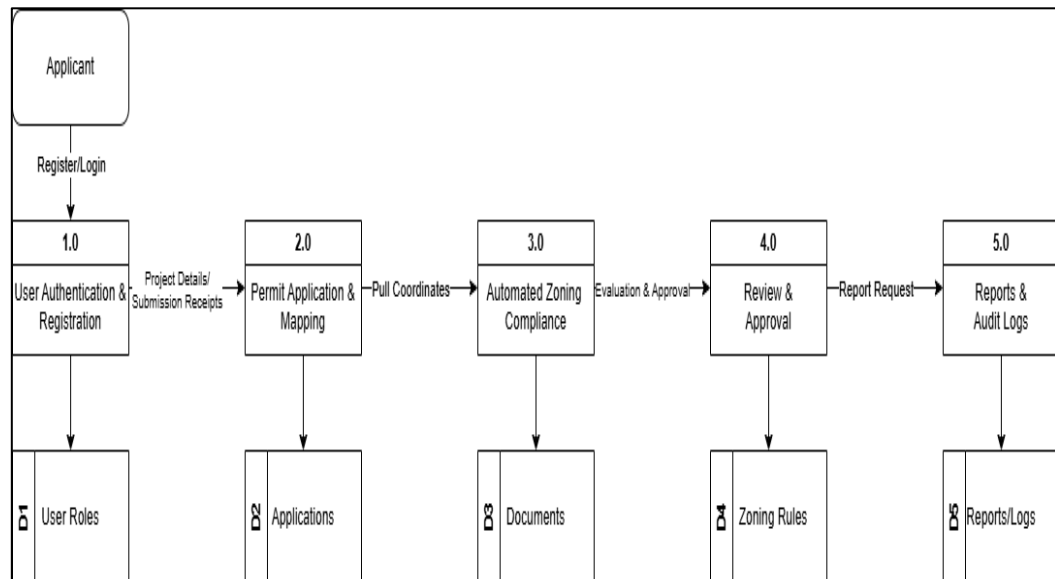


3. Application Design and Development

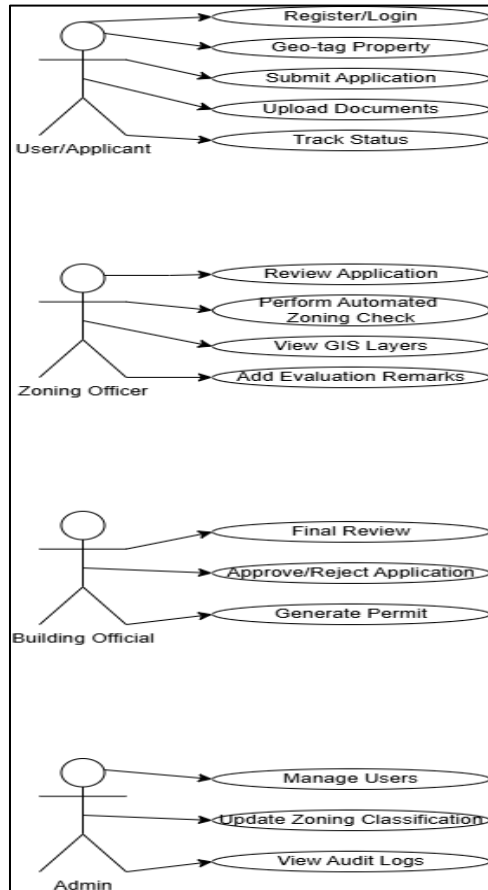
Data Flow Diagram Level 0



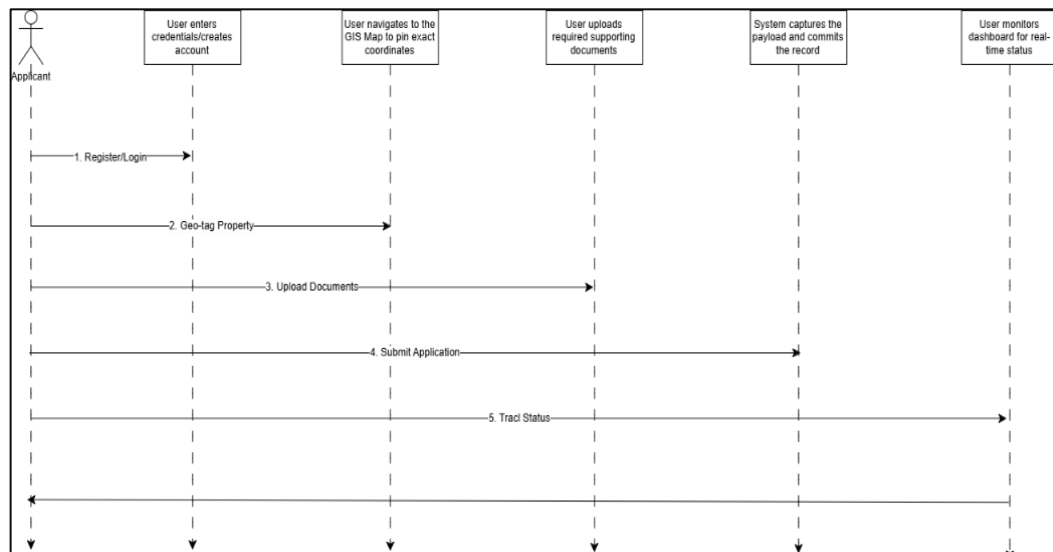
Data Flow Diagram Level 1



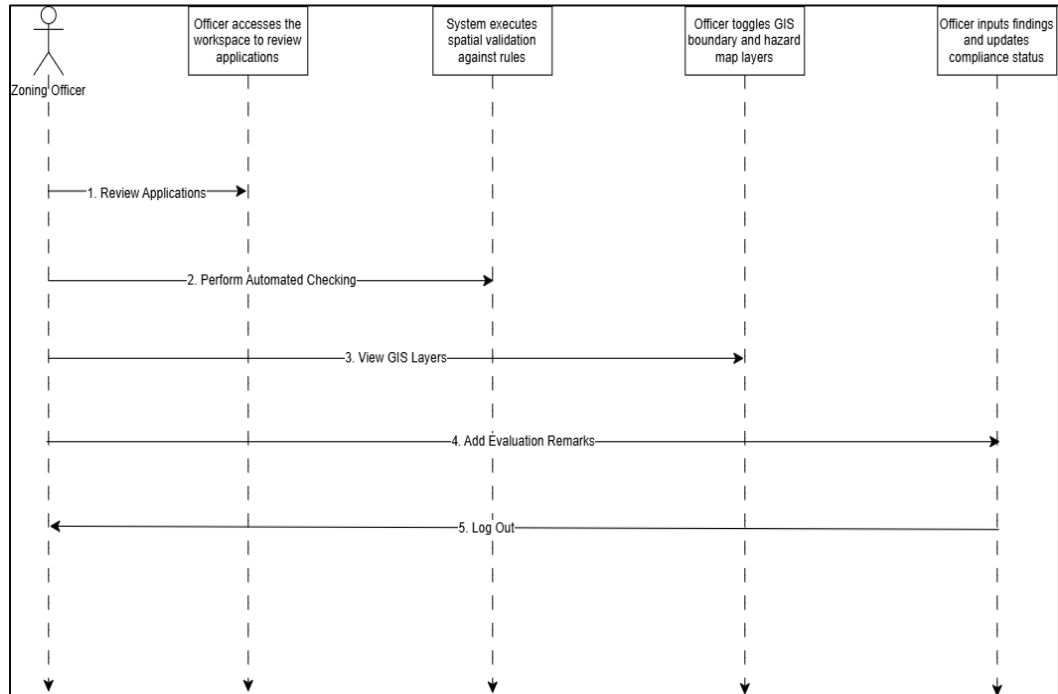
Use Case



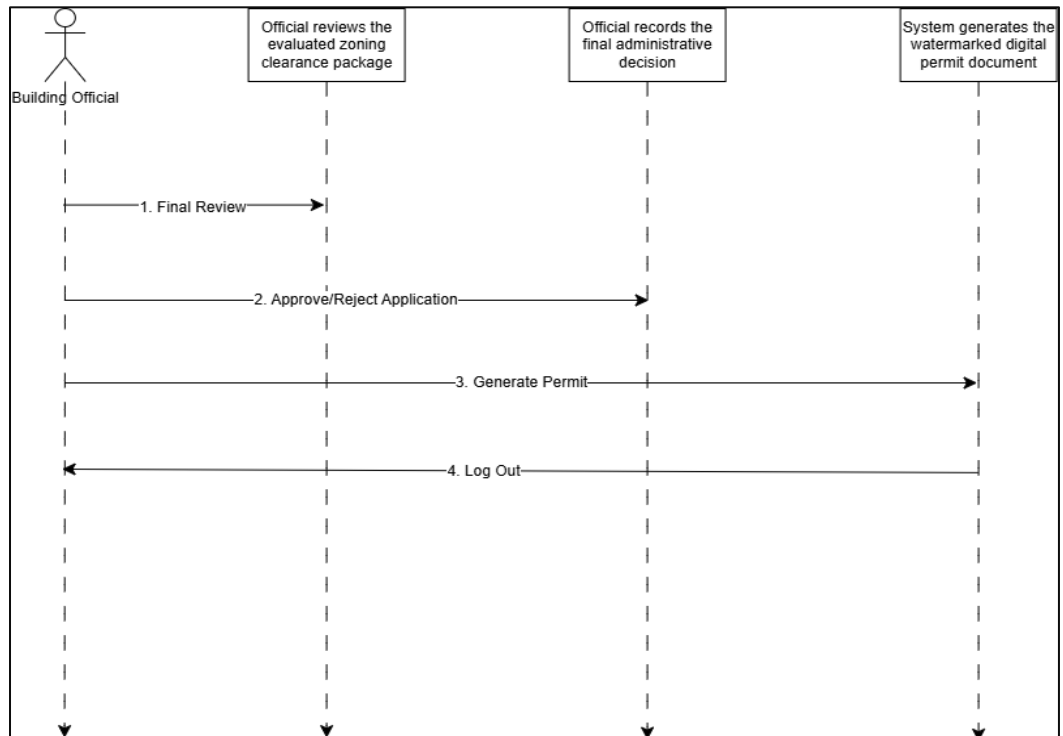
Applicant Sequence Diagram



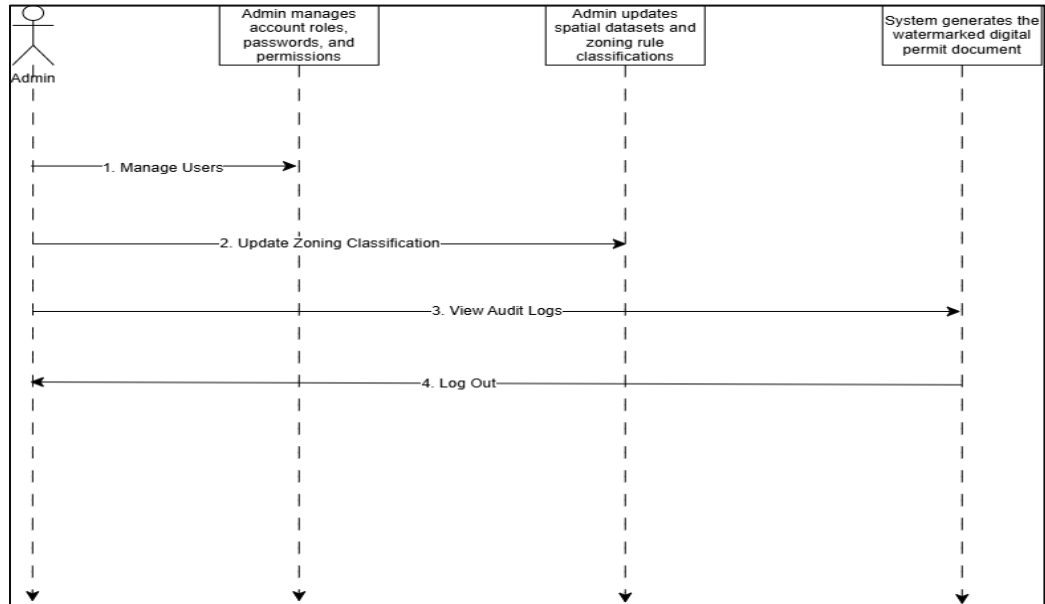
Zoning Officer Sequence Diagram



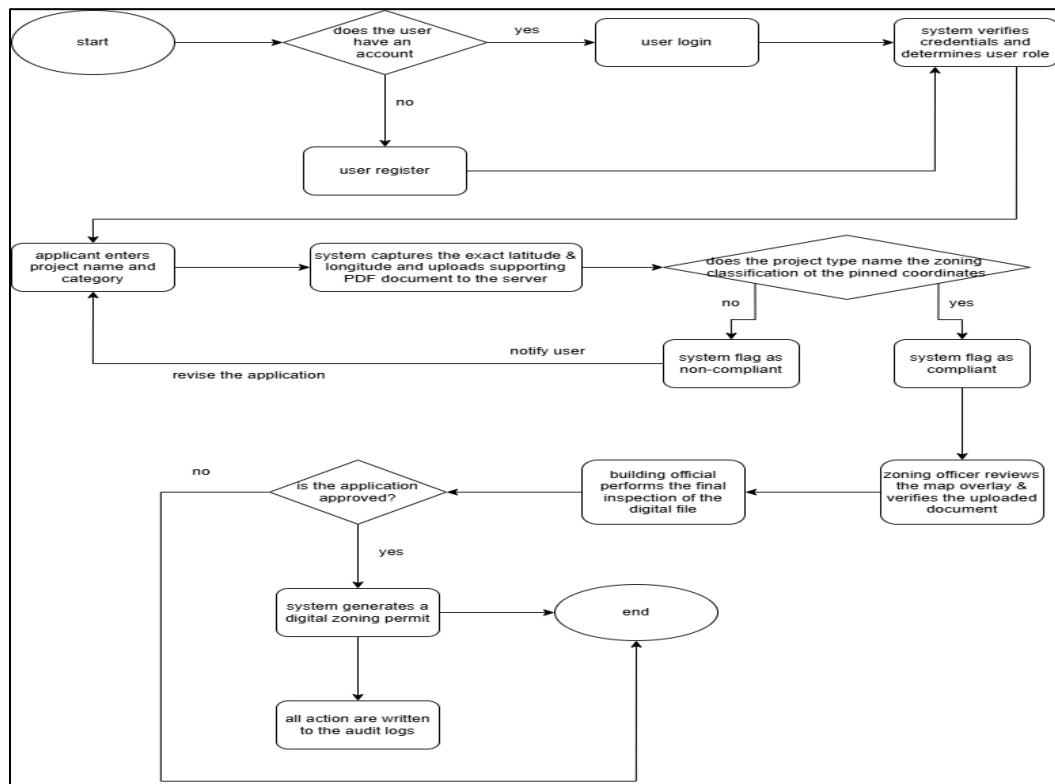
Building Officer Sequence Diagram



Admin Sequence Diagram

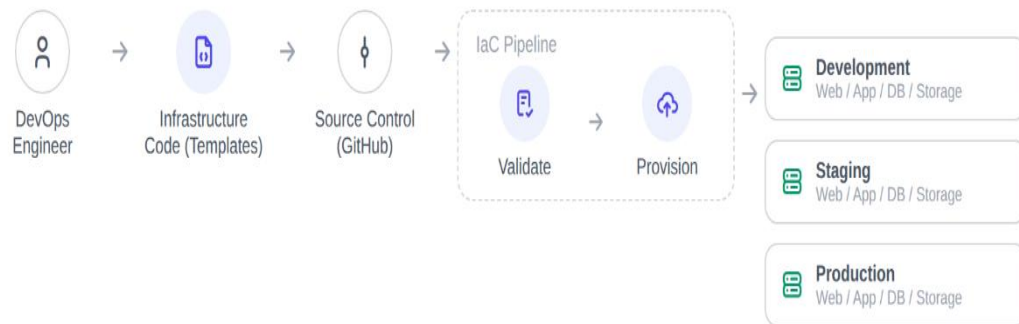


Flowchart



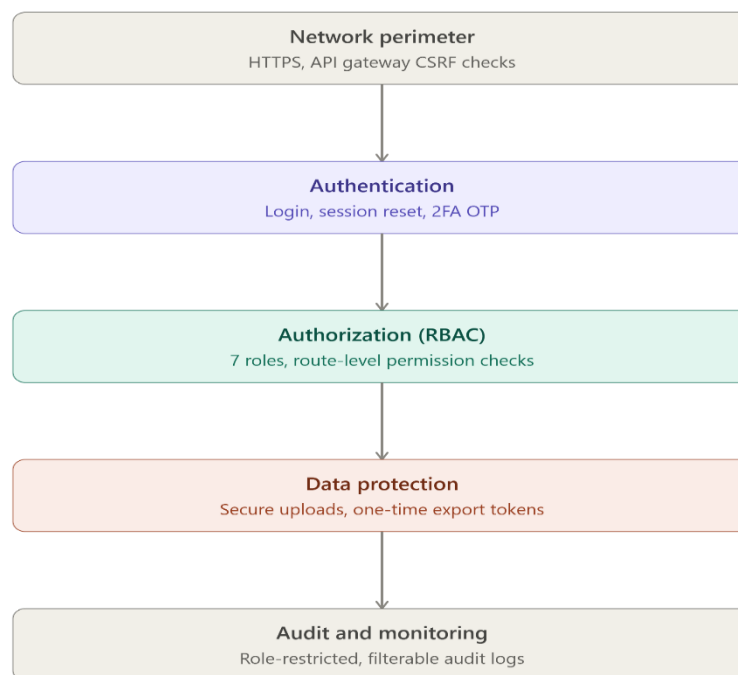
6. Deployment and Infrastructure

Infrastructure as a Code (IaC)



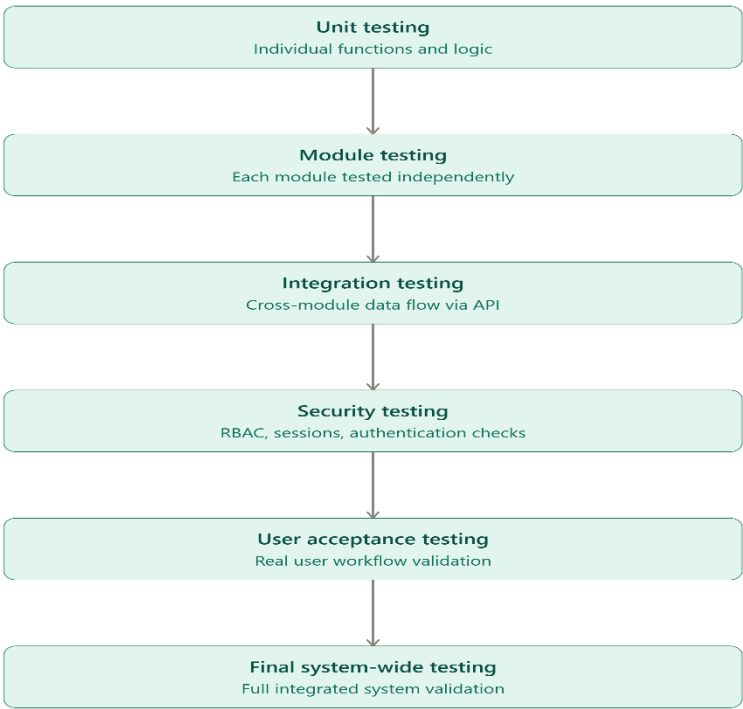
7. Security Measures:

Security Architecture Diagram



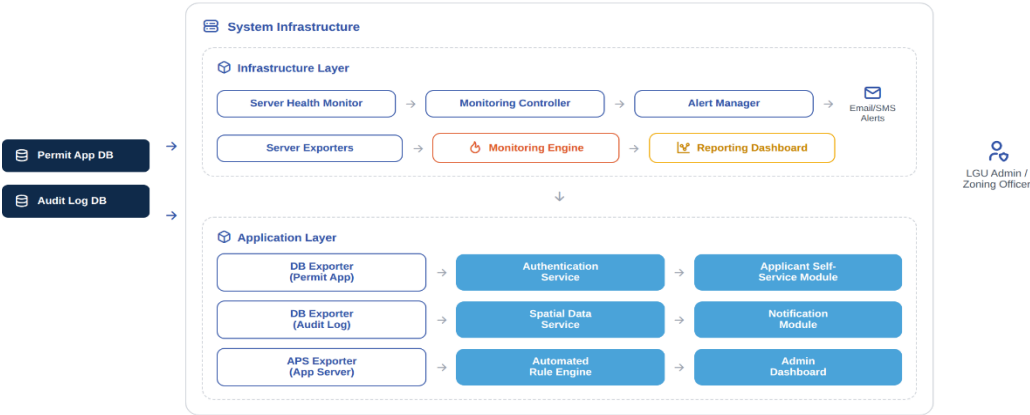
8. Testing and Quality Assurance

Testing Process Flowchart

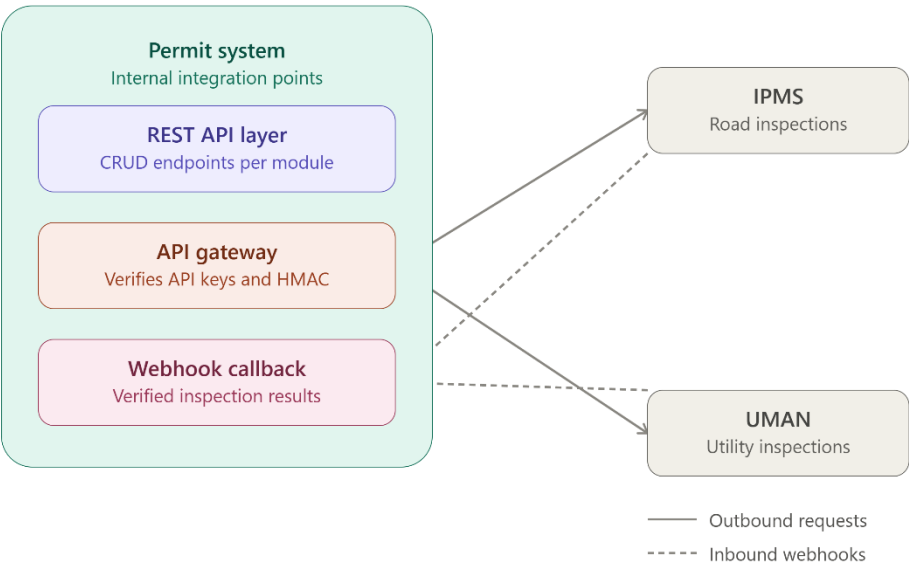


9. System Monitoring and Maintenance:

Monitoring and Alerting

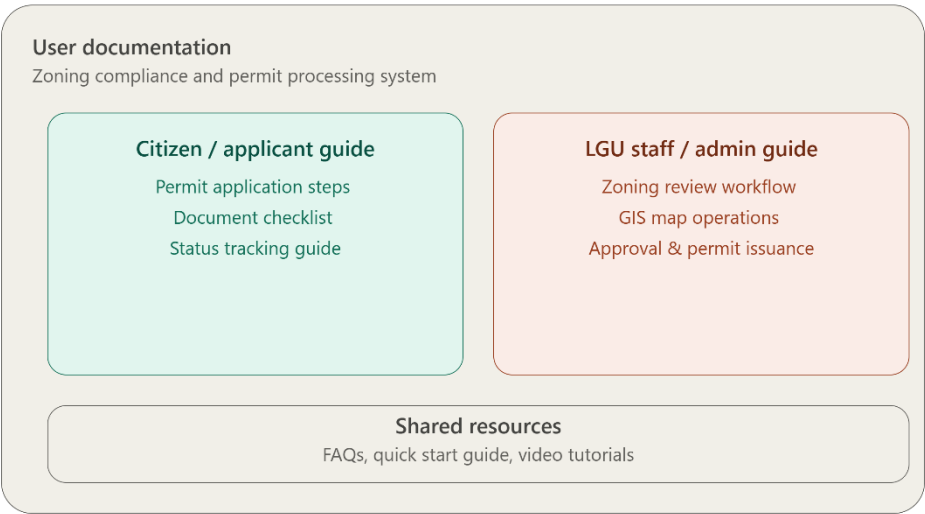


10. APIs and Integration Points:



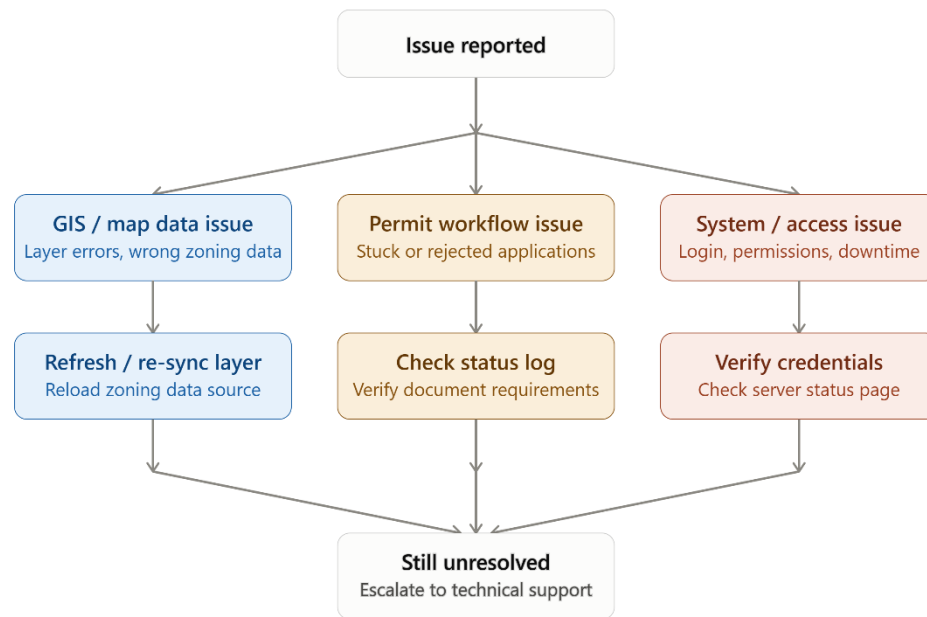
11. User Documentation:

User Documentation Structure for the Zoning GIS System



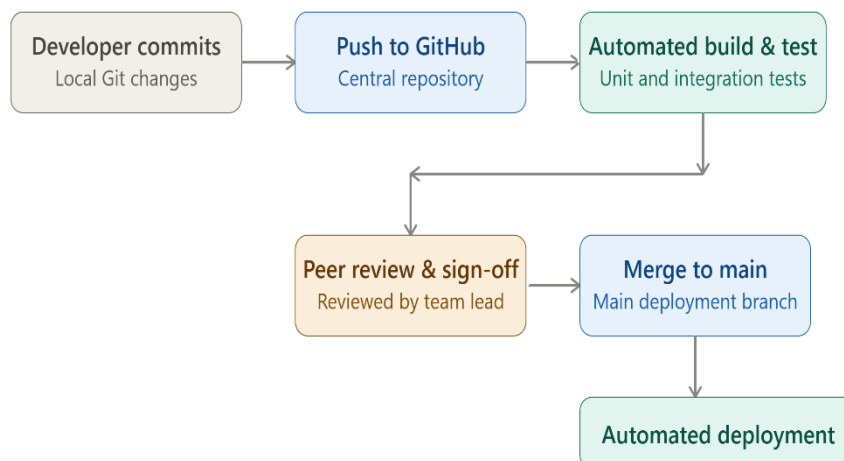
12. Known Issues and Troubleshooting

Known Issues and Troubleshooting Flow for the Zoning GIS System



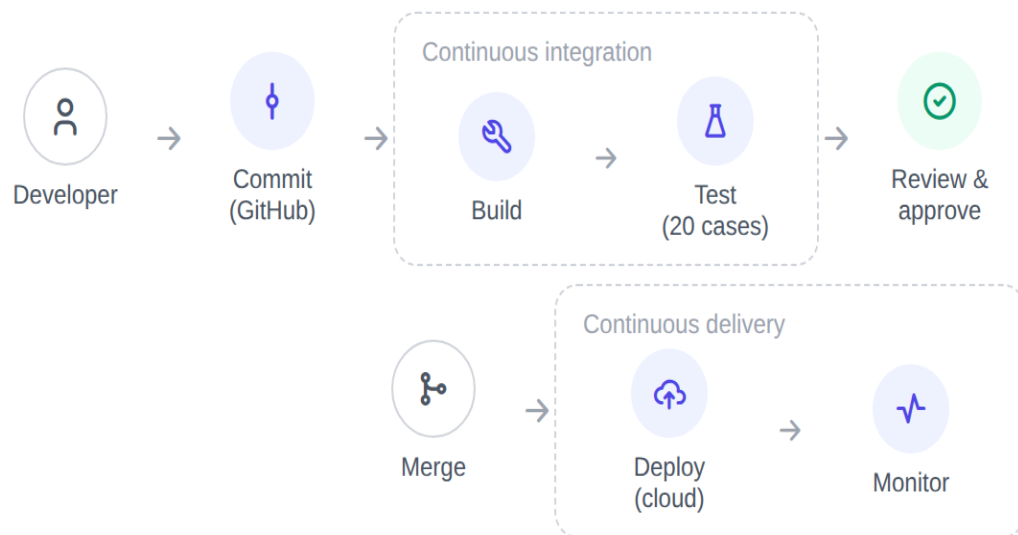
13. Version Control and Source Code Repository

Version Control and Source Code Repository Workflow

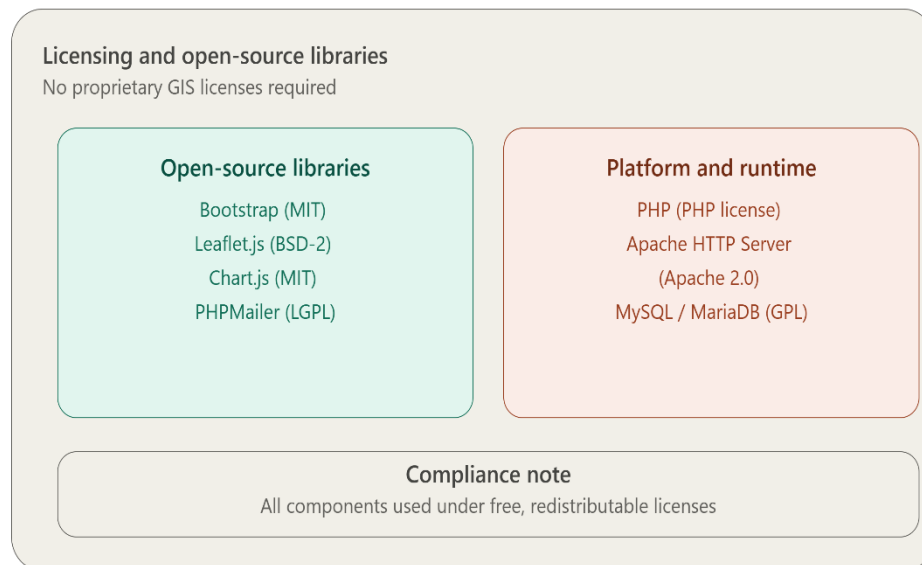


14. DevOps and Continuous Integration/Continuous Deployment (CI/CD):

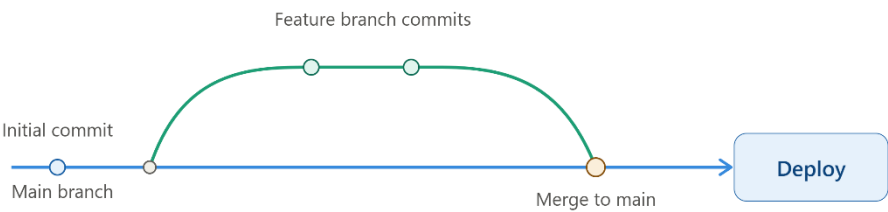
CI/CD Pipeline



15. Licensing and Open-Source Libraries:



16. Performance Metrics and Monitoring:



Appendix E: Grammarian Curriculum Vitae



Contact

Phone
0905 024 4029

Email
gwyberloux.22@gmail.com

Address
17 E Aguinaldo Street, Brgy. San Agustin,
Novaliches, Quezon City

Education

TERTIARY
Pangasinan State University- Urdaneta
City Campus
Bachelor of Secondary Education - Major
in Filipino
Leadership Awardee
2018-2022

SECONDARY
Don Ramon E. Costales Memorial Nat'l
High School
ALS Passer
2016-2017

PRIMARY
St. Albert the Great School
Salutatorian
2005-2013

Skills

Work Management

Positive Attitude

Leadership

Teaching Skills

Quick Learner

Collaborative and Cooperative

Gwyneth Berni Karloux F. Manongdo

To pursue a challenging career and to be part of a well-established school where I can share my skills and knowledge and discover more potential in me, most especially in the world of teaching.

Experience

- Teaching Internship
Pangasinan State University - Urdaneta City Campus
Urdaneta City, Pangasinan
March-June 2022
- Editor-in-Chief
Pangasinan State University-Urdaneta City Campus
Technoscope Publications
Urdaneta City, Pangasinan
Academic Year 2021-2022
- Advance ROTC Officer
Pangasinan State University - Urdaneta City Campus
Department of Military Science and Tactics
105th (PGN-E) Community Defense Center, 1RCDG, ARESCOM

Seminars and Webinars

Seminar for Basic Legislative Procedures
Pangasinan State University-Lingayen Campus
March 15, 2022

Webinar-Workshop Para Sa Mga Nagpapakadalubhasa sa Filipino
"LAYTS, KAMERA, AWRA: Ang Sining ng Pag-arte sa Entablado,
Telebisyon at Pelikula"
Pangasinan State University-Urdaneta City Campus
January 22, 2022

Reference

Ms. Richelyn Villasor
Research Coordinator, BSIT/CCS
Phone: 09465233195
Email: nylerich94@gmail.com

Appendix F: Technical Adviser Curriculum Vitae

ALDREN S. MAMATUS



Blk 20 Lot 8, Calamansian St, Capitol Park, Caloocan City, 1400



December 9, 1993



0966 630 5970 / 0976 551 9906



Drenmatus09@gmail.com



PROFILE

To obtain a position where I could effectively apply my skills and knowledge I've learned and to give an opportunity to work productively and accurately in your company.

EDUCATION

06.2025 – Present	Masters in Information Technology First City Providential College
07.2019 - 07.2021	Bestlink College of the Philippines Certification in Professional Education
06.2014 - 05.2017	Bestlink College of the Philippines Bachelor of Science in Information Technology

EXPERIENCE

08.2024 - present	Bestlink College of the Philippines College Instructor
02.2023 – 04.2024	Zapport Services Inc. VideoEditor & IT Supporting Staff
02.2021 – 07.2021	PCEXPRESS Management Trainee
06.2019 – 03.2020	Bestlink College of the Philippines College Instructor
06.2017 – 04.2019	St. Dominic Institute – Fairview Faculty Teacher
09/2017 - 05/2019	Sonshine Media Network International VideoEditor & Social Media Staff

SKILLS

- ✓ Computer literacy and familiarity with graphics software.
- ✓ To be thorough and pay attention to detail.
- ✓ The ability to work well with others.
- ✓ Knowledge of media production and communication.
- ✓ To be flexible and open to change.

SOFTWARE SKILLS

- ✓ Microsoft Office
- ✓ Adobe Photoshop
- ✓ Adobe Premiere
- ✓ Capcut
- ✓ Canva

PERSONALITY

Communicative	Punctuality
Creativity	Organized

Aldren Sevilla Mamatus
Name & Signature

Appendix G: Research Adviser Curriculum Vitae



CONTACT

09057669976

Ronaldroldan101@gmail.com

16 Esmeralda St. Brgy Tandang Sora QC

EDUCATION

2015 - 2019
BESTLINK COLLEGE OF THE PHL

- Bachelor of Science in Information Technology

2019 - 2022
BESTLINK COLLEGE OF THE PHL

- Certificate of Professional Education

2022 - Ongoing
POLYTECHNIC UNIVERSITY OF THE PHL

- Master of Information Technology (9units)

SKILLS

- Web Development (HTML, CSS, JavaScript, PHP)
- Programming (C, Java, C++)
- SQL & Database Management
- System Analysis and Design
- Hardware and Network Troubleshooting
- Scrum Master and Agile Practices
- Project Management & Leadership

RONALD G. ROLDAN JR

INFORMATION TECHNOLOGY

PROFILE

To obtain a position in a growth-oriented organization where I can utilize my expertise in IT project management, system development, and research advising to lead projects, deliver high-quality outputs on time, and contribute to the organization's efficiency and innovation.

WORK EXPERIENCE

BESTLINK COLLEGE OF THE PHILIPPINES	2019 - 2025
IT INSTRUCTOR	
BESTLINK COLLEGE OF THE PHILIPPINES	2023 - 2025
BSIT Research Coordinator	
BESTLINK COLLEGE OF THE PHILIPPINES	2019 - 2025
BSIT Research Adviser	

- Advised and presented research papers locally and internationally (Kuala Lumpur, Malaysia)
- Developed and managed multiple IT Capstone and ERP projects (HRIS, Financial, Freight, and Hospital Management Systems)
- Delivered resource speaking engagements on BSIT research agenda and project development
- Module developer and panelist for IT capstone defenses

KEY PROJECTS AND SUPERVISED

- Freight Management System
- Financial Management System
- School and Hospital Management Systems
- Airport Management System
- HRIS and Payroll Management Systems
- Transport Network Vehicle System

PERSONAL INFORMATION

Age:	28
DOB:	February 3, 1997
Nationality:	Filipino
Religion:	Christian
Civil Status:	Single
Height:	5'5"
Weight:	65 kg

CERTIFICATIONS AND TRAININGS

- AGILE DEVELOPMENT METHODOLOGY (2018)
- INFORMATION TECHNOLOGY IN MIXED REALITY (2018)
- FACULTY IN-SERVICE TRAININGS (2020, 2022, 2023, 2024)
- IN-SERVICE TRAINING FOR TEACHERS (INSET)
- FACULTY RESEARCH SEMINARS

TEACHING EXPERIENCE

- COMPUTER PROGRAMMING 1
- WEB DEVELOPMENT 1 & 2
- SYSTEM INTEGRATION AND ARCHITECTURE
- BUSINESS PROCESS MANAGEMENT
- NETWORKING 1, ITSP2 WEB SECURITY
- DATABASE MANAGEMENT SYSTEM 1
- INFORMATION MANAGEMENT
- CAPSTONE PROJECT AND RESEARCH 1 & 2

CHARACTER REFERENCES

MS. JESSICA BENARES
RMS - TEACHER
☎ 09168229335

MR. JORGE B. LUCERO
IT PROFESSOR,
BESTLINK COLLEGE OF THE PHILIPPINES
☎ 09460669897

Appendix H : Researcher's Curriculum Vitae

ALTHANIEL O. ABORDO



Camarin Road St. Barangay 178, Camarin, Caloocan City



October 10, 2003



09507073981



Abordoalthaniel05@gmail.com



PROFILE

Motivated multitasker seeking to apply my skills in a growth-oriented role. Focused on delivering warm customer service and dedicated to continuous learning and improvement.

EDUCATION

01.2023 – Present	Bestlink College of the Philippines Bachelor of Science in Information Technology
09.2021 - 07.2022	Mystical Rose School of Caloocan Information and Communications Technology.
06.2019 - 03.2020	Camarin High School Junior High School

SKILLS

- ✓ Basic Microsoft Editing
- ✓ Time Management
- ✓ Fast Learner

SOFTWARE SKILLS

- ✓ MS Word
- ✓ Canva
- ✓ PowerPoint

PERSONALITY

Disciplined	Open-minded
Kind	Reliable

Althaniel Abordo
Name & Signature

MA.CRISTINA D. BUENAFLO



BLDG G BISTEKVILLE BRGY. CULIAT, QUEZON CITY



May 20, 2004



09105218722



Crisitnabuenaflo20@gmail.com



PROFILE

Detail-oriented IT student specializing in secure, user-centric web and UI design. I leverage my technical background to build accessible interfaces and am eager to apply my creative problem-solving skills as an intern within a vital public institution.

EDUCATION

2022 – Present	Bestlink College of the Philippines Bachelor of Science in Information Technology
2020 - 2022	Bestlink College of the Philippines/Mati National Comprehensive High School Senior High School
2016 - 2020	Ernesto Rondon High School Junior High School

SKILLS

- ✓ Quality-Driven: Consistently meets objectives through self-discipline and a commitment to excellence.
- ✓ Thrives Under Pressure: Proven ability to remain calm and efficient during time-sensitive projects.
- ✓ Detail-Oriented: Leverages keen observation skills to ensure absolute accuracy in technical documentation.
- ✓ Tech-Agile: Proactive in learning and implementing emerging technologies.
- ✓ Team Contributor: Facilitates seamless workflows through clear communication and organized time management.

SOFTWARE SKILLS

- ✓ MS Office
- ✓ Capcut
- ✓ Canva

PERSONALITY

Disciplined	Approachable
Self-Motivated	Versatile

Ma.Cristina D. Buenaflor
Name & Signature

Joshua Canceran



615 Int 0 Bagbag Novaliches, Quezon City



February 24, 2000



09503787167



Joshuacanceran230@gmail.com



PROFILE

To acquire valuable knowledge and skills to complete those that I have learned from school in an actual job environment. In return, I offer my service and determination to learn and discover new things throughout the duration of my training period

EDUCATION

01.2023 – Present	Bestlink College of the Philippines Bachelor of Science in Information Technology
09.2016 - 07.2018	Electron College of the Philippines Information and Communications Technology.
06.2012 - 03.2016	Tandang Sora National High School Junior High School

SKILLS

- ✓ Basic Microsoft Editing
- ✓ Time Management
- ✓ Fast Learner

SOFTWARE SKILLS

- ✓ MS Word
- ✓ Canva
- ✓ PowerPoint

PERSONALITY

Friendly	Polite
Kind	Energetic

Joshua Canceran
Name & Signature

YUMIE MARGARETH N. DALAGAN



6835 Sto Nino St. Maligaya Park, Caloocan City



November 07, 2003



09207249702



Yumiedalagan01@gmail.com



PROFILE

Detail-oriented IT student with experience in web development, and databases. Quick learner and problem-solver seeking to contribute to an innovative team.

EDUCATION

01.2023 – Present	Bestlink College of the Philippines Bachelor of Science in Information Technology
09.2019 - 07.2021	South East-Asia Institute of Trade and Technology Information and Communications Technology.
06.2016 - 03.2018	Cielito Zamora Junior High School Junior High School

SKILLS

- ✓ Reliable, disciplined, and committed to high-quality results.
- ✓ Calm and productive under pressure or tight deadlines.
- ✓ Committed to accuracy in technical tasks and documentation.
- ✓ Flexible and quick to master new technologies.
- ✓ Strong time management and ability to work well with others.

SOFTWARE SKILLS

- ✓ VS Code
- ✓ PhpMyAdmin
- ✓ ~~Xampp~~
- ✓ ~~Capcut~~
- ✓ Canva

PERSONALITY

Analytical	Detail-Oriented
Reliable	Proactive


Yumie Margareth Dalagan
Name & Signature

Justin Michael Espino



Phase 4B Pkg 7 Blk 59 Lot 3 Bagong Silang, Caloocan City



November 25, 2001



09708350056



Justinespino12@gmail.com



PROFILE

Motivated student aiming a great future. Can utilize my foundational knowledge, learn new skills, and contribute to innovative projects.

EDUCATION

08.2024 – Present	Bestlink College of the Philippines Bachelor of Science in Information Technology
09.2020 - 07.2022	Vicente Malapitan Senior High School Information and Communications Technology.
06.2012 - 03.2018	Camarin High School Junior High School

SKILLS

- ✓ Teamwork Collaboration
- ✓ Professionalism
- ✓ Time Management

SOFTWARE SKILLS

- ✓ MS Word
- ✓ Canva
- ✓ PowerPoint

PERSONALITY

Respectful	Attentive
Kind	Humble

ROSARY MAE C. VELICARIA



Sto. Nino Meyxauayan, Bulacan



October 8, 2002



09553479538



Rosvelicaria19@gmail.com



PROFILE

Analytical Information Security major proficient in secure web development and technical manuscript refinement. I apply a rigorous mindset to basic coding and documentation to provide resourceful, high-quality solutions for a vital public institution.

EDUCATION

2022 – Present	Bestlink College of the Philippines Bachelor of Science in Information Technology
2019 - 2021	St. Mary's Academy of Sto. Nino/Meycauayan National High School Senior High School
2016 - 2020	St. Mary's Academy of Sto. Nino Junior High School

SKILLS

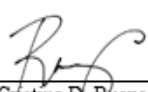
- ✓ **Integrity-Driven:** Committed to maintaining the highest ethical standards and data privacy in every technical project.
- ✓ **Quality-Focused:** Dedicated to rigorous testing and refinement to ensure all code and documentation meet professional benchmarks.
- ✓ **Continuous Learner:** Passionate about staying current with evolving IT trends and proactively upgrading technical skill sets.
- ✓ **Systematic Thinker:** Approaches complex IT tasks with a logical, step-by-step methodology to minimize errors and maximize efficiency.

SOFTWARE SKILLS

- ✓ MS Office
- ✓ Capcut
- ✓ Canva
- ✓ Xampp
- ✓ VsCode

PERSONALITY

Resourceful	Empathetic
Enthusiastic	Supportive


Ma. Cristina D. Buenaflor
Name & Signature